

UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.

Before the Honorable Clark S. Cheney
Chief Administrative Law Judge

In the Matter of

CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF

Investigation No. 337-TA-1500

**COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S MOTION FOR LEAVE TO
AMEND THE COMPLAINT AND NOTICE OF INVESTIGATION (Mtn. No. 1500-003)**

Pursuant to Commission Rule 210.14, 19 C.F.R. § 210.14, Complainant GlobalFoundries U.S. Inc. (“GlobalFoundries”) respectfully moves for leave to amend its Complaint (“Complaint”) and the Notice of Investigation (“NOI”) to add allegations that Respondent Tower Semiconductor Ltd. and its related entities (collectively, “Tower”) infringe dependent claims 8-12 and 19 of U.S. Patent No. 8,507,983 (the “Additional ’983 Claims”). A proposed amended complaint and a redline incorporating these amendments are enclosed at Exhibits A and B.

Good cause exists for amending the Complaint and the NOI to include these additional claims and Tower will suffer no prejudice from their inclusion. Good cause exists because the presence [REDACTED] was discoverable only through Tower’s confidential process documentation, which was produced for the first time in mid-August 2026. GlobalFoundries acted with diligence, reviewing Tower’s document productions, identifying the relevant disclosed features, and notifying Tower of its intent to amend within two weeks of Tower’s productions. Tower suffers no prejudice from the amendment: this Investigation remains in its early stages, with more than three months of fact discovery remaining and expert reports not due until January 2027. Moreover, the Additional ’983

[REDACTED]

Claims depend from independent claims that GlobalFoundries asserted in its Complaint, and the Additional '983 Claims have been “at issue” in this Investigation since at least July 16, 2026, when they were included in GlobalFoundries’ Domestic Industry contention interrogatory responses. Granting this Motion also serves the public interest by avoiding the need to remedy Tower’s infringement of these additional claims in a second 337 investigation.

Ground Rule 5.1 Certification

GlobalFoundries certifies it made reasonable, good faith efforts to resolve this matter with Tower prior to filing this motion, including a meet and confer on August 27, 2026, and an exchange of emails on August 28, 2026. Tower did not provide its ultimate position on the motion before filing.

I. BACKGROUND

On March 26, 2026, GlobalFoundries filed its Complaint requesting that the Commission institute an investigation into Tower’s alleged infringement of six patents, including the '983 Patent. *See* Complaint at ¶ 1.1. GlobalFoundries asserted claims 1-7, 13-18, and 20 of the '983 Patent, which relate to high-voltage semiconductor devices used in power management applications. *Id.* at ¶ 5.9.

On April 28, 2026, the Commission instituted this Investigation, defining the scope of the investigation to include the same asserted claims of the '983 Patent. *See* NOI at 2. The '983 Patent is directed to “semiconductor devices that can be used in high-voltage applications,” with the asserted claims 1, 7, 13, and 18 pertaining to high-voltage semiconductor devices having a channel with a channel length controlled by ion implantation. Complaint at ¶¶ 5.11-5.12. The Additional

[REDACTED]

[REDACTED]

'983 Claims depend from the already-asserted '983 claims¹ and recite specific tilt angle ion implantation features. Claims 8 and 19 recite that the ion implantation comprises a tilt angle implant; claims 9 and 11 specify a tilt angle between about 1 degree and 45 degrees; and claims 10 and 12 recite multiple or quad tilt angle implants. *See* Ex. C ('983 Patent) at Cl. 8-19.

Prior to filing its Complaint, GlobalFoundries had access only to publicly available information derived from teardowns of a device made using Tower's 180nm process. The technical details necessary to assert the Additional '983 Claims, *i.e.*, [REDACTED], [REDACTED], were not publicly available and could not be discerned through teardown imaging. Unlike surface-level process characteristics, [REDACTED] are embedded deep within a foundry's proprietary manufacturing recipes and cannot be detected through reverse engineering.

On July 13, 2026, Tower made its first production of 11 substantive technical documents, including a single process flow on the secure computer relating to [REDACTED] [REDACTED]), but had not yet produced the corresponding design manuals for any of its [REDACTED]. Process flow documents are highly detailed documents that set out the step-by-step sequence of the semiconductor manufacturing process. Design manuals set out the basic design rules and act as a blueprint for multiple processes. GlobalFoundries promptly reviewed these materials the following day to prepare its responses to Tower's infringement contention interrogatories. None of the materials produced by that date [REDACTED] [REDACTED]. On July 16, 2026, by which time Tower had produced a grand total of 23 technical documents on its secure computer (11 of which were produced on July 16),

¹ Specifically, Additional '983 Claims 8-12 depend from already-asserted independent Claim 1, and Additional '983 Claim 19 depends from already-asserted independent Claim 13.

[REDACTED]

GlobalFoundries served its initial responses to Tower’s infringement and domestic industry contention interrogatories, including infringement claim charts for asserted claims 1, 7, 13, and 18 of the ’983 Patent and domestic industry claim charts that included the Additional ’983 Claims among others.

Nearly a month later, on August 11, 2026, Tower produced a design manual corresponding to its [REDACTED]. Ex. D [REDACTED] at TOWER_1500_00041811. Upon review, GlobalFoundries discovered that [REDACTED]. Specifically, the design manual included illustrations revealing that [REDACTED]—a feature of which GlobalFoundries was previously unaware. *E.g., id.* at -14861. To investigate this newly discovered feature, GlobalFoundries’ technical expert re-reviewed the relevant process flows on Tower’s secure computer from August 17 to 19, 2026. Combined with its knowledge from the [REDACTED] design rule manual, GlobalFoundries learned for the first time several aspects of [REDACTED], including: [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Altogether, these newfound discoveries point to the possibility of a [REDACTED].

On August 26, 2026—within one week of completing its analysis—GlobalFoundries promptly notified Tower of its intent to amend its Complaint to add the Additional ’983 Claims. The parties met and conferred the next day. GlobalFoundries explained its position and further

[REDACTED]

informed Tower that a detailed recipe and/or implant program of record, which Tower has still yet to produce, is further necessary for GlobalFoundries to determine whether the [REDACTED]

II. LEGAL STANDARD

Commission Rule 210.14 provides that “[a]fter an investigation has been instituted, the complaint or notice of investigation may be amended only by leave of the Commission for good cause shown and upon such conditions as are necessary to avoid prejudicing the public interest and the rights of the parties to the investigation.” 19 C.F.R. § 210.14(b)(1). Good cause exists where a complainant obtains new information during discovery that was not previously known. *Certain Variable Speed Wind Turbine Generators and Components Thereof*, Inv. 337-TA-1218, Order No. 10 at 2 (Dec. 2, 2020) (“The record supports a conclusion that [complainant] gained confidential information supporting allegations of infringement of the Additional Claims after it filed the complaint.”); *see also Certain Electronic Imaging Devices*, Inv. 337-TA-850, Order No. 10 at 2 (Oct. 26, 2012); *Certain Cartridges for Electronic Nicotine Delivery Systems and Components Thereof*, Inv. 337-TA-1141, Order No. 29 at 2 (May 15, 2019). Moreover, consolidation of all relevant claims in a single investigation, including previously unasserted dependent claims, “tip[s] the scales” in favor of amendment. *See, e.g., Certain Light-Emitting Diode Products, Fixtures, and Components Thereof*, Inv. 337-TA-1213, Order No. 13 at 3 (Jan. 8, 2021); *Certain Semiconductor Chips with DRAM Circuitry, and Modules and Products Containing Same*, Inv. 337-TA-819, Order No. 7 at 2 (May 10, 2012) (“[T]he public interest [is] served by the inclusion of all relevant claims in a single investigation and for that reason good cause exists for [a] proposed amendment.”). No prejudice exists where “adding the Additional Claims to this investigation will not expand the scope of accused products” at issue in the investigation, and

[REDACTED]

“ample time remains for the parties to conduct discovery.” *Variable Speed Wind Turbine Generators*, Inv. 337-TA-1218, Order No. 10 at 2 (Dec. 2, 2020).

III. ARGUMENT

A. Good Cause Exists to Amend the Complaint to Add the Additional '983 Claims.

GlobalFoundries seeks leave to add dependent claims 8-12 and 19 of the '983 Patent to its infringement allegations. Good cause exists for GlobalFoundries' amendment for at least three independent reasons.

First, the information supporting infringement of the Additional '983 Claims was not available to GlobalFoundries when it filed its Complaint. [REDACTED] [REDACTED] are not visible from teardown imaging of the Accused Products. GlobalFoundries could have only identified them by reviewing Tower's confidential process documentation, and more specifically Tower's design manuals that were produced in mid-August 2026, more than four months after GlobalFoundries filed its Complaint. *See Certain Wind Turbine Generators*, Inv. 337-TA-1218, Order No. 10 (finding good cause where “confidential documents confirming infringement of these claims were not produced until” well after the complaint was filed). When GlobalFoundries reviewed the design manuals that Tower produced in mid-August for its [REDACTED], it learned that those [REDACTED] [REDACTED] that Tower had previously produced documents about. Those [REDACTED] indicated the possibility of [REDACTED] [REDACTED]. The design manuals combined with the corresponding process flows indicated more specifically: [REDACTED]

[REDACTED]

[REDACTED],

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Second, GlobalFoundries acted diligently in reviewing Tower’s document productions and identifying these additional claims. GlobalFoundries reviewed Tower’s technical documentation on a rolling basis as they were produced (primarily on the secure computer). It was not until Tower’s August 11, 2026 production that GlobalFoundries first saw a diagram in a design manual corresponding to its [REDACTED]

[REDACTED] feature. With this new information from the design manual serving as a roadmap, GlobalFoundries worked quickly with its expert to confirm this feature in the relevant (highly detailed) process flow documents on Tower’s secure computer over the course of the following week, and then GlobalFoundries notified Tower of its intent to amend on August 26, 2026.² All of this occurred within the span of two weeks. GlobalFoundries continues to seek production of additional technical documentation to further develop its infringement case (for the Additional ’983 Claims and the already asserted claims), but did not wish to delay in bringing this motion while waiting for further discovery.

Third, amendment will promote efficiency by enabling all relevant claims to be adjudicated in a single Investigation. Processes that fall within the scope of the Additional ’983 Claims necessarily relate closely to claims 1, 7, 13, and 18, from which the Additional ’983 Claims depend. Denying this motion would require GlobalFoundries to “file a new complaint . . . with the exact

² Despite GlobalFoundries’ repeated objections to Towers’ production of the design manuals on its secure computer, Tower continued to produce only self-selected “relevant” pages electronically and insisted that GlobalFoundries could only review the full versions on its secure computer. To date, Tower has not provided any reason why the pages available only on the secure computer warrant higher confidentiality protection than the “relevant” pages it produced electronically.

[REDACTED]

same parties and similar technology,” resulting in “a greater waste of resources for all involved, including [Tower], [GlobalFoundries], and the Commission.” *Certain Light-Emitting Diode Products*, Inv. 337-TA-1213, Order No. 13 at 3; *Certain Display Devices Including Digital Televisions and Monitors*, Inv. 337-TA-713, Order No. 16, 2010 WL 4783430, at *3 (Sept. 10, 2010). It would be more efficient to adjudicate whether Tower infringes these dependent claims in the same Investigation as the independent claim upon which they depend than to require the parties and the Commission to consider similar arguments about the same processes in a separate one. Commission ALJs have recognized that such efficiency concerns “tip[] the scales” in favor of amendment. *Certain Light-Emitting Diode Products*, Inv. No. 337-TA-1213, Order No. 13 at 3 (granting amendment because “[t]ipping the scales on this record is the public interest in adjudicating all relevant claims as efficiently as possible”) (citing *Certain Semiconductor Chips with DRAM Circuitry, and Modules and Products Containing Same*, Inv. 337-TA-819, Order No. 7 at 2 (May 10, 2012)). Here, where GlobalFoundries diligently pursued information that was not available except through discovery, the case for amendment is straightforward, and leave should be granted.

B. Tower Will Suffer No Prejudice.

Tower will not be prejudiced if leave to amend is granted. To begin, “[p]rejudice to the parties is limited in circumstances where the investigation is in its early stages.” *Certain Radiotherapy Systems and Treatment Planning Software, and Components Thereof*, Inv. 337-TA-968, Order No. 12 at 3 (Mar. 9, 2016). This Investigation is in its early stages: fact discovery does not close until December 4, 2026, more than three months from this motion, the parties have not scheduled or taken any depositions, and initial expert reports are not due until January 2027. Tower has sufficient time to conduct necessary fact discovery to build its defense relating to the

[REDACTED]

Additional '983 Claims. This is particularly so where the amendment would not result in a meaningful expansion of the Investigation. *See Certain Variable Speed Wind Turbine Generators*, Inv. No. 337-TA-1218, Comm'n Determination Not to Review ID (Dec. 21, 2020). Where newly asserted "dependent claims only implicate a few additional limitations and are directed to the same accused products that were identified in the complaint[,] . . . the prejudice to [a respondent] is minimal." *LED Lighting Devices, LED Power Supplies, and Components Thereof*, Inv. 337-TA-1081, Order No. 29 at 4 (Feb. 26, 2018). Here, the Additional '983 Claims depend on claims 1, 7, and 13, which are already in the Investigation. GlobalFoundries asserted them for infringement in its Complaint. And Tower presumably has been searching for prior art for the '983 Patent already. It thus cannot claim prejudice on this ground. *Certain Light-Emitting Diode Products*, Inv. 337-TA-1213, Order No. 13 at 3 (finding the "several weeks remaining in discovery and the months remaining before trial are sufficient to cure any 'minimal' prejudice . . . in preparing [a] defense.>").

Neither can Tower claim ignorance or unfair surprise. The Additional '983 Claims have been in issue in this investigation for weeks. These dependent claims, with the addition of claim 19, were charted in GlobalFoundries' Domestic Industry contention interrogatory responses served on July 16, 2026—six weeks before this motion. *See* Ex. E (Excerpts from Exhibit B2 to GlobalFoundries' July 16, 2026 1st Supp. Responses to Tower's 1st Set of Interrogatories). Tower thus has had ample opportunity to carefully analyze these claims, assess their validity, and prepare defenses. For that same reason, Tower cannot claim prejudice to its claim construction briefing, since Tower had more than a month's notice to argue terms for the Additional '983 Claims in its opening claim construction brief after receiving GlobalFoundries' Domestic Industry contention interrogatory responses. The proposed amendment therefore need not impact the claim construction process which is currently ongoing. The proposed amendment will also not affect

[REDACTED]

any of the other deadlines in the Procedural Schedule. GlobalFoundries proposes that it serve its initial contentions for the Additional '983 Claims on September 23 (after the parties file the claim construction replies on September 3 and the *Markman* hearing on September 17). Then, Tower can serve its invalidity contentions within a reasonable time thereafter. Under this proposal, there is sufficient time before the next substantive deadlines in the Procedural Schedule: the October 14 deadline to file tentative lists of witnesses for trial, and notice of prior art, (2) the November 19 deadline to serve final contention interrogatory responses on issues for which the party bears the burden of proof, and (3) the December 4 deadline to serve final contention interrogatory responses on issues for which the party does not bear the burden of proof. Tower presumably has already been searching for prior art, and Tower has sufficient time to develop any additional evidence it needs for its defense before the notice of prior art and final contention interrogatory deadlines months away. *LED Lighting Devices*, Order No. 29 at 4. And even if Tower needed some amount of additional time, there is space in the schedule to adjust these deadlines without needing to adjust the April 12-16, 2027 Evidentiary Hearing and November 30, 2027 Target Date.

IV. CONCLUSION

For the foregoing reasons, GlobalFoundries respectfully requests that the Court issue an Initial Determination granting leave to amend the Complaint and the Notice of Investigation to add dependent claims 8-12 and 19 of the '983 Patent.

Word Count Certificate

Pursuant to Ground Rule 1.7, the undersigned certifies that this Motion complies with the word limitation set forth in the Ground Rule 5.3. The certified word count includes footnotes and excludes caption, tables of contents or authorities, words in figures, the word count certificate, signature block, and the certificate of service.

Certified word count: 2862.

Dated: August 31, 2026

Respectfully submitted,

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CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **PUBLIC VERSION OF COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S MOTION FOR LEAVE TO AMEND THE COMPLAINT AND NOTICE OF INVESTIGATION** were served on September 8, 2026 as follows:

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EXHIBIT A

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**[PROPOSED] AMENDED COMPLAINT OF GLOBALFOUNDRIES U.S. INC., UNDER
SECTION 337 OF THE TARIFF ACT OF 1930, AS AMENDED**

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TABLE OF CONTENTS

I.	INTRODUCTION	1
II.	COMPLAINANT	6
III.	PROPOSED RESPONDENTS.....	6
IV.	THE TECHNOLOGY AND PRODUCTS-AT-ISSUE	7
V.	THE ASSERTED PATENTS AND NON-TECHNICAL DESCRIPTION OF THE INVENTIONS.....	7
A.	U.S. Patent No. 8,330,235.....	8
	1. Identification of the Patent and Ownership by GlobalFoundries.....	8
	2. Non-Technical Description of the Patent	8
B.	U.S. Patent No. 8,507,983.....	9
	1. Identification of the Patent and Ownership by GlobalFoundries.....	9
	2. Non-Technical Description of the Patent	9
C.	U.S. Patent No. 9,093,425.....	10
	1. Identification of the Patent and Ownership by GlobalFoundries.....	10
	2. Non-Technical Description of the Patent	11
D.	U.S. Patent No. 9,865,546.....	12
	1. Identification of the Patent and Ownership by GlobalFoundries.....	12
	2. Non-Technical Description of the Patent	12
E.	U.S. Patent No. 10,062,748.....	13
	1. Identification of the Patent and Ownership by GlobalFoundries.....	13
	2. Non-Technical Description of the Patent	13
F.	U.S. Patent No. 10,707,167.....	14
	1. Identification of the Patent and Ownership by GlobalFoundries.....	14
	2. Non-Technical Description of the Patent	14

VI.	UNLAWFUL AND UNFAIR ACTS OF RESPONDENTS.....	15
A.	Infringement of U.S. Patent No. 8,330,235	15
B.	Infringement of U.S. Patent No. 8,507,983	15
C.	Infringement of U.S. Patent No. 9,093,425	16
D.	Infringement of U.S. Patent No. 9,865,546	16
E.	Infringement of U.S. Patent No. 10,062,748	17
F.	Infringement of U.S. Patent No. 10,707,167	17
VII.	SPECIFIC INSTANCES OF UNFAIR IMPORTATION AND SALE	17
A.	The Qorvo PAC22140 Evaluation Kit	18
B.	The Qorvo QM81026 RF Power Tracker	19
C.	The Denso DNSRR004.....	19
VIII.	HARMONIZED TARIFF SCHEDULE ITEM NUMBERS.....	20
IX.	THE DOMESTIC INDUSTRY.....	20
A.	The Economic Prong of the Domestic Industry Requirement Is Satisfied.....	21
B.	The Technical Prong of the Domestic Industry Requirement Is Satisfied	21
X.	RELATED LITIGATION	22
XI.	RELIEF REQUESTED.....	23

TABLE OF EXHIBITS

EXHIBIT	DOCUMENT
1.	Copy of U.S. Patent No. 8,330,235
2.	Copy of U.S. Patent No. 8,507,983
3.	Copy of U.S. Patent No. 9,093,425
4.	Copy of U.S. Patent No. 9,865,546
5.	Copy of U.S. Patent No. 10,062,748
6.	Copy of U.S. Patent No. 10,707,167
7.	Copies of Assignment Documents for U.S. Patent No. 8,330,235
8.	Copies of Assignment Documents for U.S. Patent No. 8,507,983
9.	Copies of Assignment Documents for U.S. Patent No. 9,093,425
10.	Copies of Assignment Documents for U.S. Patent No. 9,865,546
11.	Copies of Assignment Documents for U.S. Patent No. 10,062,748
12.	Copies of Assignment Documents for U.S. Patent No. 10,707,167
13.	GlobalFoundries, “Aerospace, Defense and Critical Infrastructure,” https://gf.com/markets/aerospace-and-defense/
14.	Nov. 20, 2024 GlobalFoundries Press Release, “GlobalFoundries and U.S. Department of Commerce Announce Award Agreement on CHIPS Act Funding for Essential Chip Manufacturing,” https://gf.com/gf-press-release/globalfoundries-and-u-s-department-of-commerce-announce-award-agreement-on-chips-act-funding-for-essential-chip-manufacturing/
15.	GlobalFoundries July 1, 2015 Press Release, “GlobalFoundries Completes Acquisition of IBM Microelectronics Business,” https://gf.com/gf-press-release/globalfoundries-completes-acquisition-ibm-microelectronics-business/
16.	GlobalFoundries Feb. 11, 2026 News Release, “GlobalFoundries Reports Fourth Quarter 2025 and Fiscal Year 2025 Financial Results,” https://investors.gf.com/news-releases/news-release-details/globalfoundries-reports-fourth-quarter-2025-and-fiscal-year-2025
17.	GlobalFoundries Apr. 26, 2021 Press Release, “GLOBALFOUNDRIES Moves Corporate Headquarters to its Most Advanced Semiconductor Manufacturing Facility in New York,” https://gf.com/gf-press-release/globalfoundries-moves-corporate-headquarters-its-most-advanced-semiconductor/
18.	GlobalFoundries June 4, 2025 Press Release, “GlobalFoundries Announces \$16B U.S. Investment to Reshore Essential Chip Manufacturing and Accelerate AI Growth,” https://gf.com/gf-press-release/globalfoundries-announces-16b-u-s-investment-to-reshore-essential-chip-manufacturing-and-accelerate-ai-growth/
19.	Tower Feb. 11, 2026 Press Release, “Tower Semiconductor Reports Record Revenue for the Fourth Quarter of 2025,” https://towersemi.com/2026/02/11/02112026/
20.	Q1 2025 Tower Semiconductor Ltd Earnings Call on May 14, 2025, https://ir.towersemi.com/static-files/4dbeca76-e387-4bd7-aad0-1daf4607282a
21.	Intel Aug. 16, 2023 Newsroom Release, “Intel Announces Termination of Tower Semiconductor Acquisition,” https://newsroom.intel.com/corporate/intel-news-aug-2023
22.	MacroTrends, “Tower Semiconductor Research and Development Expenses

EXHIBIT	DOCUMENT
	2012-2025-TSEM,” https://www.macrotrends.net/stocks/charts/TSEM/tower-semiconductor/research-development-expenses
23.	Insights, “Tower Semiconductor Patents - Insights & Stats,” https://insights.greyb.com/tower-semiconductor-patents/
24.	List of Foreign Counterpart Patents and Foreign and Domestic Patent Applications for the Asserted Patents
25.	CONFIDENTIAL: List of Licensees
26.	Receipts Relating to the ’235, ’425, ’546, and ’167 Patents’ Representative Accused Product (Qorvo QM81026)
27.	Photographs Relating to the ’235, ’425, ’546, and ’167 Patents’ Representative Accused Product (Qorvo QM81026)
28.	Infringement Claim Chart for U.S. Patent No. 8,330,235 (Qorvo QM81026)
29.	Receipts Relating to the ’983 and ’748 Patents’ Representative Accused Product (Qorvo PAC22140)
30.	Photographs Relating to the ’983 and ’748 Patents’ Representative Accused Product (Qorvo PAC22140)
31.	Infringement Claim Chart for U.S. Patent No. 8,507,983 (Qorvo PAC22140)
32.	Receipts Relating to the ’425, ’546, and ’167 Patents’ Representative Accused Product (Denso DNSRR004)
33.	Photographs Relating to the ’425, ’546, and ’167 Patents’ Representative Accused Product (Denso DNSRR004)
34.	Infringement Claim Chart for U.S. Patent No. 9,093,425 (Qorvo QM81026)
35.	Infringement Claim Chart for U.S. Patent No. 9,093,425 (Denso DNSRR004)
36.	Infringement Claim Chart for U.S. Patent No. 9,865,546 (Qorvo QM81026)
37.	Infringement Claim Chart for U.S. Patent No. 9,865,546 (Denso DNSRR004)
38.	Infringement Claim Chart for U.S. Patent No. 10,062,748 (Qorvo PAC22140)
39.	Infringement Claim Chart for U.S. Patent No. 10,707,167 (Qorvo QM81026)
40.	Infringement Claim Chart for U.S. Patent No. 10,707,167 (Denso DNSRR004)
41.	Qorvo Product List, https://www.qorvo.com/products/product-list?categoryID=ca0128
42.	Qorvo March 19, 2018 News Release, “TowerJazz and Active-Semi Announce Ramp to Production of Motor Driver IC on 200V SOI Technology,” https://www.qorvo.com/newsroom/news/2018/towerjazz-and-active-semi-announce-ramp-to-production-of-motor-driver-ic-on-200v-soi-technology
43.	Qorvo Apr. 10, 2019 News Release, “Qorvo to Acquire Active-Semi International,” https://www.qorvo.com/newsroom/news/2019/qorvo-to-acquire-active-semi-international
44.	SiliconValley.com, “A Pillar of Specialty Silicon: A Comprehensive Deep Dive into Tower Semiconductor (TSEM),” https://markets.financialcontent.com/siliconvalley/article/predictstreet-2025-12-29-a-pillar-of-specialty-silicon-a-comprehensive-deep-dive-into-tower-semiconductor-tsem
45.	CONFIDENTIAL: Excerpts from WiSol Teardown Report
46.	CONFIDENTIAL: Tower Guard Ring Comparison

EXHIBIT	DOCUMENT
47.	Tower Aug, 14, 2017 News Release, “TowerJazz Announces DENSO Corporation utilized its Advanced 0.18um SiGe Technology to Develop a 24GHz Rear and Side Radar Sensor,” https://ir.towersemi.com/news-releases/news-release-details/towerjazz-announces-denso-corporation-utilized-its-advanced
48.	Yolegroup, “Denso DNSRR004 Short Range Radar”, https://medias.yolegroup.com/uploads/2019/12/SP19500-Yole-SYSTEM-Denso-DNSRR004-Blind-Spot-Radar-Flyer.pdf
49.	CONFIDENTIAL: Domestic Industry Exhibit
50.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 8,330,235
51.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 8,507,983
52.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 9,093,425
53.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 9,865,546
54.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 10,062,748
55.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 10,707,167

TABLE OF APPENDICES

APPENDIX	DOCUMENT
A.	Copy of the Prosecution History for U.S. Patent No. 8,330,235
B.	Copy of the Prosecution History for U.S. Patent No. 8,507,983
C.	Copy of the Prosecution History for U.S. Patent No. 9,093,425
D.	Copy of the Prosecution History for U.S. Patent No. 9,865,546
E.	Copy of the Prosecution History for U.S. Patent No. 10,062,748
F.	Copy of the Prosecution History for U.S. Patent No. 10,707,167
G.	Copy of the Prosecution History for the Priority Applications for U.S. Patent No. 8,330,235 (Application No. 12/366,378)
H.	Copy of the Prosecution Histories for the Priority Applications for U.S. Patent No. 8,507,983 (Application Nos. 12/390,509 and 12/500,620)

I. INTRODUCTION

1.1 GlobalFoundries U.S. Inc. (“GlobalFoundries” or “Complainant”) requests that the United States International Trade Commission institute an Investigation pursuant to Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337 (“Section 337”), to remedy the unlawful importation, sale for importation, and/or sale after importation of certain semiconductor devices, products containing same, and components thereof by Tower Semiconductor Ltd., (Israel); Tower Partners Semiconductor, Ltd. (Japan); Tower Semiconductor Italy S.R.L. (Italy); Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively “Tower” or “the Proposed Respondents”). Tower’s products infringe certain claims (“the Asserted Claims”) of U.S. Patent Nos. 8,330,235 (“the ’235 Patent”); 8,507,983 (“the ’983 Patent”); 9,093,425 (“the ’425 Patent”); 9,865,546 (“the ’546 Patent”); 10,062,748 (“the ’748 Patent”); and 10,707,167 (“the ’167 Patent”) (collectively, “the Asserted Patents”) as follows:

U.S. Patent No.	Asserted Claims ¹
8,330,235	1 , 5-9, 10 , 11, 12, 15-19
8,507,983	1-7 , 8-12, 13-18 , 19, 20
9,093,425	1-9
9,865,546	1-7 , 8-14
10,062,748	1 , 2-8, 9 , 10, 11 , 12, 14, 16-18
10,707,167	1-6

Pursuant to Commission Rule 212.(a)(9)(i), copies of the Asserted Patents are included as **Exhibits 1-6**². GlobalFoundries holds all rights, title, and interest in and to the Asserted Patents. Pursuant to Commission Rule 212.(a)(9)(ii), copies of the Assignment Records for each of the Asserted Patents are included as **Exhibits 7-12**.

¹ Bolded claim numbers designate independent claims.

² Certified copies of the patents, assignments and prosecution histories at issue have been requested from the USPTO and will be provided to the Commission upon receipt.

1.2 GlobalFoundries is a leading American semiconductor manufacturer, a trusted supplier to the U.S. Department of War³ and a major recipient of federal investment under the CHIPS and Science Act.⁴ GlobalFoundries brings this action against Tower for infringement of GlobalFoundries' U.S. patents relating to its industry-leading RF and power management manufacturing process technologies. For over a decade, Tower has been engaged in a systematic strategy of freeloading, spending very little on capacity expansion (nearly none in the U.S.) and even less on R&D. Instead, it profits by leveraging the patented American innovation of others, like GlobalFoundries, and selling at lower prices into the market. GF brings this lawsuit to put an end to Tower's practices.

1.3 Semiconductor manufacturing today requires some of the most expensive and sophisticated factories on the planet, utilizing highly technical and complex nano-scale process technologies and costly manufacturing tools. Entire global industries, such as healthcare, artificial intelligence, automotive, and defense industries, depend on the availability and technological capability of the semiconductor products these factories and tools produce. This is why semiconductor chips are at the forefront of today's great global technological competition between the United States and China. It is imperative, therefore, for nations and companies to protect innovation in semiconductor process technologies, like the ones that GlobalFoundries invests in and creates, and that Tower has unlawfully used and profited from at GlobalFoundries' expense.

1.4 GlobalFoundries is the most advanced pure-play foundry in the U.S. and Europe

³ **Exhibit 13**, GlobalFoundries, "Aerospace, Defense and Critical Infrastructure," <https://gf.com/markets/aerospace-and-defense/>.

⁴ **Exhibit 14**, Nov. 20, 2024 GlobalFoundries Press Release, "GlobalFoundries and U.S. Department of Commerce Announce Award Agreement on CHIPS Act Funding for Essential Chip Manufacturing," <https://gf.com/gf-press-release/globalfoundries-and-u-s-department-of-commerce-announce-award-agreement-on-chips-act-funding-for-essential-chip-manufacturing/>.

and employs thousands of people in the U.S. and worldwide. While other companies were abandoning semiconductor manufacturing in the U.S., GlobalFoundries bucked this trend by investing billions of dollars on advanced technology and research in the United States. GlobalFoundries originated from the semiconductor manufacturing arm of another leading U.S. semiconductor company, Advanced Micro Devices, in 2009, and expanded globally through acquisition and organic investment. A major \$15 billion U.S. investment in its 300 acre facility known as Fab 8 in Malta, New York, is the leading-edge of GlobalFoundries' significant U.S. investments. GlobalFoundries broke ground for that state of the art facility in 2009 and produces vanguard technology from that location for customers worldwide. Another major U.S. acquisition took place in 2015 when GlobalFoundries acquired IBM's microelectronic facilities and personnel in Burlington, Vermont, a facility that became known as Fab 9. GlobalFoundries acquired not just IBM's facilities and technologists, but also approximately 16,000 IBM patents and applications; decades of experience and expertise in semiconductor development, device expertise, design, and manufacturing; and an expanded manufacturing footprint. The acquisition cemented GlobalFoundries' position as a global leader in world-class semiconductor manufacturing and advanced process technologies.⁵

1.5 GlobalFoundries employs over 4,500 American workers and has invested billions of dollars over decades in domestic semiconductor manufacturing, including advanced radio frequency (RF) and power management platform technologies critical to national security, aerospace, defense, and communications infrastructure. GlobalFoundries' foundational technologies are steeped in IBM and AMD heritage, two storied American companies that, prior

⁵ **Exhibit 15**, GlobalFoundries July 1, 2015 Press Release, "GlobalFoundries Completes Acquisition of IBM Microelectronics Business," <https://gf.com/gf-press-release/globalfoundries-completes-acquisition-ibm-microelectronics-business/>.

to GlobalFoundries also spent billions in R&D on semiconductor manufacturing technology in the United States. Building on this technology foundation, GlobalFoundries invests over \$500 million annually in R&D and technology enablement,⁶ and employs thousands of engineers, to sustain its innovation leadership. GF has invested more than \$15 billion in its Fab 8 facility over the last decade to support innovation and manufacturing capacity. In 2021, the company doubled its planned investment to expand global capacity, with \$500 million targeted for Malta, NY alone.⁷ As a major CHIPS recipient, GlobalFoundries is committed to an additional \$16 billion in investment in its US fabs.⁸ All the while, GlobalFoundries has been creating and patenting hundreds of world class innovations annually to build and secure its technology differentiation leadership and rights.

1.6 In stark contrast to GlobalFoundries, Tower is a foundry business that has chronically underinvested in research and development relative to its U.S. peers. With a base of operations in Israel, and plants in Japan, Europe and the United States, Tower generated a little over \$1.5 billion in 2025,⁹ and forecasts revenue will grow beyond \$2 billion.¹⁰ In 2023, it failed

⁶ **Exhibit 16**, GlobalFoundries Feb. 11, 2026 News Release, “GlobalFoundries Reports Fourth Quarter 2025 and Fiscal Year 2025 Financial Results,” <https://investors.gf.com/news-releases/news-release-details/globalfoundries-reports-fourth-quarter-2025-and-fiscal-year-2025>.

⁷ **Exhibit 17**, GlobalFoundries Apr. 26, 2021 Press Release, “GLOBALFOUNDRIES Moves Corporate Headquarters to its Most Advanced Semiconductor Manufacturing Facility in New York,” <https://gf.com/gf-press-release/globalfoundries-moves-corporate-headquarters-its-most-advanced-semiconductor/>.

⁸ **Exhibit 18**, GlobalFoundries June 4, 2025 Press Release, “GlobalFoundries Announces \$16B U.S. Investment to Reshore Essential Chip Manufacturing and Accelerate AI Growth,” <https://gf.com/gf-press-release/globalfoundries-announces-16b-u-s-investment-to-reshore-essential-chip-manufacturing-and-accelerate-ai-growth/>.

⁹ **Exhibit 19**, Tower Feb. 11, 2026 Press Release, “Tower Semiconductor Reports Record Revenue for the Fourth Quarter of 2025,” <https://towersemi.com/2026/02/11/02112026/>

¹⁰ **Exhibit 20**, Q1 2025 Tower Semiconductor Ltd Earnings Call on May 14, 2025, <https://ir.towersemi.com/static-files/4dbeca76-e387-4bd7-aad0-1daf4607282a> at p. 4.

to merge with Intel and received \$353 million from Intel as a break-up fee,¹¹ spending most of that to fuel its foreign operations where the vast majority of its employees are located. More recently, Tower stated that its partnership with Intel has floundered and that it is in mediation with the U.S. chipmaker.¹² Historically, Tower has been a marginal foundry that struggled at times to survive, priding itself on a lowest-investment model. It has often bought old assets cheaply to avoid meaningful capex investments in its facilities, spending far less than the multiple billions required to compete lawfully. Tower's history of investment in R&D is even more meager. Historic figures show R&D expenses of less than \$90 million annually.¹³ Tower generally files fewer than 15 patent applications annually and holds fewer than 350 granted patents globally.¹⁴

1.7 Rather than investing at levels commensurate with the technological demands of advanced RF integration, Tower profits by offering products that incorporate innovations protected by GF's U.S. patents. Tower undercuts market pricing in the RF and power management segments, capturing share through low pricing strategies made possible by avoiding, via infringement, the cost of the original R&D borne by GlobalFoundries. That strategy should not be permitted to continue to provide rewards going forward.

1.8 GlobalFoundries seeks a limited exclusion order barring from entry into the United States infringing semiconductor devices, products containing same, and components thereof manufactured by or on behalf of, or imported by or on behalf of, Tower. GlobalFoundries also

¹¹ **Exhibit 21**, Intel Aug. 16, 2023 Newsroom Release, "Intel Announces Termination of Tower Semiconductor Acquisition," <https://newsroom.intel.com/corporate/intel-news-aug-2023>

¹² **Exhibit 19**, Tower Feb. 11, 2026 Press Release, "Tower Semiconductor Reports Record Revenue for the Fourth Quarter of 2025," <https://towersemi.com/2026/02/11/02112026/>

¹³ **Exhibit 22**, MacroTrends, "Tower Semiconductor Research and Development Expenses 2012-2025-TSEM," <https://www.macrotrends.net/stocks/charts/TSEM/tower-semiconductor/research-development-expenses>.

¹⁴ **Exhibit 23**, Insights, "Tower Semiconductor Patents - Insights & Stats," <https://insights.greyb.com/tower-semiconductor-patents/>.

seeks cease and desist orders prohibiting the sale for importation, importation, sale after importation, distribution, offer for sale, promotion, marketing, advertising, testing, demonstration, warehousing inventory for distribution, solicitation of sales, programming, repairing, maintenance, use, transfer, and other commercial activity relating to infringing semiconductor devices, products containing same, and components thereof. GlobalFoundries also seeks imposition of a bond during the Presidential Review Period pursuant to 19 U.S.C. § 1337(j).

1.9 As further set forth below, pursuant to Commission Rule 210.12(a)(6)(i), a domestic industry as defined in 19 U.S.C. § 1337(a)(2) and (3) exists in the United States in view of GlobalFoundries' domestic activities relating to the technology of the Asserted Patents and products that practice the Asserted Patents.

II. COMPLAINANT

2.1 Complainant GlobalFoundries U.S. Inc. is a Delaware Corporation with a principal place of business at 400 Stonebreak Road Ext., Malta, NY 12020.

III. PROPOSED RESPONDENTS

3.1 The Proposed Respondents are Tower Semiconductor Ltd., (Israel); Tower Partners Semiconductor, Ltd. (Japan); Tower Semiconductor Italy S.R.L. (Italy); Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively "Tower" or "the Proposed Respondents"). On information and belief, each of the named Tower entities imports into the United States, sells for importation into the United States, and/or sells in the United States after importation certain semiconductor devices, products containing same, and components thereof. Tower's semiconductor devices may be and are used in many products, including, without limitation, evaluation kits, garden tools, automotive products, and mobile devices which are imported into the United States as described below.

IV. THE TECHNOLOGY AND PRODUCTS-AT-ISSUE

4.1 This investigation generally concerns semiconductor devices and/or methods of manufacturing semiconductor devices. Descriptions of the specific technological features and advancements of each of the asserted patents are set forth below in Section V.

4.2 Tower imports, sells for importation, and/or sells within the United States after importation certain semiconductor devices that incorporate these functionalities.

4.3 Pursuant to 19 C.F.R. § 210.12(a)(12), the category of products accused are certain semiconductor devices manufactured using Tower's RF, power management, BCD, logic, SiGe, and BiCMOS processes, including wafers and chips, and components thereof.

4.4 Tower imports, sells for importation, and/or sells within the United States, after importation, semiconductor devices that can be and are incorporated into products including, but not limited to, the Qorvo PAC22140 Evaluation Kit, the Denso DNSRR004 Blind Spot Monitor Sensor, and the Qorvo QM81026 RF Power Tracker.

4.5 The Accused Products directly or indirectly infringe one or more of the Asserted Claims by employing the above functionalities protected by GlobalFoundries' innovations.

V. THE ASSERTED PATENTS AND NON-TECHNICAL DESCRIPTION OF THE INVENTIONS¹⁵

5.1 Pursuant to Commission Rule 210.12(c)(1), copies of the prosecution histories for the Asserted Patents have been submitted with this Complaint as **Appendices A-F**. Pursuant to Commission Rule 210.12(c)(2), copies of the prosecution histories for any priority applications for the Asserted Patents also have been submitted with this Complaint as **Appendices G-H**.

5.2 Pursuant to Commission Rule 210.12(a)(9)(v), a list of each foreign patent, each

¹⁵ All non-technical descriptions of and categorizations of the patents are for the purposes of providing general information only and should not be interpreted as a representation regarding the scope or applicability of any patent, or the requirements or meaning of any patent claims.

foreign or domestic patent application, and each foreign or domestic patent application that has been denied, abandoned or withdrawn, corresponding to each involved U.S. patent, with an indication of the prosecution status of each such patent application is attached as **Exhibit 24**.

5.3 Pursuant to Commission Rule 210.12(a)(9), a list of entities believed to be licensed to the Asserted Patents is attached as **Confidential Exhibit 25**.

A. U.S. Patent No. 8,330,235

1. Identification of the Patent and Ownership by GlobalFoundries

5.4 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 8,330,235 titled “METHOD TO REDUCE MOL DAMAGE ON NISI” on December 11, 2012. *See Exhibit 1* (’235 Patent). The ’235 Patent issued from Application No. 13/096,511, filed on April 28, 2011. *Id.*

5.5 GlobalFoundries is asserting claims 1, 5-9, 10, 11, 12, and 15-19 of the ’235 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.6 The Asserted Claims of the ’235 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’235 Patent on February 5, 2029. GlobalFoundries holds all rights, title, and interest in the ’235 Patent with full rights to enforce the same. **Exhibit 7** (’235 Patent Assignments).

2. Non-Technical Description of the Patent

5.7 The ’235 Patent is generally directed to transistor devices with nickel silicide layers that are formulated to prevent degradation upon removal of overlying stress liners. ’235 Patent at Abstract. Conventional semiconductor devices include a refractory metal (e.g., nickel) deposited over exposed surface features to provide resistance to certain etchants. **Exhibit 1** (’235 Patent) at 1:19-28. Thermal processing (e.g., thermal annealing) causes the refractory metal to react with

the underlying silicon to form electrically conductive silicide layer portions. *Id.* at 1:41-56. Some subsequent etching processes, such as reactive ion etching, can result in degradation of the silicide layer. *Id.* at 2:50-62. Nickel silicide is commonly doped with platinum to improve the material and electrical properties. *Id.* at 7:13-14. However, platinum also increases the resistivity of nickel silicide. The '235 Patent describes techniques for effective shielding of a nickel silicide layer while addressing a problematic increase in resistivity by compositionally grading a platinum-doped nickel silicide layer. *Id.* at 7:19-26. For example, the '235 Patent describes nickel silicide layers have a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. *Id.* at Abstract.

B. U.S. Patent No. 8,507,983

1. Identification of the Patent and Ownership by GlobalFoundries

5.8 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 8,507,983 titled “HIGH VOLTAGE DEVICE” on August 13, 2013. *See Exhibit 2* ('983 Patent). The '983 Patent issued from Application No. 13/550571, filed on July 16, 2012. *Id.*

5.9 GlobalFoundries is asserting claims 1-7, 8-12, 13-18, and 19, 20 of the '983 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.10 The Asserted Claims of the '983 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '983 Patent on February 23, 2029. GlobalFoundries holds all rights, title, and interest in the '983 Patent with full rights to enforce the same. **Exhibit 8** ('983 Patent Assignments).

2. Non-Technical Description of the Patent

5.11 The '983 Patent is generally directed to semiconductor devices that can be used in high-voltage applications, which commonly include lateral double-diffused (LD) transistors. “For high performance LD transistors, low drain-to-source on-resistance ($R_{ds(on)}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability.” **Exhibit 2**, '983 Patent at 1:17-21. “To achieve low $R_{ds(on)}$, the channel of the high [voltage] LD transistor should be as short as possible.” *Id.* at 1:21-22.

5.12 The '983 Patent thus describes a high voltage device that has a gate within an active device region, along with a “doped channel well disposed in the substrate adjacent to a first edge of the gate.” *Id.* at 1:45-49. “The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate.” *Id.* at 1:49-51. “The first edge of the gate and channel edge defines an effective channel length of the device,” and the “effective channel length is self-aligned to the gate.” *Id.* at 1:51-53. “With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided [and] [v]ery short effective channel lengths with very low R_{ds} can be achieved, enabling high current to pass through a small area.” *Id.* at 6:1-4. The claimed device can also include a “drift isolation region” that is disposed in “a doped drift well adjacent to a second edge of the gate.” *Id.* at 2:9-11. “The drift isolation region [] can improve breakdown voltage of the HV device due to high voltage.” *Id.* at 5:17-19. The device can also have diffusion regions adjacent to the gate, which “serve as, for example, source and drain of the transistor.” *Id.* at 3:55-57.

C. U.S. Patent No. 9,093,425

1. Identification of the Patent and Ownership by GlobalFoundries

5.13 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 9,093,425 titled “SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS” on July 28, 2015. *See Exhibit 3* ('425

Patent). The '425 Patent issued from Application No. 14/177481, filed on February 11, 2014. *Id.*

5.14 GlobalFoundries is asserting claims 1-9 of the '425 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.15 The Asserted Claims of the '425 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '425 Patent on February 11, 2034. GlobalFoundries holds all rights, title, and interest in the '425 Patent with full rights to enforce the same. **Exhibit 9** ('425 Patent Assignments).

2. Non-Technical Description of the Patent

5.16 The '425 Patent is generally directed to self-aligned liners formed on metal semiconductor alloy contacts. “[M]etal semiconductor alloy containing contacts” can suffer from defects including “(1) [e]ncroachment of the metal semiconductor alloy which typically increases leakage through the source/drain junction” and “(2) [s]potty (non-uniform) metal semiconductor alloys which may cause unwanted etching of the metal semiconductor alloy contact during a subsequently performed etch.” **Exhibit 3**, '425 Patent at 1:47-54. “These two defects are either formed during [or enhanced by] a subsequently performed etching process.” *Id.* at 1:55-57.

5.17 The '425 Patent describes a semiconductor structure where “[m]etal semiconductor alloy contacts are provided on each of a source region and a drain region” and “[a] transition metal is then deposited on each of the metal semiconductor alloy contacts.” '425 patent at 1:62-65. “[T]he deposited transition metal reacts ... with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact.” *Id.* at 1:66-2:3. “The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.” *Id.* at 2:7-11.

D. U.S. Patent No. 9,865,546

1. Identification of the Patent and Ownership by GlobalFoundries

5.18 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 9,865,546 titled “CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME” on January 9, 2018. *See Exhibit 4* (’546 Patent). The ’546 Patent issued from Application No. 14/729,298, filed on June 3, 2015. *Id.*

5.19 GlobalFoundries is asserting claims 1-7 and 8-14 of the ’546 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.20 The Asserted Claims of the ’546 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’546 Patent on July 3, 2035. GlobalFoundries holds all rights, title, and interest in the ’546 Patent with full rights to enforce the same. **Exhibit 10** (’546 Patent Assignments).

2. Non-Technical Description of the Patent

5.21 The ’546 Patent is generally directed to formation of a contact in a dielectric layer over a semiconductor substrate. **Exhibit 4**, (’546 Patent) at Abstract. More specifically, the ’546 Patent relates to forming an intermix region at an interface of a bottom portion of a contact and an upper portion of a semiconductor substrate. *Id.* at 1:7-13. For example, the ’546 Patent discloses techniques for depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate (e.g., the source and/or drain). *Id.* at 2:43-49. Unreacted (non-diffused) portions of the first liner layer can then be removed before forming the contact. *Id.* at 2:49-51. Such techniques may, for example, improve contact resistance while avoiding issues caused by increased metal thickness in contact openings. *Id.* at 2:51-54.

E. U.S. Patent No. 10,062,748

1. Identification of the Patent and Ownership by GlobalFoundries

5.22 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 10,062,748 titled “SEGMENTED GUARD-RING AND CHIP EDGE SEALS” on August 28, 2018. *See Exhibit 5* (’748 Patent). The ’748 Patent issued from Application No. 15/443276, filed on February 27, 2017. *Id.*

5.23 GlobalFoundries is asserting claims 1-12, 14, and 16-18 of the ’748 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.24 The Asserted Claims of the ’748 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’748 Patent on February 27, 2037. GlobalFoundries holds all rights, title, and interest in the ’748 Patent with full rights to enforce the same. **Exhibit 11** (’748 Patent Assignments).

2. Non-Technical Description of the Patent

5.25 The ’748 Patent is generally directed to “segmented guard-ring and chip edge seals.” **Exhibit 5**, ’748 Patent at 1:6-8. Guard rings are useful in many technologies, including “low-k dielectric, porous applications” in which the guard ring can help “prevent moisture ingress.” *Id.* at 1:25-28. However, “the use of a segmented guard ring can cause reliability fails in these low-k dielectric, porous applications.” *Id.* at 1:28-31. The ’748 Patent “provides segmented guard-ring and chip edge seals with low-K dielectric material in middle of line (MOL) and/or back end of the line (BEOL) processing.” *Id.* at 2:23-27. These structures can be compatible with the low-K dielectric material which “can be a non-porous or porous polymer, carbon doped oxide, or oxide or combinations thereof.” *Id.* at 3:35-37.

F. U.S. Patent No. 10,707,167

1. Identification of the Patent and Ownership by GlobalFoundries

5.26 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 10,707,167 titled “CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME” on July 7, 2020. *See Exhibit 6* (’167 Patent). The ’167 Patent issued from Application No. 15/826,939, filed on November 30, 2017. *Id.*

5.27 GlobalFoundries is asserting claims 1-6 of the ’167 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.28 The Asserted Claims of the ’167 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’167 Patent on May 30, 2036. GlobalFoundries holds all rights, title, and interest in the ’167 Patent with full rights to enforce the same. **Exhibit 12** (’167 Patent Assignments).

2. Non-Technical Description of the Patent

5.29 The ’167 Patent is generally directed to formation of a contact in a dielectric layer over a semiconductor substrate. **Exhibit 6** (’167 Patent) at Abstract. More specifically, the ’167 Patent relates to forming an intermix region at an interface of a bottom portion of a contact and an upper portion of a semiconductor substrate. *Id.* at 1:9-15. For example, the ’167 Patent discloses techniques for depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate (e.g., the source and/or drain). *Id.* at 2:46-51. Unreacted (non-diffused) portions of the first liner layer can then be removed before forming the contact. *Id.* at 2:51-53. Such techniques may, for example, improve contact resistance while avoiding issues caused by increased metal thickness in contact openings. *Id.* at 2:53-56.

VI. UNLAWFUL AND UNFAIR ACTS OF RESPONDENTS

6.1 The unlawful and unfair acts of Tower identified below are exemplary and should not be considered in any way limiting.

6.2 GlobalFoundries reserves the right to identify additional infringing products in light of further discovery. GlobalFoundries has not yet had the benefit of discovery, and thus its identification of specific models or types of products below is not intended to limit the scope of the investigation. Any remedy should extend to all infringing products.

A. Infringement of U.S. Patent No. 8,330,235

6.3 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '235 Patent, literally or under the doctrine of equivalents. The '235 Accused Products include at least devices manufactured using Tower's RF processes RF processes (including its 65nm RF process). For example, the Tower-made Qorvo QM81026 Power Tracker (the "'235 Representative Accused Product") infringes at least claims 1, 5-9, 10, 11, 12, and 15-19 of the '235 Patent. *See Exhibits 26-27.*

6.4 A claim chart comparing the independent claims of the '235 Patent to the '235 Representative Accused Product is attached as **Exhibit 28**.

B. Infringement of U.S. Patent No. 8,507,983

6.5 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '983 Patent, literally or under the doctrine of equivalents. The '983 Accused Products include at least products manufactured using Tower's BCD processes (including its TS18SOI process). For example, the Tower-made PA22BZdie in the Qorvo PAC22140 Evaluation Kit (the "'983 Representative Accused Product") infringes at least claims 1-7, 8-12, 13-18, 19, and 20 of the '983 Patent. *See*

Exhibits 29-30.

6.6 A claim chart comparing the independent claims of the '983 Patent to the '983 Tower Representative Accused Product is attached as **Exhibit 31**.

C. Infringement of U.S. Patent No. 9,093,425

6.7 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '425 Patent, literally or under the doctrine of equivalents. The '425 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'425 Representative Accused Products") infringe at least claims 1-9 of the '425 Patent. *See Exhibits 32-33 and Exhibits 26-27.*

6.8 Claim charts comparing the independent claim of the '425 Patent to the Tower '425 Representative Accused Products are attached as **Exhibit 34** and **Exhibit 35**.

D. Infringement of U.S. Patent No. 9,865,546

6.9 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '546 Patent, literally or under the doctrine of equivalents. The '546 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'546 Representative Accused Products") infringe at least claims 1-14 of the '546 Patent. *See Exhibits 32-33 and Exhibits 26-27.*

6.10 Claim charts comparing the independent claims of the '546 Patent to the Tower '546

Representative Accused Products are attached as **Exhibit 36** and **Exhibit 37**.

E. Infringement of U.S. Patent No. 10,062,748

6.11 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '748 Patent, literally or under the doctrine of equivalents. The '748 Accused Products include at least products manufactured using Tower's BCD processes (including its TS18SOI process). For example, the Tower-made PA22BZdie in the Qorvo PAC22140 Evaluation Kit (the "'748 Representative Accused Product") infringes at least claims 1-12, 14, and 16-18 of the '748 Patent. *See Exhibits 29-30*.

6.12 A claim chart comparing the independent claims of the '748 Patent to the Tower '748 Representative Accused Product is attached as **Exhibit 38**.

F. Infringement of U.S. Patent No. 10,707,167

6.13 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '167 Patent, literally or under the doctrine of equivalents. The '167 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'167 Representative Accused Products") infringe at least claims 1-6 of the '167 Patent. *See Exhibits 32-33 and Exhibits 26-27*.

6.14 Claim charts comparing the independent claim of the '167 Patent to the Tower '167 Representative Accused Products are attached as **Exhibit 39** and **Exhibit 40**.

VII. SPECIFIC INSTANCES OF UNFAIR IMPORTATION AND SALE

7.1 On information and belief, Tower is importing, selling for importation, and/or

selling within the United States after importation, certain semiconductor devices, products containing same, and components thereof.

7.2 The specific instances set forth below are representative examples of Tower's unlawful importation, sale for importation, and/or sales within the United States after importation of infringing products.

A. The Qorvo PAC22140 Evaluation Kit

7.3 Qorvo offers a family of products under the name Power Application Controller (PAC).¹⁶ In 2018, semiconductor company Active-Semi announced that it was relying on Tower to manufacture a PAC product referred to as PAC5232.¹⁷ In 2019, Qorvo acquired Active-Semi.¹⁸ Qorvo remains one of Tower's major customers.¹⁹ On information and belief, Tower continues to manufacture Qorvo's PAC family of products.

7.4 The Qorvo PAC22140 chip is part of Qorvo's family of PAC products. **Exhibit 41** at 2. This chip, which includes an infringing PA22BZ die manufactured by Tower, is a component of the Qorvo PAC22140 Evaluation Kit ("Qorvo PAC22140EVK1") that is imported into and sold in the United States. *See Exhibits 29-30. Exhibit 29* is a set of receipts showing the purchase of a Qorvo PAC22140EVK1 in the United States from Mouser Electronics on March 13, 2026. These

¹⁶ **Exhibit 41**, Qorvo Product List at 2-5, <https://www.qorvo.com/products/product-list?categoryID=ca0128>.

¹⁷ **Exhibit 42**, Qorvo March 19, 2018 News Release, "TowerJazz and Active-Semi Announce Ramp to Production of Motor Driver IC on 200V SOI Technology," <https://www.qorvo.com/newsroom/news/2018/towerjazz-and-active-semi-announce-ramp-to-production-of-motor-driver-ic-on-200v-soi-technology>.

¹⁸ **Exhibit 43**, Qorvo Apr. 10, 2019 News Release, "Qorvo to Acquire Active-Semi International," <https://www.qorvo.com/newsroom/news/2019/qorvo-to-acquire-active-semi-international>.

¹⁹ *See, e.g., Exhibit 44*, SiliconValley.com, "A Pillar of Specialty Silicon: A Comprehensive Deep Dive into Tower Semiconductor (TSEM)," <https://markets.financialcontent.com/siliconvalley/article/predictstreet-2025-12-29-a-pillar-of-specialty-silicon-a-comprehensive-deep-dive-into-tower-semiconductor-tsem>.

receipts show that an import tariff was applied to the order, specifically in the amount of \$56.25, confirming that the Qorvo PAC22140EVK1 was imported into the United States. **Exhibit 29**. The purchased product arrived in an Active-Semi box, labeled with the Qorvo PAC22140EVK1 product name. **Exhibit 30** (photographs of the purchased PAC22140EVK1) at 2.

B. The Qorvo QM81026 RF Power Tracker

7.5 The Qorvo QM81026 RF Power Tracker chip (“Qorvo QM81026 RF chip”) is a component of a smartphone, which is assembled in India and is imported into and sold in the United States. *See Exhibits 26-27*. A receipt showing purchase of a smartphone in the United States on February 28, 2026 is attached as **Exhibit 26**. Qorvo is a significant Tower customer. **Exhibit 44**. On information and belief, the Qorvo QM81026 RF chip is manufactured by Tower. For example, the guard ring structures in the accused Qorvo QM81026 RF chip are identical to the guard ring structures found in other RF chips known to be Tower-manufactured, such as the Wisol CL301F1 LNA. *See Confidential Exhibits 45-46*. Therefore, on information and belief, Tower manufactures the Qorvo QM81026 RF chip.

C. The Denso DNSRR004

7.6 In 2017 Tower announced that it used its technology to develop Denso’s rear and side radar sensor.²⁰ The Denso DNSRR004 is a rear and side radar sensor for detecting vehicles in adjacent lanes.²¹ The Denso DNSRR004 is a component of the Toyota Blind Spot Monitor Sensor Module (“Toyota Blind Spot Monitor Sensor”) and includes an infringing Monolithic

²⁰ **Exhibit 47**, Tower Aug. 14, 2017 News Release, “TowerJazz Announces DENSO Corporation utilized its Advanced 0.18um SiGe Technology to Develop a 24GHz Rear and Side Radar Sensor,” <https://ir.towersemi.com/news-releases/news-release-details/towerjazz-announces-denso-corporation-utilized-its-advanced>.

²¹ **Exhibit 48**, Yolegroup, “Denso DNSRR004 Short Range Radar”, <https://medias.yolegroup.com/uploads/2019/12/SP19500-Yole-SYSTEM-Denso-DNSRR004-Blind-Spot-Radar-Flyer.pdf>.

Microwave Integrated Circuit (MMIC) manufactured using SiGe CMOS Technology from Tower. **Exhibit 48** at 1. The Toyota Blind Spot Monitor Sensor is imported into and sold in the United States. *See Exhibits 32-33*. A receipt showing purchase of a Toyota Blind Spot Monitor Sensor in the United States from Jim Coleman Toyota in Maryland on October 13, 2025 is attached as **Exhibit 32**. Photographs of the purchased Toyota Blind Spot Monitor Sensor show that the Denso DNSRR004 is made in Japan. **Exhibit 33** at 1-2. On information and belief, Tower continues to manufacture Denso's products.

VIII. HARMONIZED TARIFF SCHEDULE ITEM NUMBERS

8.1 On information and belief, the Harmonized Tariff Schedule of the United States item number under which the infringing semiconductor devices may be imported into the United States may be at least 8541.59.0040 (unmounted chips, dice and wafers). This classification are intended for illustration only and are not intended to be restrictive of the accused products or of products subject to the relief requested.

IX. THE DOMESTIC INDUSTRY

9.1 In accordance with Section 337(a)(2) and (a)(3), a domestic industry exists in the United States in connection with the Asserted Patents by virtue of the activities within the United States of GlobalFoundries, including manufacture, assembly, research, product development, and application engineering, sales and marketing. GlobalFoundries' significant investment in plant and equipment, and significant employment of labor and/or capital in the United States with respect to the Domestic Industry Products is described in greater detail below. The Asserted Patents are practiced by at least the following Domestic Industry Products:

U.S. Patent No.	Domestic Industry Products / Processes
8,330,235	Products manufactured using GlobalFoundries' 130CBIC process
8,507,983	Products manufactured using GlobalFoundries' CMHV7SF process
9,093,425	Products manufactured using GlobalFoundries' 130CBIC process
9,865,546	GlobalFoundries' 130CBIC process

U.S. Patent No.	Domestic Industry Products / Processes
10,707,167	Products manufactured using GlobalFoundries' 130CBIC process
10,062,748	Products manufactured using GlobalFoundries' CSOI8SW process

A. The Economic Prong of the Domestic Industry Requirement Is Satisfied

9.2 A domestic industry under subparts (A) and (B) of Section 337(a)(3) exists by virtue of GlobalFoundries' activities within the United States, including manufacture, research, product development, applications engineering, sales, and marketing.

9.3 GlobalFoundries has been manufacturing products in the United States since 2012, and it owns or leases numerous domestic facilities in the United States. GlobalFoundries' United States facilities locations include Austin, Texas (design center); Burlington, Vermont (manufacturing center); Dallas, Texas (design and engineering center); Malta, New York (manufacturing center); Raleigh, North Carolina (design center); and Santa Clara, California (design center and customer support). The Domestic Industry Products in this Investigation are manufactured at GlobalFoundries' Burlington, and Singapore facilities, and they are supported by research, product development, applications engineering, sales, and marketing activities undertaken throughout GlobalFoundries' United States facilities. GlobalFoundries' (i) quantitatively and qualitatively significant investment plant and equipment, and (ii) quantitatively and qualitatively significant employment of labor or capital with respect to the Domestic Industry Products is described in greater detail in **Confidential Exhibit 49**.

B. The Technical Prong of the Domestic Industry Requirement Is Satisfied

9.4 In accordance with Section 337(a)(2) and (a)(3), a domestic industry exists in the United States in connection with the Asserted Patents.

9.5 Claim charts and explanatory information for products and processes that currently practice at least one exemplary valid claim of the Asserted Patents accompany this Complaint as

Confidential Exhibits 50-55.²²

9.6 GlobalFoundries reserves its right to update its list of domestic industry products, including in light of additional discovery.

9.7 **Confidential Exhibit 50** includes a claim chart that shows how the '235 Patent Domestic Industry Products practice at least claim 1 of the '235 Patent.

9.8 **Confidential Exhibit 51** includes a claim chart that shows how the '983 Patent Domestic Industry Products practice at least claim 1 of the '983 Patent.

9.9 **Confidential Exhibit 52** includes a claim chart that shows how the '425 Patent Domestic Industry Products practice at least claim 1 of the '425 Patent.

9.10 **Confidential Exhibit 53** includes a claim chart that shows how the '546 Patent Domestic Industry Process practices at least claim 1 of the '546 Patent.

9.11 **Confidential Exhibit 54** includes a claim chart that shows how the '748 Patent Domestic Industry Products practice at least claim 1 of the '748 Patent.

9.12 **Confidential Exhibit 55** includes a claim chart that shows how the '167 Patent Domestic Industry Products practice at least claim 1 of the '167 Patent.

X. RELATED LITIGATION

10.1 The Asserted Patents are the subject of a civil action being concurrently filed in the U.S. District Court for the Western District of Texas.

10.2 Pursuant to Commission Rule 210.12(a)(5), GlobalFoundries states that, other than the litigation specified above, to GlobalFoundries' knowledge, the alleged unfair methods of competition and unfair acts, or the subject matter thereof, are not, and have not been, the subject of any court or agency litigation, or of any arbitration.

²² GlobalFoundries reserves its right to rely on additional claims of the Asserted Patents to support the technical prong of its domestic industry.

XI. RELIEF REQUESTED

11.1 WHEREFORE, by reason of the foregoing, GlobalFoundries respectfully requests that the U.S. International Trade Commission:

11.2 Institute an immediate investigation pursuant to Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337(a)(1)(B)(i) and (b)(1) with respect to violations of Section 337 based upon the importation, sale for importation, and sale after importation into the United States by the Proposed Respondents of certain semiconductor devices, products containing same, and components thereof, that infringe one or more of the asserted claims of U.S. Patent Nos. 8,330,235; 8,507,983; 9,093,425; 9,865,546; 10,062,748; and 10,707,167.

11.3 Find a violation of Section 337 based on said unlawful acts;

11.4 Issue a permanent limited exclusion order under 19 U.S.C. § 1337(d)(1) barring from entry into the United States infringing certain semiconductor devices, products containing same, and components thereof manufactured by or on behalf of, or imported by or on behalf of, Tower or its affiliates;

11.5 Issue permanent cease and desist orders, under 19 U.S.C. § 1337(f), directing the Tower Respondents to cease and desist from the sale for importation, importation, sale after importation, distribution, offering for sale, promoting, marketing, advertising, testing, demonstrating, warehousing inventory for distribution, solicitation of sales, programming, repairing, maintaining, using, transferring, and other commercial activity relating to infringing certain semiconductor devices, products containing same, and components thereof;

11.6 Impose a bond upon the Proposed Respondents who continue to import infringing articles, including certain semiconductor devices, products containing same, and components thereof, during the 60-day Presidential review period pursuant to 19 U.S.C. § 1337(j); and

11.7 Grant such other and further relief as the Commission deems just and proper based

on the facts determined by the investigation and the authority of the Commission.

Dated: August 31, 2026

Respectfully submitted,

/s/

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EXHIBIT B

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA- -1500

**[PROPOSED] AMENDED COMPLAINT OF GLOBALFOUNDRIES U.S. INC., UNDER
SECTION 337 OF THE TARIFF ACT OF 1930, AS AMENDED**

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TABLE OF CONTENTS

I.	INTRODUCTION.....	1
II.	COMPLAINANT.....	6
III.	PROPOSED RESPONDENTS.....	6
IV.	THE TECHNOLOGY AND PRODUCTS-AT-ISSUE.....	7
V.	THE ASSERTED PATENTS AND NON-TECHNICAL DESCRIPTION OF THE INVENTIONS.....	7
A.	U.S. Patent No. 8,330,235.....	8
	1. Identification of the Patent and Ownership by GlobalFoundries.....	8
	2. Non-Technical Description of the Patent.....	8
B.	U.S. Patent No. 8,507,983.....	9
	1. Identification of the Patent and Ownership by GlobalFoundries.....	9
	2. Non-Technical Description of the Patent.....	9
C.	U.S. Patent No. 9,093,425.....	10
	1. Identification of the Patent and Ownership by GlobalFoundries.....	10
	2. Non-Technical Description of the Patent.....	11
D.	U.S. Patent No. 9,865,546.....	12
	1. Identification of the Patent and Ownership by GlobalFoundries.....	12
	2. Non-Technical Description of the Patent.....	12
E.	U.S. Patent No. 10,062,748.....	13
	1. Identification of the Patent and Ownership by GlobalFoundries.....	13
	2. Non-Technical Description of the Patent.....	13
F.	U.S. Patent No. 10,707,167.....	14
	1. Identification of the Patent and Ownership by GlobalFoundries.....	14
	2. Non-Technical Description of the Patent.....	14

VI.	UNLAWFUL AND UNFAIR ACTS OF RESPONDENTS.....	15
A.	Infringement of U.S. Patent No. 8,330,235.....	15
B.	Infringement of U.S. Patent No. 8,507,983.....	15
C.	Infringement of U.S. Patent No. 9,093,425.....	16
D.	Infringement of U.S. Patent No. 9,865,546.....	16
E.	Infringement of U.S. Patent No. 10,062,748.....	17
F.	Infringement of U.S. Patent No. 10,707,167.....	17
VII.	SPECIFIC INSTANCES OF UNFAIR IMPORTATION AND SALE.....	17
A.	The Qorvo PAC22140 Evaluation Kit.....	18
B.	The Qorvo QM81026 RF Power Tracker.....	19
C.	The Denso DNSRR004.....	19
VIII.	HARMONIZED TARIFF SCHEDULE ITEM NUMBERS.....	20
IX.	THE DOMESTIC INDUSTRY.....	20
A.	The Economic Prong of the Domestic Industry Requirement Is Satisfied.....	21
B.	The Technical Prong of the Domestic Industry Requirement Is Satisfied.....	21
X.	RELATED LITIGATION.....	22
XI.	RELIEF REQUESTED.....	23

TABLE OF EXHIBITS

EXHIBIT	DOCUMENT
1.	Copy of U.S. Patent No. 8,330,235
2.	Copy of U.S. Patent No. 8,507,983
3.	Copy of U.S. Patent No. 9,093,425
4.	Copy of U.S. Patent No. 9,865,546
5.	Copy of U.S. Patent No. 10,062,748
6.	Copy of U.S. Patent No. 10,707,167
7.	Copies of Assignment Documents for U.S. Patent No. 8,330,235
8.	Copies of Assignment Documents for U.S. Patent No. 8,507,983
9.	Copies of Assignment Documents for U.S. Patent No. 9,093,425
10.	Copies of Assignment Documents for U.S. Patent No. 9,865,546
11.	Copies of Assignment Documents for U.S. Patent No. 10,062,748
12.	Copies of Assignment Documents for U.S. Patent No. 10,707,167
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22.	MacroTrends, “Tower Semiconductor Research and Development Expenses 2012-2025-TSEM,” https://www.macrotrends.net/stocks/charts/TSEM/tower-semiconductor/research-development-expenses
23.	Insights, “Tower Semiconductor Patents - Insights & Stats,” https://insights.greyb.com/tower-semiconductor-patents/
24.	List of Foreign Counterpart Patents and Foreign and Domestic Patent Applications for the Asserted Patents
25.	CONFIDENTIAL: List of Licensees
26.	Receipts Relating to the ’235, ’425, ’546, and ’167 Patents’ Representative Accused Product (Qorvo QM81026)
27.	Photographs Relating to the ’235, ’425, ’546, and ’167 Patents’ Representative Accused Product (Qorvo QM81026)
28.	Infringement Claim Chart for U.S. Patent No. 8,330,235 (Qorvo QM81026)
29.	Receipts Relating to the ’983 and ’748 Patents’ Representative Accused Product (Qorvo PAC22140)
30.	Photographs Relating to the ’983 and ’748 Patents’ Representative Accused Product (Qorvo PAC22140)
31.	Infringement Claim Chart for U.S. Patent No. 8,507,983 (Qorvo PAC22140)
32.	Receipts Relating to the ’425, ’546, and ’167 Patents’ Representative Accused Product (Denso DNSRR004)
33.	Photographs Relating to the ’425, ’546, and ’167 Patents’ Representative Accused Product (Denso DNSRR004)
34.	Infringement Claim Chart for U.S. Patent No. 9,093,425 (Qorvo QM81026)
35.	Infringement Claim Chart for U.S. Patent No. 9,093,425 (Denso DNSRR004)
36.	Infringement Claim Chart for U.S. Patent No. 9,865,546 (Qorvo QM81026)
37.	Infringement Claim Chart for U.S. Patent No. 9,865,546 (Denso DNSRR004)
38.	Infringement Claim Chart for U.S. Patent No. 10,062,748 (Qorvo PAC22140)
39.	Infringement Claim Chart for U.S. Patent No. 10,707,167 (Qorvo QM81026)
40.	Infringement Claim Chart for U.S. Patent No. 10,707,167 (Denso DNSRR004)
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44.	SiliconValley.com, “A Pillar of Specialty Silicon: A Comprehensive Deep Dive into Tower Semiconductor (TSEM),”

EXHIBIT	DOCUMENT
	https://markets.financialcontent.com/siliconvalley/article/predictstreet-2025-12-29-a-pillar-of-specialty-silicon-a-comprehensive-deep-dive-into-tower-semiconductor-tsem
45.	CONFIDENTIAL: Excerpts from WiSol Teardown Report
46.	CONFIDENTIAL: Tower Guard Ring Comparison
47.	Tower Aug, 14, 2017 News Release, “TowerJazz Announces DENSO Corporation utilized its Advanced 0.18um SiGe Technology to Develop a 24GHz Rear and Side Radar Sensor,” https://ir.towersemi.com/news-releases/news-release-details/towerjazz-announce-s-denso-corporation-utilized-its-advanced
48.	Yolegroup, “Denso DNSRR004 Short Range Radar”, https://medias.yolegroup.com/uploads/2019/12/SP19500-Yole-SYSTEM-Denso-DNSRR004-Blind-Spot-Radar-Flyer.pdf
49.	CONFIDENTIAL: Domestic Industry Exhibit
50.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 8,330,235
51.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 8,507,983
52.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 9,093,425
53.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 9,865,546
54.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 10,062,748
55.	CONFIDENTIAL: Domestic Industry Claim Chart for U.S. Patent No. 10,707,167

TABLE OF APPENDICES

APPENDIX	DOCUMENT
A.	Copy of the Prosecution History for U.S. Patent No. 8,330,235
B.	Copy of the Prosecution History for U.S. Patent No. 8,507,983
C.	Copy of the Prosecution History for U.S. Patent No. 9,093,425
D.	Copy of the Prosecution History for U.S. Patent No. 9,865,546
E.	Copy of the Prosecution History for U.S. Patent No. 10,062,748
F.	Copy of the Prosecution History for U.S. Patent No. 10,707,167
G.	Copy of the Prosecution History for the Priority Applications for U.S. Patent No. 8,330,235 (Application No. 12/366,378)
H.	Copy of the Prosecution Histories for the Priority Applications for U.S. Patent No. 8,507,983 (Application Nos. 12/390,509 and 12/500,620)

I. INTRODUCTION

1.1 GlobalFoundries U.S. Inc. (“GlobalFoundries” or “Complainant”) requests that the United States International Trade Commission institute an Investigation pursuant to Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337 (“Section 337”), to remedy the unlawful importation, sale for importation, and/or sale after importation of certain semiconductor devices, products containing same, and components thereof by Tower Semiconductor Ltd., (Israel); Tower Partners Semiconductor, Ltd. (Japan); Tower Semiconductor Italy S.R.L. (Italy); Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively “Tower” or “the Proposed Respondents”). Tower’s products infringe certain claims (“the Asserted Claims”) of U.S. Patent Nos. 8,330,235 (“the ’235 Patent”); 8,507,983 (“the ’983 Patent”); 9,093,425 (“the ’425 Patent”); 9,865,546 (“the ’546 Patent”); 10,062,748 (“the ’748 Patent”); and 10,707,167 (“the ’167 Patent”) (collectively, “the Asserted Patents”) as follows:

U.S. Patent No.	Asserted Claims ¹
8,330,235	1 , 5-9, 10 , 11, 12, 15-19
8,507,983	1-7 , 8-12 , 13-18 , 19 , 20
9,093,425	1-9
9,865,546	1-7 , 8-14
10,062,748	1 , 2-8, 9 , 10, 11 , 12, 14, 16-18
10,707,167	1-6

Pursuant to Commission Rule 212.(a)(9)(i), copies of the Asserted Patents are included as **Exhibits 1-6**². GlobalFoundries holds all rights, title, and interest in and to the Asserted Patents.

Pursuant to Commission Rule 212.(a)(9)(ii), copies of the Assignment Records for each of the Asserted Patents are included as **Exhibits 7-12**.

¹ Bolded claim numbers designate independent claims.

² Certified copies of the patents, assignments and prosecution histories at issue have been requested from the USPTO and will be provided to the Commission upon receipt.

1.2 GlobalFoundries is a leading American semiconductor manufacturer, a trusted supplier to the U.S. Department of War³ and a major recipient of federal investment under the CHIPS and Science Act.⁴ GlobalFoundries brings this action against Tower for infringement of GlobalFoundries' U.S. patents relating to its industry-leading RF and power management manufacturing process technologies. For over a decade, Tower has been engaged in a systematic strategy of freeloading, spending very little on capacity expansion (nearly none in the U.S.) and even less on R&D. Instead, it profits by leveraging the patented American innovation of others, like GlobalFoundries, and selling at lower prices into the market. GF brings this lawsuit to put an end to Tower's practices.

1.3 Semiconductor manufacturing today requires some of the most expensive and sophisticated factories on the planet, utilizing highly technical and complex nano-scale process technologies and costly manufacturing tools. Entire global industries, such as healthcare, artificial intelligence, automotive, and defense industries, depend on the availability and technological capability of the semiconductor products these factories and tools produce. This is why semiconductor chips are at the forefront of today's great global technological competition between the United States and China. It is imperative, therefore, for nations and companies to protect innovation in semiconductor process technologies, like the ones that GlobalFoundries invests in and creates, and that Tower has unlawfully used and profited from at GlobalFoundries' expense.

³ **Exhibit 13**, GlobalFoundries, "Aerospace, Defense and Critical Infrastructure," <https://gf.com/markets/aerospace-and-defense/>.

⁴ **Exhibit 14**, Nov. 20, 2024 GlobalFoundries Press Release, "GlobalFoundries and U.S. Department of Commerce Announce Award Agreement on CHIPS Act Funding for Essential Chip Manufacturing," <https://gf.com/gf-press-release/globalfoundries-and-u-s-department-of-commerce-announce-award-agreement-on-chips-act-funding-for-essential-chip-manufacturing/>.

1.4 GlobalFoundries is the most advanced pure-play foundry in the U.S. and Europe and employs thousands of people in the U.S. and worldwide. While other companies were abandoning semiconductor manufacturing in the U.S., GlobalFoundries bucked this trend by investing billions of dollars on advanced technology and research in the United States. GlobalFoundries originated from the semiconductor manufacturing arm of another leading U.S. semiconductor company, Advanced Micro Devices, in 2009, and expanded globally through acquisition and organic investment. A major \$15 billion U.S. investment in its 300 acre facility known as Fab 8 in Malta, New York, is the leading-edge of GlobalFoundries' significant U.S. investments. GlobalFoundries broke ground for that state of the art facility in 2009 and produces vanguard technology from that location for customers worldwide. Another major U.S. acquisition took place in 2015 when GlobalFoundries acquired IBM's microelectronic facilities and personnel in Burlington, Vermont, a facility that became known as Fab 9. GlobalFoundries acquired not just IBM's facilities and technologists, but also approximately 16,000 IBM patents and applications; decades of experience and expertise in semiconductor development, device expertise, design, and manufacturing; and an expanded manufacturing footprint. The acquisition cemented GlobalFoundries' position as a global leader in world-class semiconductor manufacturing and advanced process technologies.⁵

1.5 GlobalFoundries employs over 4,500 American workers and has invested billions of dollars over decades in domestic semiconductor manufacturing, including advanced radio frequency (RF) and power management platform technologies critical to national security,

⁵ **Exhibit 15**, GlobalFoundries July 1, 2015 Press Release, "GlobalFoundries Completes Acquisition of IBM Microelectronics Business," <https://gf.com/gf-press-release/globalfoundries-completes-acquisition-ibm-microelectronics-business/>.

aerospace, defense, and communications infrastructure. GlobalFoundries' foundational technologies are steeped in IBM and AMD heritage, two storied American companies that, prior to GlobalFoundries also spent billions in R&D on semiconductor manufacturing technology in the United States. Building on this technology foundation, GlobalFoundries invests over \$500 million annually in R&D and technology enablement,⁶ and employs thousands of engineers, to sustain its innovation leadership. GF has invested more than \$15 billion in its Fab 8 facility over the last decade to support innovation and manufacturing capacity. In 2021, the company doubled its planned investment to expand global capacity, with \$500 million targeted for Malta, NY alone.⁷ As a major CHIPS recipient, GlobalFoundries is committed to an additional \$16 billion in investment in its US fabs.⁸ All the while, GlobalFoundries has been creating and patenting hundreds of world class innovations annually to build and secure its technology differentiation leadership and rights.

1.6 In stark contrast to GlobalFoundries, Tower is a foundry business that has chronically underinvested in research and development relative to its U.S. peers. With a base of operations in Israel, and plants in Japan, Europe and the United States, Tower generated a little

⁶ **Exhibit 16**, GlobalFoundries Feb. 11, 2026 News Release, "GlobalFoundries Reports Fourth Quarter 2025 and Fiscal Year 2025 Financial Results," <https://investors.gf.com/news-releases/news-release-details/globalfoundries-reports-fourth-quarter-2025-and-fiscal-year-2025>.

⁷ **Exhibit 17**, GlobalFoundries Apr. 26, 2021 Press Release, "GLOBALFOUNDRIES Moves Corporate Headquarters to its Most Advanced Semiconductor Manufacturing Facility in New York," <https://gf.com/gf-press-release/globalfoundries-moves-corporate-headquarters-its-most-advanced-semiconductor/>.

⁸ **Exhibit 18**, GlobalFoundries June 4, 2025 Press Release, "GlobalFoundries Announces \$16B U.S. Investment to Reshore Essential Chip Manufacturing and Accelerate AI Growth," <https://gf.com/gf-press-release/globalfoundries-announces-16b-u-s-investment-to-reshore-essential-chip-manufacturing-and-accelerate-ai-growth/>.

over \$1.5 billion in 2025,⁹ and forecasts revenue will grow beyond \$2 billion.¹⁰ In 2023, it failed to merge with Intel and received \$353 million from Intel as a break-up fee,¹¹ spending most of that to fuel its foreign operations where the vast majority of its employees are located. More recently, Tower stated that its partnership with Intel has floundered and that it is in mediation with the U.S. chipmaker.¹² Historically, Tower has been a marginal foundry that struggled at times to survive, priding itself on a lowest-investment model. It has often bought old assets cheaply to avoid meaningful capex investments in its facilities, spending far less than the multiple billions required to compete lawfully. Tower's history of investment in R&D is even more meager. Historic figures show R&D expenses of less than \$90 million annually.¹³ Tower generally files fewer than 15 patent applications annually and holds fewer than 350 granted patents globally.¹⁴

1.7 Rather than investing at levels commensurate with the technological demands of advanced RF integration, Tower profits by offering products that incorporate innovations protected by GF's U.S. patents. Tower undercuts market pricing in the RF and power management segments, capturing share through low pricing strategies made possible by

⁹ **Exhibit 19**, Tower Feb. 11, 2026 Press Release, "Tower Semiconductor Reports Record Revenue for the Fourth Quarter of 2025," <https://towersemi.com/2026/02/11/02112026/>

¹⁰ **Exhibit 20**, Q1 2025 Tower Semiconductor Ltd Earnings Call on May 14, 2025, <https://ir.towersemi.com/static-files/4dbeca76-e387-4bd7-aad0-1daf4607282a> at p. 4.

¹¹ **Exhibit 21**, Intel Aug. 16, 2023 Newsroom Release, "Intel Announces Termination of Tower Semiconductor Acquisition," <https://newsroom.intel.com/corporate/intel-news-aug-2023>

¹² **Exhibit 19**, Tower Feb. 11, 2026 Press Release, "Tower Semiconductor Reports Record Revenue for the Fourth Quarter of 2025," <https://towersemi.com/2026/02/11/02112026/>

¹³ **Exhibit 22**, MacroTrends, "Tower Semiconductor Research and Development Expenses 2012-2025-TSEM," <https://www.macrotrends.net/stocks/charts/TSEM/tower-semiconductor/research-development-expenses>.

¹⁴ **Exhibit 23**, Insights, "Tower Semiconductor Patents - Insights & Stats," <https://insights.greyb.com/tower-semiconductor-patents/>.

avoiding, via infringement, the cost of the original R&D borne by GlobalFoundries. That strategy should not be permitted to continue to provide rewards going forward.

1.8 GlobalFoundries seeks a limited exclusion order barring from entry into the United States infringing semiconductor devices, products containing same, and components thereof manufactured by or on behalf of, or imported by or on behalf of, Tower. GlobalFoundries also seeks cease and desist orders prohibiting the sale for importation, importation, sale after importation, distribution, offer for sale, promotion, marketing, advertising, testing, demonstration, warehousing inventory for distribution, solicitation of sales, programming, repairing, maintenance, use, transfer, and other commercial activity relating to infringing semiconductor devices, products containing same, and components thereof. GlobalFoundries also seeks imposition of a bond during the Presidential Review Period pursuant to 19 U.S.C. § 1337(j).

1.9 As further set forth below, pursuant to Commission Rule 210.12(a)(6)(i), a domestic industry as defined in 19 U.S.C. § 1337(a)(2) and (3) exists in the United States in view of GlobalFoundries' domestic activities relating to the technology of the Asserted Patents and products that practice the Asserted Patents.

II. COMPLAINANT

2.1 Complainant GlobalFoundries U.S. Inc. is a Delaware Corporation with a principal place of business at 400 Stonebreak Road Ext., Malta, NY 12020.

III. PROPOSED RESPONDENTS

3.1 The Proposed Respondents are Tower Semiconductor Ltd., (Israel); Tower Partners Semiconductor, Ltd. (Japan); Tower Semiconductor Italy S.R.L. (Italy); Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings,

Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively “Tower” or “the Proposed Respondents”). On information and belief, each of the named Tower entities imports into the United States, sells for importation into the United States, and/or sells in the United States after importation certain semiconductor devices, products containing same, and components thereof. Tower’s semiconductor devices may be and are used in many products, including, without limitation, evaluation kits, garden tools, automotive products, and mobile devices which are imported into the United States as described below.

IV. THE TECHNOLOGY AND PRODUCTS-AT-ISSUE

4.1 This investigation generally concerns semiconductor devices and/or methods of manufacturing semiconductor devices. Descriptions of the specific technological features and advancements of each of the asserted patents are set forth below in Section V.

4.2 Tower imports, sells for importation, and/or sells within the United States after importation certain semiconductor devices that incorporate these functionalities.

4.3 Pursuant to 19 C.F.R. § 210.12(a)(12), the category of products accused are certain semiconductor devices manufactured using Tower’s RF, power management, BCD, logic, SiGe, and BiCMOS processes, including wafers and chips, and components thereof.

4.4 Tower imports, sells for importation, and/or sells within the United States, after importation, semiconductor devices that can be and are incorporated into products including, but not limited to, the Qorvo PAC22140 Evaluation Kit, the Denso DNSRR004 Blind Spot Monitor Sensor, and the Qorvo QM81026 RF Power Tracker.

4.5 The Accused Products directly or indirectly infringe one or more of the Asserted Claims by employing the above functionalities protected by GlobalFoundries’ innovations.

V. THE ASSERTED PATENTS AND NON-TECHNICAL DESCRIPTION OF THE INVENTIONS¹⁵

¹⁵ All non-technical descriptions of and categorizations of the patents are for the purposes of providing general information only and should not be interpreted as a representation regarding the

5.1 Pursuant to Commission Rule 210.12(c)(1), copies of the prosecution histories for the Asserted Patents have been submitted with this Complaint as **Appendices A-F**. Pursuant to Commission Rule 210.12(c)(2), copies of the prosecution histories for any priority applications for the Asserted Patents also have been submitted with this Complaint as **Appendices G-H**.

5.2 Pursuant to Commission Rule 210.12(a)(9)(v), a list of each foreign patent, each foreign or domestic patent application, and each foreign or domestic patent application that has been denied, abandoned or withdrawn, corresponding to each involved U.S. patent, with an indication of the prosecution status of each such patent application is attached as **Exhibit 24**.

5.3 Pursuant to Commission Rule 210.12(a)(9), a list of entities believed to be licensed to the Asserted Patents is attached as **Confidential Exhibit 25**.

A. U.S. Patent No. 8,330,235

1. Identification of the Patent and Ownership by GlobalFoundries

5.4 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 8,330,235 titled “METHOD TO REDUCE MOL DAMAGE ON NISI” on December 11, 2012. *See Exhibit 1* (’235 Patent). The ’235 Patent issued from Application No. 13/096,511, filed on April 28, 2011. *Id.*

5.5 GlobalFoundries is asserting claims 1, 5-9, 10, 11, 12, and 15-19 of the ’235 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.6 The Asserted Claims of the ’235 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’235 Patent on February 5, 2029. GlobalFoundries

providing general information only and should not be interpreted as a representation regarding the scope or applicability of any patent, or the requirements or meaning of any patent claims.

holds all rights, title, and interest in the '235 Patent with full rights to enforce the same. **Exhibit 7** ('235 Patent Assignments).

2. Non-Technical Description of the Patent

5.7 The '235 Patent is generally directed to transistor devices with nickel silicide layers that are formulated to prevent degradation upon removal of overlying stress liners. '235 Patent at Abstract. Conventional semiconductor devices include a refractory metal (e.g., nickel) deposited over exposed surface features to provide resistance to certain etchants. **Exhibit 1** ('235 Patent) at 1:19-28. Thermal processing (e.g., thermal annealing) causes the refractory metal to react with the underlying silicon to form electrically conductive silicide layer portions. *Id.* at 1:41-56. Some subsequent etching processes, such as reactive ion etching, can result in degradation of the silicide layer. *Id.* at 2:50-62. Nickel silicide is commonly doped with platinum to improve the material and electrical properties. *Id.* at 7:13-14. However, platinum also increases the resistivity of nickel silicide. The '235 Patent describes techniques for effective shielding of a nickel silicide layer while addressing a problematic increase in resistivity by compositionally grading a platinum-doped nickel silicide layer. *Id.* at 7:19-26. For example, the '235 Patent describes nickel silicide layers have a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. *Id.* at Abstract.

B. U.S. Patent No. 8,507,983

1. Identification of the Patent and Ownership by GlobalFoundries

5.8 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 8,507,983 titled "HIGH VOLTAGE DEVICE" on August 13, 2013. *See Exhibit 2* ('983 Patent). The '983 Patent issued from Application No. 13/550571,

filed on July 16, 2012. *Id.*

5.9 GlobalFoundries is asserting claims 1-7, [8-12](#), 13-18, and [19](#), 20 of the '983 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.10 The Asserted Claims of the '983 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '983 Patent on February 23, 2029. GlobalFoundries holds all rights, title, and interest in the '983 Patent with full rights to enforce the same. **Exhibit 8** ('983 Patent Assignments).

2. Non-Technical Description of the Patent

5.11 The '983 Patent is generally directed to semiconductor devices that can be used in high-voltage applications, which commonly include lateral double-diffused (LD) transistors. “For high performance LD transistors, low drain-to-source on-resistance ($R_{ds(on)}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability.” **Exhibit 2**, '983 Patent at 1:17-21. “To achieve low $R_{ds(on)}$, the channel of the high [voltage] LD transistor should be as short as possible.” *Id.* at 1:21-22.

5.12 The '983 Patent thus describes a high voltage device that has a gate within an active device region, along with a “doped channel well disposed in the substrate adjacent to a first edge of the gate.” *Id.* at 1:45-49. “The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate.” *Id.* at 1:49-51. “The first edge of the gate and channel edge defines an effective channel length of the device,” and the “effective channel length is self-aligned to the gate.” *Id.* at 1:51-53. “With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided [and] [v]ery short effective channel lengths with very low R_{ds} can be achieved, enabling high current to pass through a

small area.” *Id.* at 6:1-4. The claimed device can also include a “drift isolation region” that is disposed in “a doped drift well adjacent to a second edge of the gate.” *Id.* at 2:9-11. “The drift isolation region [] can improve breakdown voltage of the HV device due to high voltage.” *Id.* at 5:17-19. The device can also have diffusion regions adjacent to the gate, which “serve as, for example, source and drain of the transistor.” *Id.* at 3:55-57.

C. U.S. Patent No. 9,093,425

1. Identification of the Patent and Ownership by GlobalFoundries

5.13 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 9,093,425 titled “SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS” on July 28, 2015. *See Exhibit 3* (’425 Patent). The ’425 Patent issued from Application No. 14/177481, filed on February 11, 2014. *Id.*

5.14 GlobalFoundries is asserting claims 1-9 of the ’425 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.15 The Asserted Claims of the ’425 Patent are valid, enforceable, and currently in full force and effect until the expiration of the ’425 Patent on February 11, 2034. GlobalFoundries holds all rights, title, and interest in the ’425 Patent with full rights to enforce the same. **Exhibit 9** (’425 Patent Assignments).

2. Non-Technical Description of the Patent

5.16 The ’425 Patent is generally directed to self-aligned liners formed on metal semiconductor alloy contacts. “[M]etal semiconductor alloy containing contacts” can suffer from defects including “(1) [e]ncroachment of the metal semiconductor alloy which typically increases leakage through the source/drain junction” and “(2) [s]potty (non-uniform) metal semiconductor alloys which may cause unwanted etching of the metal semiconductor alloy

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contact during a subsequently performed etch.” **Exhibit 3**, ’425 Patent at 1:47-54. “These two defects are either formed during [or enhanced by] a subsequently performed etching process.” *Id.* at 1:55-57.

5.17 The ’425 Patent describes a semiconductor structure where “[m]etal semiconductor alloy contacts are provided on each of a source region and a drain region” and “[a] transition metal is then deposited on each of the metal semiconductor alloy contacts.” ’425 patent at 1:62-65. “[T]he deposited transition metal reacts ... with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact.” *Id.* at 1:66-2:3. “The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.” *Id.* at 2:7-11.

D. U.S. Patent No. 9,865,546

1. Identification of the Patent and Ownership by GlobalFoundries

5.18 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 9,865,546 titled “CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME” on January 9, 2018. *See Exhibit 4* (’546 Patent). The ’546 Patent issued from Application No. 14/729,298, filed on June 3, 2015. *Id.*

5.19 GlobalFoundries is asserting claims 1-7 and 8-14 of the ’546 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional claims.

5.20 The Asserted Claims of the '546 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '546 Patent on July 3, 2035. GlobalFoundries holds all rights, title, and interest in the '546 Patent with full rights to enforce the same. **Exhibit 10** ('546 Patent Assignments).

2. Non-Technical Description of the Patent

5.21 The '546 Patent is generally directed to formation of a contact in a dielectric layer over a semiconductor substrate. **Exhibit 4**, ('546 Patent) at Abstract. More specifically, the '546 Patent relates to forming an intermix region at an interface of a bottom portion of a contact and an upper portion of a semiconductor substrate. *Id.* at 1:7-13. For example, the '546 Patent discloses techniques for depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate (e.g., the source and/or drain). *Id.* at 2:43-49. Unreacted (non-diffused) portions of the first liner layer can then be removed before forming the contact. *Id.* at 2:49-51. Such techniques may, for example, improve contact resistance while avoiding issues caused by increased metal thickness in contact openings. *Id.* at 2:51-54.

E. U.S. Patent No. 10,062,748

1. Identification of the Patent and Ownership by GlobalFoundries

5.22 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 10,062,748 titled "SEGMENTED GUARD-RING AND CHIP EDGE SEALS" on August 28, 2018. *See Exhibit 5* ('748 Patent). The '748 Patent issued from Application No. 15/443276, filed on February 27, 2017. *Id.*

5.23 GlobalFoundries is asserting claims 1-12, 14, and 16-18 of the '748 Patent in this Investigation. Further investigation and discovery may lead to the assertion of additional

claims.

5.24 The Asserted Claims of the '748 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '748 Patent on February 27, 2037. GlobalFoundries holds all rights, title, and interest in the '748 Patent with full rights to enforce the same. **Exhibit 11** ('748 Patent Assignments).

2. Non-Technical Description of the Patent

5.25 The '748 Patent is generally directed to “segmented guard-ring and chip edge seals.” **Exhibit 5**, '748 Patent at 1:6-8. Guard rings are useful in many technologies, including “low-k dielectric, porous applications” in which the guard ring can help “prevent moisture ingress.” *Id.* at 1:25-28. However, “the use of a segmented guard ring can cause reliability fails in these low-k dielectric, porous applications.” *Id.* at 1:28-31. The '748 Patent “provides segmented guard-ring and chip edge seals with low-K dielectric material in middle of line (MOL) and/or back end of the line (BEOL) processing.” *Id.* at 2:23-27. These structures can be compatible with the low-K dielectric material which “can be a non-porous or porous polymer, carbon doped oxide, or oxide or combinations thereof.” *Id.* at 3:35-37.

F. U.S. Patent No. 10,707,167

1. Identification of the Patent and Ownership by GlobalFoundries

5.26 The United States Patent and Trademark Office, after full and fair examination, duly and legally issued U.S. Patent No. 10,707,167 titled “CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME” on July 7, 2020. *See Exhibit 6* ('167 Patent). The '167 Patent issued from Application No. 15/826,939, filed on November 30, 2017. *Id.*

5.27 GlobalFoundries is asserting claims 1-6 of the '167 Patent in this Investigation.

Further investigation and discovery may lead to the assertion of additional claims.

5.28 The Asserted Claims of the '167 Patent are valid, enforceable, and currently in full force and effect until the expiration of the '167 Patent on May 30, 2036. GlobalFoundries holds all rights, title, and interest in the '167 Patent with full rights to enforce the same. **Exhibit 12** ('167 Patent Assignments).

2. Non-Technical Description of the Patent

5.29 The '167 Patent is generally directed to formation of a contact in a dielectric layer over a semiconductor substrate. **Exhibit 6** ('167 Patent) at Abstract. More specifically, the '167 Patent relates to forming an intermix region at an interface of a bottom portion of a contact and an upper portion of a semiconductor substrate. *Id.* at 1:9-15. For example, the '167 Patent discloses techniques for depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate (e.g., the source and/or drain). *Id.* at 2:46-51. Unreacted (non-diffused) portions of the first liner layer can then be removed before forming the contact. *Id.* at 2:51-53. Such techniques may, for example, improve contact resistance while avoiding issues caused by increased metal thickness in contact openings. *Id.* at 2:53-56.

VI. UNLAWFUL AND UNFAIR ACTS OF RESPONDENTS

6.1 The unlawful and unfair acts of Tower identified below are exemplary and should not be considered in any way limiting.

6.2 GlobalFoundries reserves the right to identify additional infringing products in light of further discovery. GlobalFoundries has not yet had the benefit of discovery, and thus its identification of specific models or types of products below is not intended to limit the scope of the investigation. Any remedy should extend to all infringing products.

A. Infringement of U.S. Patent No. 8,330,235

6.3 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '235 Patent, literally or under the doctrine of equivalents. The '235 Accused Products include at least devices manufactured using Tower's RF processes RF processes (including its 65nm RF process). For example, the Tower-made Qorvo QM81026 Power Tracker (the "'235 Representative Accused Product") infringes at least claims 1, 5-9, 10, 11, 12, and 15-19 of the '235 Patent. *See Exhibits 26-27.*

6.4 A claim chart comparing the independent claims of the '235 Patent to the '235 Representative Accused Product is attached as **Exhibit 28**.

B. Infringement of U.S. Patent No. 8,507,983

6.5 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '983 Patent, literally or under the doctrine of equivalents. The '983 Accused Products include at least products manufactured using Tower's BCD processes (including its TS18SOI process). For example, the Tower-made PA22BZdie in the Qorvo PAC22140 Evaluation Kit (the "'983 Representative Accused Product") infringes at least claims 1-7, [8-12](#), 13-18, [19](#), and 20 of the '983 Patent. *See Exhibits 29-30.*

6.6 A claim chart comparing the independent claims of the '983 Patent to the '983 Tower Representative Accused Product is attached as **Exhibit 31**.

C. Infringement of U.S. Patent No. 9,093,425

6.7 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '425 Patent, literally or

under the doctrine of equivalents. The '425 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'425 Representative Accused Products") infringe at least claims 1-9 of the '425 Patent. *See Exhibits 32-33 and Exhibits 26-27.*

6.8 Claim charts comparing the independent claim of the '425 Patent to the Tower '425 Representative Accused Products are attached as **Exhibit 34** and **Exhibit 35**.

D. Infringement of U.S. Patent No. 9,865,546

6.9 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '546 Patent, literally or under the doctrine of equivalents. The '546 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'546 Representative Accused Products") infringe at least claims 1-14 of the '546 Patent. *See Exhibits 32-33 and Exhibits 26-27.*

6.10 Claim charts comparing the independent claims of the '546 Patent to the Tower '546 Representative Accused Products are attached as **Exhibit 36** and **Exhibit 37**.

E. Infringement of U.S. Patent No. 10,062,748

6.11 On information and belief, Tower imports, sells for importation, and/or sells after

importation into the United States, Accused Products that infringe the '748 Patent, literally or under the doctrine of equivalents. The '748 Accused Products include at least products manufactured using Tower's BCD processes (including its TS18SOI process). For example, the Tower-made PA22BZdie in the Qorvo PAC22140 Evaluation Kit (the "'748 Representative Accused Product") infringes at least claims 1-12, 14, and 16-18 of the '748 Patent. *See Exhibits 29-30.*

6.12 A claim chart comparing the independent claims of the '748 Patent to the Tower '748 Representative Accused Product is attached as **Exhibit 38**.

F. Infringement of U.S. Patent No. 10,707,167

6.13 On information and belief, Tower imports, sells for importation, and/or sells after importation into the United States, Accused Products that infringe the '167 Patent, literally or under the doctrine of equivalents. The '167 Accused Products include at least products manufactured using Tower's SiGe and BiCMOS processes (including the Tower SBC18 process), as well as its RF processes (including its 65nm RF process). For example, the Tower-manufactured M7DN25PS die in the DENSO DNSRR004 Blind Spot Monitor Sensor and the Tower-made Qorvo QM81026 Power Tracker (the "'167 Representative Accused Products") infringe at least claims 1-6 of the '167 Patent. *See Exhibits 32-33 and Exhibits 26-27.*

6.14 Claim charts comparing the independent claim of the '167 Patent to the Tower '167 Representative Accused Products are attached as **Exhibit 39** and **Exhibit 40**.

VII. SPECIFIC INSTANCES OF UNFAIR IMPORTATION AND SALE

7.1 On information and belief, Tower is importing, selling for importation, and/or selling within the United States after importation, certain semiconductor devices, products

containing same, and components thereof.

7.2 The specific instances set forth below are representative examples of Tower's unlawful importation, sale for importation, and/or sales within the United States after importation of infringing products.

A. The Qorvo PAC22140 Evaluation Kit

7.3 Qorvo offers a family of products under the name Power Application Controller (PAC).¹⁶ In 2018, semiconductor company Active-Semi announced that it was relying on Tower to manufacture a PAC product referred to as PAC5232.¹⁷ In 2019, Qorvo acquired Active-Semi.¹⁸ Qorvo remains one of Tower's major customers.¹⁹ On information and belief, Tower continues to manufacture Qorvo's PAC family of products.

7.4 The Qorvo PAC22140 chip is part of Qorvo's family of PAC products. **Exhibit 41** at 2. This chip, which includes an infringing PA22BZ die manufactured by Tower, is a component of the Qorvo PAC22140 Evaluation Kit ("Qorvo PAC22140EVK1") that is imported into and sold in the United States. *See Exhibits 29-30*. **Exhibit 29** is a set of receipts showing the purchase of a Qorvo PAC22140EVK1 in the United States from Mouser Electronics on March 13, 2026. These receipts show that an import tariff was applied to the order, specifically

¹⁶ **Exhibit 41**, Qorvo Product List at 2-5, <https://www.qorvo.com/products/product-list?categoryID=ca0128>.

¹⁷ **Exhibit 42**, Qorvo March 19, 2018 News Release, "TowerJazz and Active-Semi Announce Ramp to Production of Motor Driver IC on 200V SOI Technology," <https://www.qorvo.com/newsroom/news/2018/towerjazz-and-active-semi-announce-ramp-to-production-of-motor-driver-ic-on-200v-soi-technology>.

¹⁸ **Exhibit 43**, Qorvo Apr. 10, 2019 News Release, "Qorvo to Acquire Active-Semi International," <https://www.qorvo.com/newsroom/news/2019/qorvo-to-acquire-active-semi-international>.

¹⁹ *See, e.g.*, **Exhibit 44**, SiliconValley.com, "A Pillar of Specialty Silicon: A Comprehensive Deep Dive into Tower Semiconductor (TSEM)," <https://markets.financialcontent.com/siliconvalley/article/predictstreet-2025-12-29-a-pillar-of-specialty-silicon-a-comprehensive-deep-dive-into-tower-semiconductor-tsem>.

in the amount of \$56.25, confirming that the Qorvo PAC22140EVK1 was imported into the United States. **Exhibit 29.** The purchased product arrived in an Active-Semi box, labeled with the Qorvo PAC22140EVK1 product name. **Exhibit 30** (photographs of the purchased PAC22140EVK1) at 2.

B. The Qorvo QM81026 RF Power Tracker

7.5 The Qorvo QM81026 RF Power Tracker chip (“Qorvo QM81026 RF chip”) is a component of a smartphone, which is assembled in India and is imported into and sold in the United States. *See Exhibits 26-27.* A receipt showing purchase of a smartphone in the United States on February 28, 2026 is attached as **Exhibit 26.** Qorvo is a significant Tower customer. **Exhibit 44.** On information and belief, the Qorvo QM81026 RF chip is manufactured by Tower. For example, the guard ring structures in the accused Qorvo QM81026 RF chip are identical to the guard ring structures found in other RF chips known to be Tower-manufactured, such as the Wisol CL301F1 LNA. *See Confidential Exhibits 45-46.* Therefore, on information and belief, Tower manufactures the Qorvo QM81026 RF chip.

C. The Denso DNSRR004

7.6 In 2017 Tower announced that it used its technology to develop Denso’s rear and side radar sensor.²⁰ The Denso DNSRR004 is a rear and side radar sensor for detecting vehicles in adjacent lanes.²¹ The Denso DNSRR004 is a component of the Toyota Blind Spot

²⁰ **Exhibit 47**, Tower Aug. 14, 2017 News Release, “TowerJazz Announces DENSO Corporation utilized its Advanced 0.18um SiGe Technology to Develop a 24GHz Rear and Side Radar Sensor,” <https://ir.towersemi.com/news-releases/news-release-details/towerjazz-announces-denso-corporation-utilized-its-advanced>.

²¹ **Exhibit 48**, Yolegroup, “Denso DNSRR004 Short Range Radar”, <https://medias.yolegroup.com/uploads/2019/12/SP19500-Yole-SYSTEM-Denso-DNSRR004-Blind-Spot-Radar-Flyer.pdf>.

Monitor Sensor Module (“Toyota Blind Spot Monitor Sensor”) and includes an infringing Monolithic Microwave Integrated Circuit (MMIC) manufactured using SiGe CMOS Technology from Tower. **Exhibit 48** at 1. The Toyota Blind Spot Monitor Sensor is imported into and sold in the United States. *See Exhibits 32-33*. A receipt showing purchase of a Toyota Blind Spot Monitor Sensor in the United States from Jim Coleman Toyota in Maryland on October 13, 2025 is attached as **Exhibit 32**. Photographs of the purchased Toyota Blind Spot Monitor Sensor show that the Denso DNSRR004 is made in Japan. **Exhibit 33** at 1-2. On information and belief, Tower continues to manufacture Denso’s products.

VIII. HARMONIZED TARIFF SCHEDULE ITEM NUMBERS

8.1 On information and belief, the Harmonized Tariff Schedule of the United States item number under which the infringing semiconductor devices may be imported into the United States may be at least 8541.59.0040 (unmounted chips, dice and wafers). This classification are intended for illustration only and are not intended to be restrictive of the accused products or of products subject to the relief requested.

IX. THE DOMESTIC INDUSTRY

9.1 In accordance with Section 337(a)(2) and (a)(3), a domestic industry exists in the United States in connection with the Asserted Patents by virtue of the activities within the United States of GlobalFoundries, including manufacture, assembly, research, product development, and application engineering, sales and marketing. GlobalFoundries’ significant investment in plant and equipment, and significant employment of labor and/or capital in the United States with respect to the Domestic Industry Products is described in greater detail below. The Asserted Patents are practiced by at least the following Domestic Industry Products:

U.S. Patent No.	Domestic Industry Products / Processes
8,330,235	Products manufactured using GlobalFoundries’ 130CBIC process
8,507,983	Products manufactured using GlobalFoundries’ CMHV7SF process

U.S. Patent No.	Domestic Industry Products / Processes
9,093,425	Products manufactured using GlobalFoundries' 130CBIC process
9,865,546	GlobalFoundries' 130CBIC process
10,707,167	Products manufactured using GlobalFoundries' 130CBIC process
10,062,748	Products manufactured using GlobalFoundries' CSOI8SW process

A. The Economic Prong of the Domestic Industry Requirement Is Satisfied

9.2 A domestic industry under subparts (A) and (B) of Section 337(a)(3) exists by virtue of GlobalFoundries' activities within the United States, including manufacture, research, product development, applications engineering, sales, and marketing.

9.3 GlobalFoundries has been manufacturing products in the United States since 2012, and it owns or leases numerous domestic facilities in the United States. GlobalFoundries' United States facilities locations include Austin, Texas (design center); Burlington, Vermont (manufacturing center); Dallas, Texas (design and engineering center); Malta, New York (manufacturing center); Raleigh, North Carolina (design center); and Santa Clara, California (design center and customer support). The Domestic Industry Products in this Investigation are manufactured at GlobalFoundries' Burlington, and Singapore facilities, and they are supported by research, product development, applications engineering, sales, and marketing activities undertaken throughout GlobalFoundries' United States facilities. GlobalFoundries' (i) quantitatively and qualitatively significant investment plant and equipment, and (ii) quantitatively and qualitatively significant employment of labor or capital with respect to the Domestic Industry Products is described in greater detail in **Confidential Exhibit 49**.

B. The Technical Prong of the Domestic Industry Requirement Is Satisfied

9.4 In accordance with Section 337(a)(2) and (a)(3), a domestic industry exists in the United States in connection with the Asserted Patents.

9.5 Claim charts and explanatory information for products and processes that currently practice at least one exemplary valid claim of the Asserted Patents accompany this Complaint as **Confidential Exhibits 50-55**.²²

9.6 GlobalFoundries reserves its right to update its list of domestic industry products, including in light of additional discovery.

9.7 **Confidential Exhibit 50** includes a claim chart that shows how the '235 Patent Domestic Industry Products practice at least claim 1 of the '235 Patent.

9.8 **Confidential Exhibit 51** includes a claim chart that shows how the '983 Patent Domestic Industry Products practice at least claim 1 of the '983 Patent.

9.9 **Confidential Exhibit 52** includes a claim chart that shows how the '425 Patent Domestic Industry Products practice at least claim 1 of the '425 Patent.

9.10 **Confidential Exhibit 53** includes a claim chart that shows how the '546 Patent Domestic Industry Process practices at least claim 1 of the '546 Patent.

9.11 **Confidential Exhibit 54** includes a claim chart that shows how the '748 Patent Domestic Industry Products practice at least claim 1 of the '748 Patent.

9.12 **Confidential Exhibit 55** includes a claim chart that shows how the '167 Patent Domestic Industry Products practice at least claim 1 of the '167 Patent.

X. RELATED LITIGATION

10.1 The Asserted Patents are the subject of a civil action being concurrently filed in the U.S. District Court for the Western District of Texas.

10.2 Pursuant to Commission Rule 210.12(a)(5), GlobalFoundries states that, other

²² GlobalFoundries reserves its right to rely on additional claims of the Asserted Patents to support the technical prong of its domestic industry.

than the litigation specified above, to GlobalFoundries' knowledge, the alleged unfair methods of competition and unfair acts, or the subject matter thereof, are not, and have not been, the subject of any court or agency litigation, or of any arbitration.

XI. RELIEF REQUESTED

11.1 WHEREFORE, by reason of the foregoing, GlobalFoundries respectfully requests that the U.S. International Trade Commission:

11.2 Institute an immediate investigation pursuant to Section 337 of the Tariff Act of 1930, as amended, 19 U.S.C. § 1337(a)(1)(B)(i) and (b)(1) with respect to violations of Section 337 based upon the importation, sale for importation, and sale after importation into the United States by the Proposed Respondents of certain semiconductor devices, products containing same, and components thereof, that infringe one or more of the asserted claims of U.S. Patent Nos. 8,330,235; 8,507,983; 9,093,425; 9,865,546; 10,062,748; and 10,707,167.

11.3 Find a violation of Section 337 based on said unlawful acts;

11.4 Issue a permanent limited exclusion order under 19 U.S.C. § 1337(d)(1) barring from entry into the United States infringing certain semiconductor devices, products containing same, and components thereof manufactured by or on behalf of, or imported by or on behalf of, Tower or its affiliates;

11.5 Issue permanent cease and desist orders, under 19 U.S.C. § 1337(f), directing the Tower Respondents to cease and desist from the sale for importation, importation, sale after importation, distribution, offering for sale, promoting, marketing, advertising, testing, demonstrating, warehousing inventory for distribution, solicitation of sales, programming, repairing, maintaining, using, transferring, and other commercial activity relating to infringing certain semiconductor devices, products containing same, and components thereof;

11.6 Impose a bond upon the Proposed Respondents who continue to import infringing articles, including certain semiconductor devices, products containing same, and components thereof, during the 60-day Presidential review period pursuant to 19 U.S.C. § 1337(j); and

11.7 Grant such other and further relief as the Commission deems just and proper based on the facts determined by the investigation and the authority of the Commission.

Dated: ~~March 26~~August 31, 2026

Respectfully submitted,

/s/ ~~Bert C. Reiser~~

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Counsel for Complainant
~~GlobalFoundries U.S. Inc.~~

VERIFICATION

~~I, Adam Noah, declare in accordance with 19 C.F.R. §§ 210.4 and 210.12(a), under penalty of perjury, that the following statements are true:~~

~~1. I am VP and Chief IP Counsel of GlobalFoundries U.S. Inc., and I am duly authorized to verify this Complaint under Section 337 of the Tariff Act of 1930, as Amended ("Complaint");~~

~~2. I have read the Complaint and am aware of its contents;~~

~~3. To the best of my knowledge, information, and belief, formed after an inquiry reasonable under the circumstances, (a) the claims and other legal contentions in the Complaint are warranted by existing law or by a non-frivolous argument for the extension, modification, or reversal of existing law or the establishment of new law, and (b) the allegations and other factual contentions in the Complaint have evidentiary support or, if specifically so identified, are likely to have evidentiary support after a reasonable opportunity for further investigation or discovery; and~~

~~4. The Complaint is not being presented for any improper purpose, such as to harass or to cause unnecessary delay or needless increase in the cost of the Investigation or related proceedings.~~

~~5. I declare under penalty of perjury under the laws of the United States that the foregoing is true and correct. Executed on March 25, 2026 in Amherst, Massachusetts.~~

Adam Noah
VP and Chief IP Counsel
GlobalFoundries U.S. Inc.

EXHIBIT C



US008507983B2

(12) **United States Patent**
Zhang et al.

(10) **Patent No.:** **US 8,507,983 B2**
(45) **Date of Patent:** **Aug. 13, 2013**

(54) **HIGH VOLTAGE DEVICE**

(56) **References Cited**

(75) Inventors: **Guowei Zhang**, Singapore (SG);
Purakh Raj Verma, Singapore (SG)

(73) Assignee: **GLOBALFOUNDRIES Singapore Pte. Ltd.**, Singapore (SG)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **13/550,571**

(22) Filed: **Jul. 16, 2012**

(65) **Prior Publication Data**

US 2012/0280318 A1 Nov. 8, 2012

Related U.S. Application Data

(60) Division of application No. 12/500,620, filed on Jul. 10, 2009, now Pat. No. 8,222,130, which is a continuation-in-part of application No. 12/390,509, filed on Feb. 23, 2009, now Pat. No. 8,053,319.

(51) **Int. Cl.**
H01L 29/76 (2006.01)

(52) **U.S. Cl.**
USPC **257/335; 257/E29.256**

(58) **Field of Classification Search**
USPC **257/335, 346, E29.256**
See application file for complete search history.

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* cited by examiner

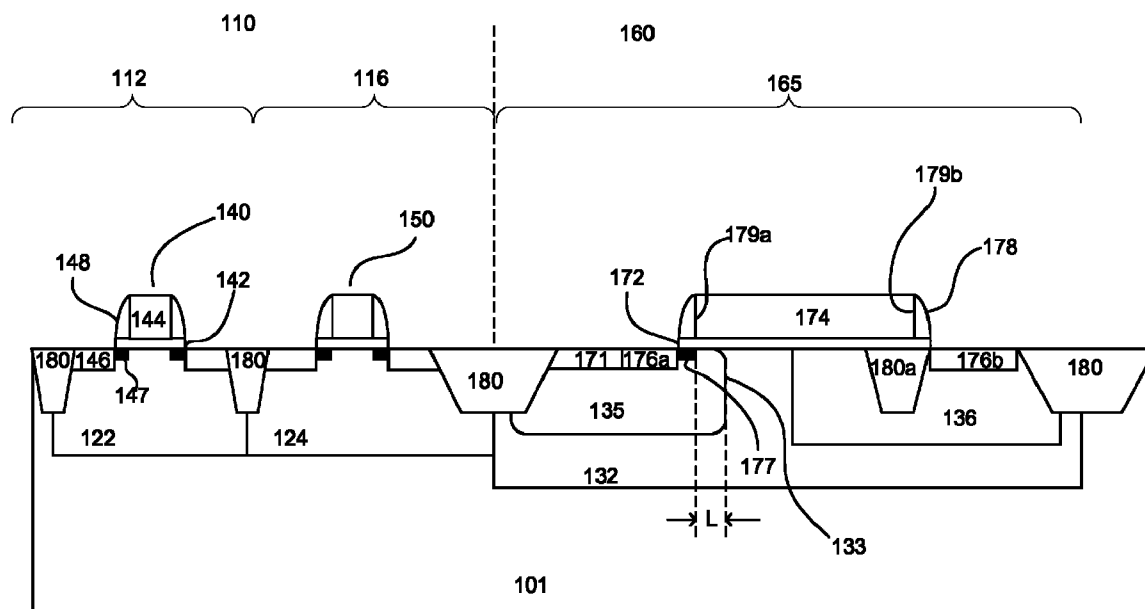
Primary Examiner — Quoc Hoang

(74) *Attorney, Agent, or Firm* — Horizon IP Pte Ltd

(57) **ABSTRACT**

A device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. A doped drift well adjacent to a second edge of the gate is also included.

20 Claims, 29 Drawing Sheets



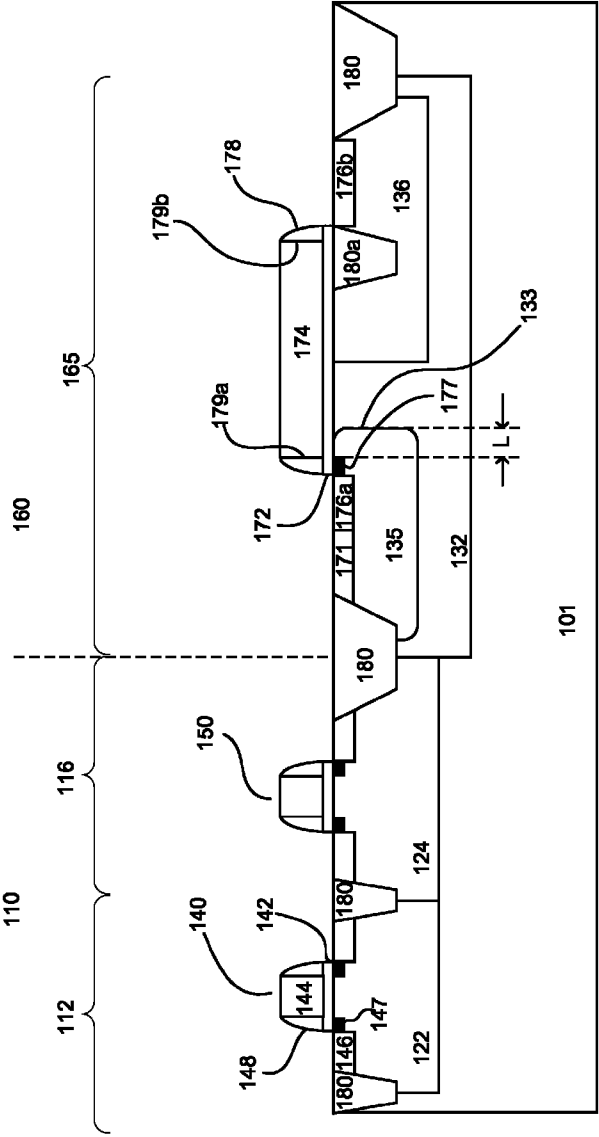


Fig. 1

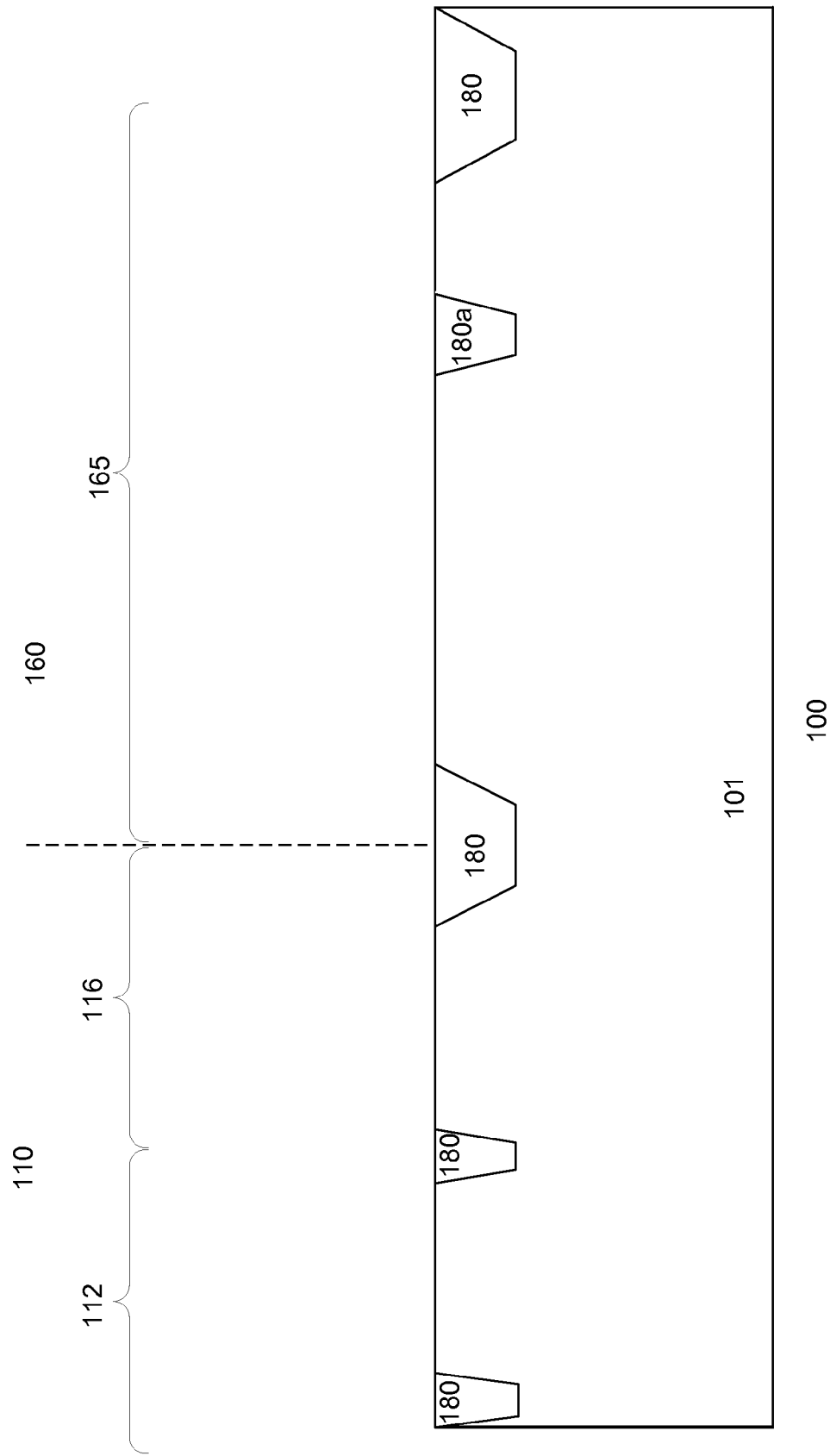


Fig. 2a

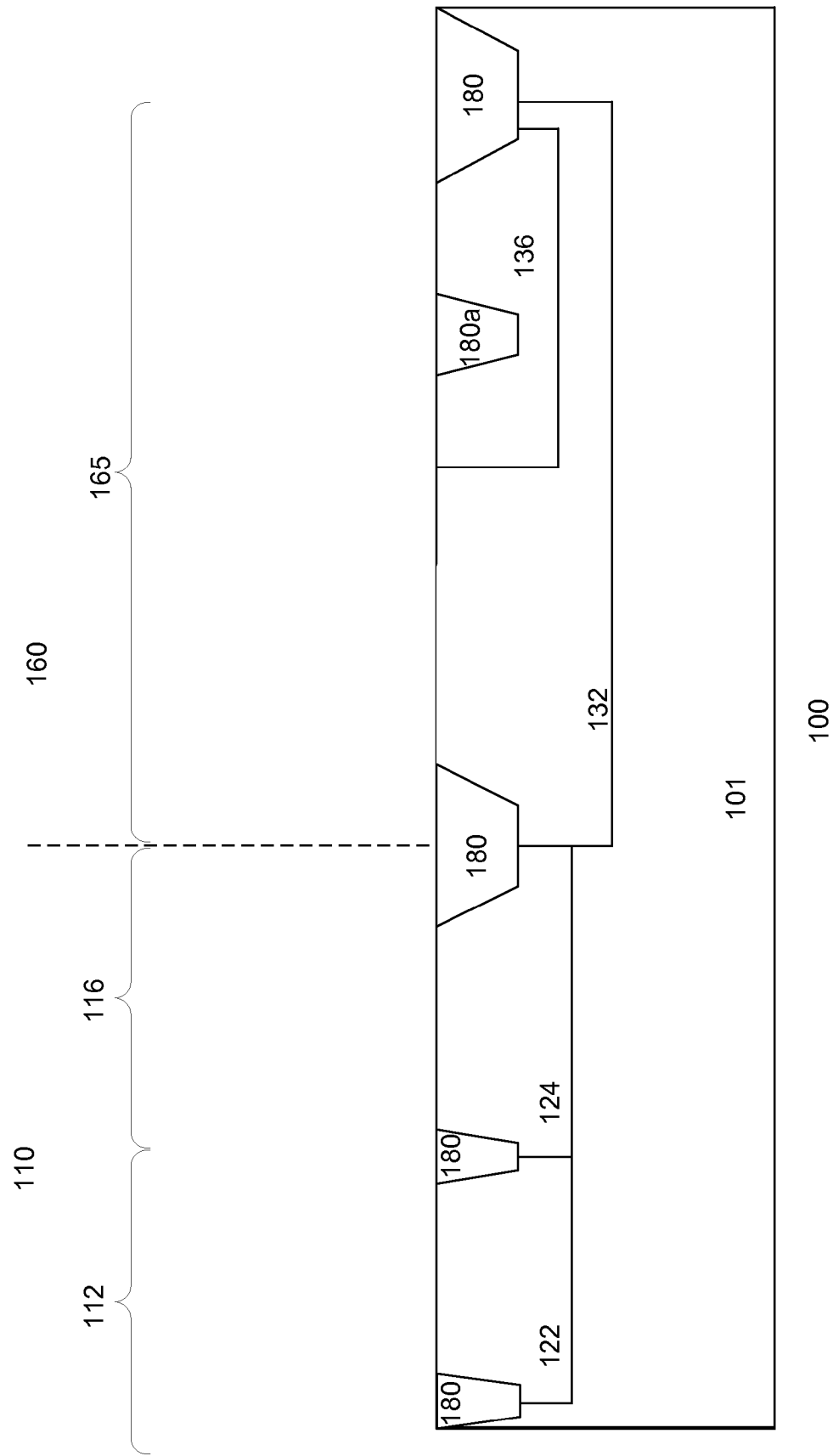


Fig. 2b

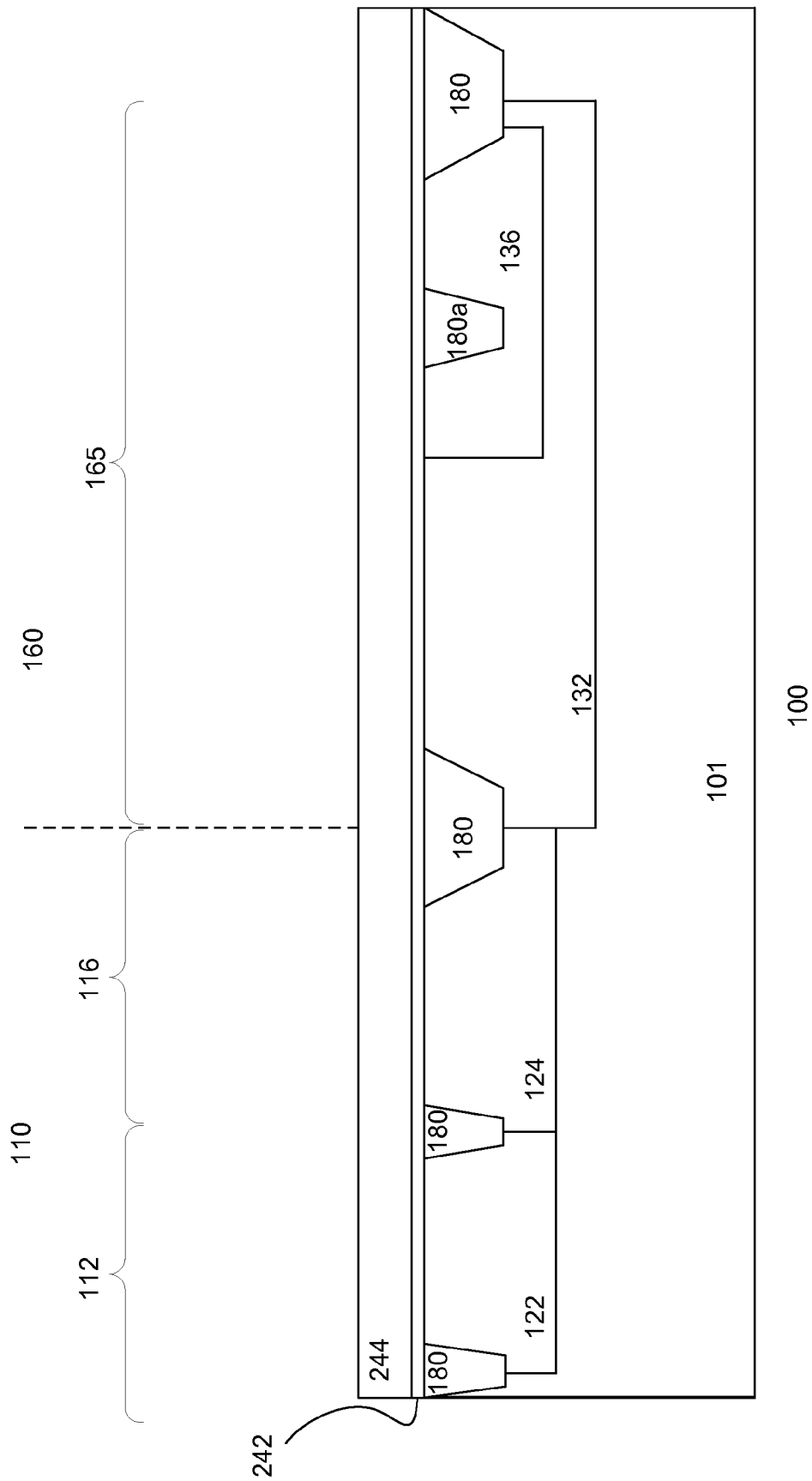


Fig. 2c

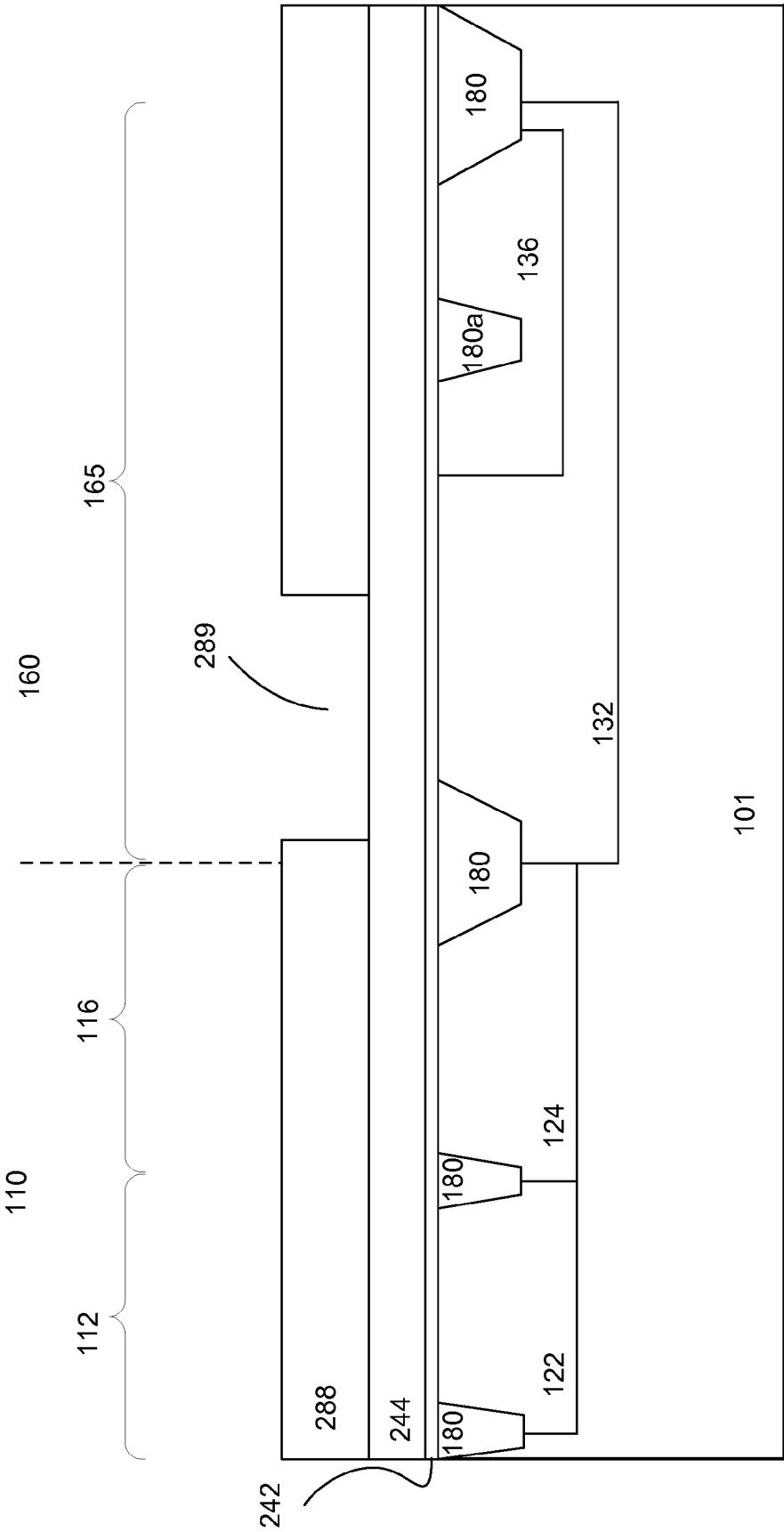


Fig. 2d

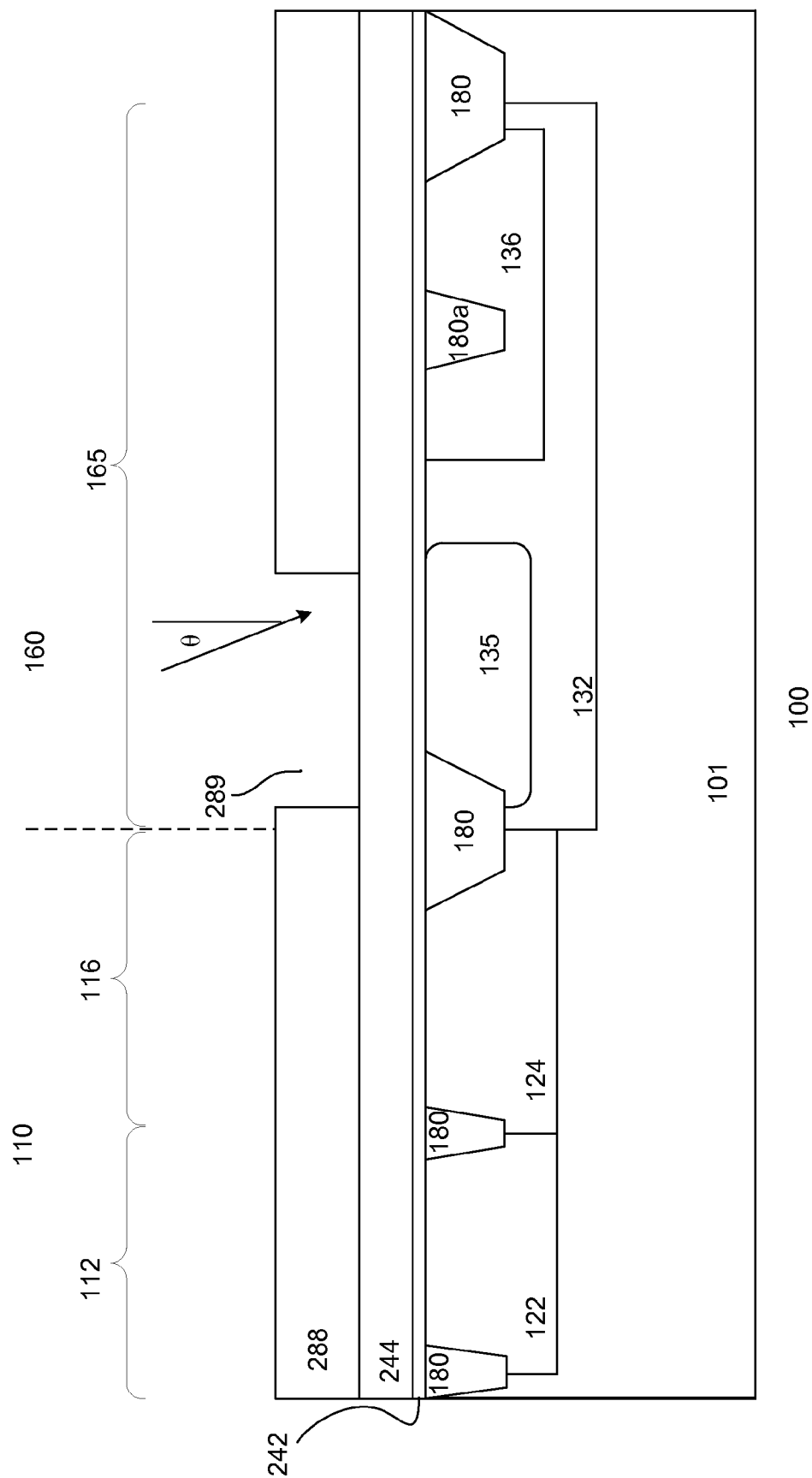


Fig. 2e

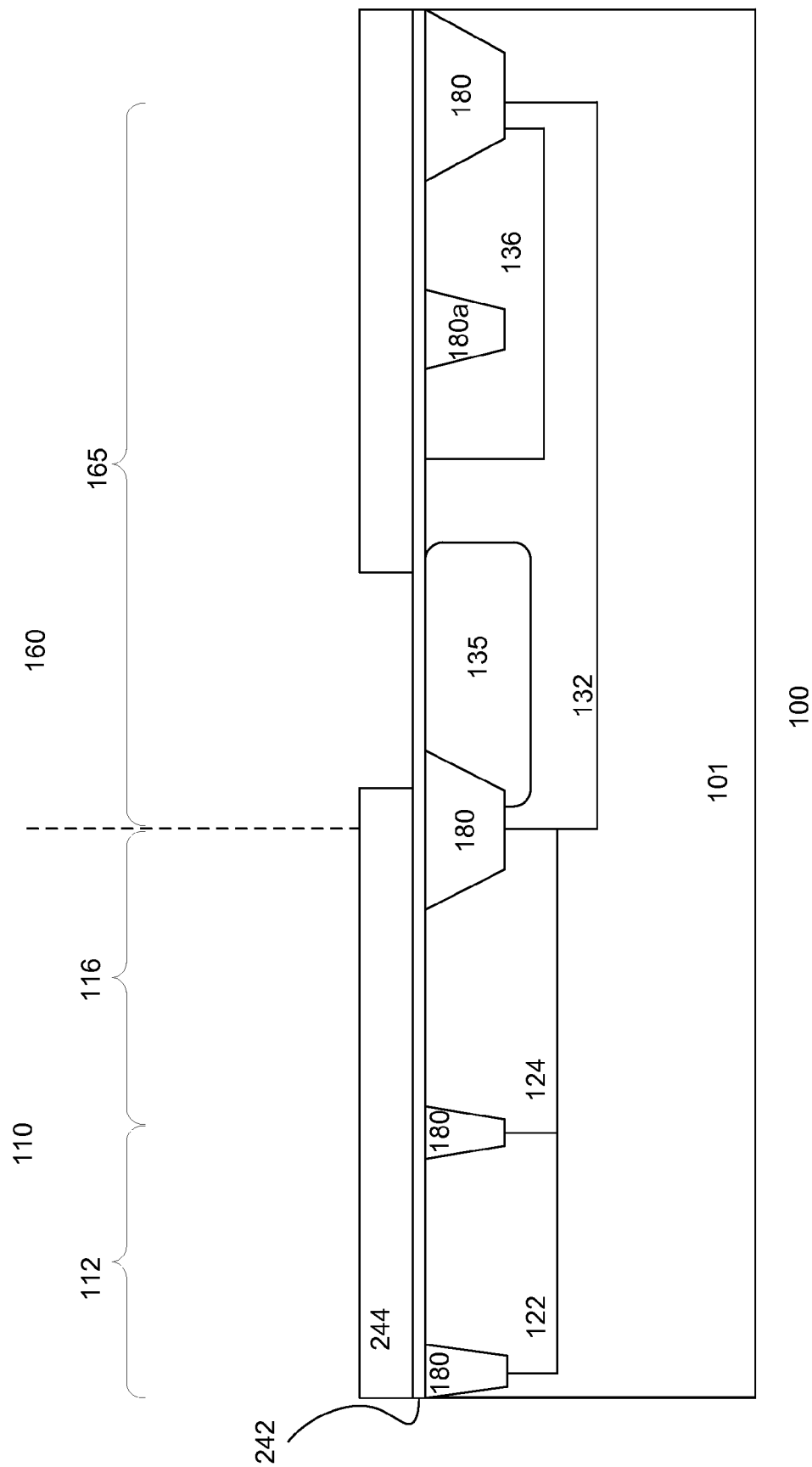


Fig. 2f

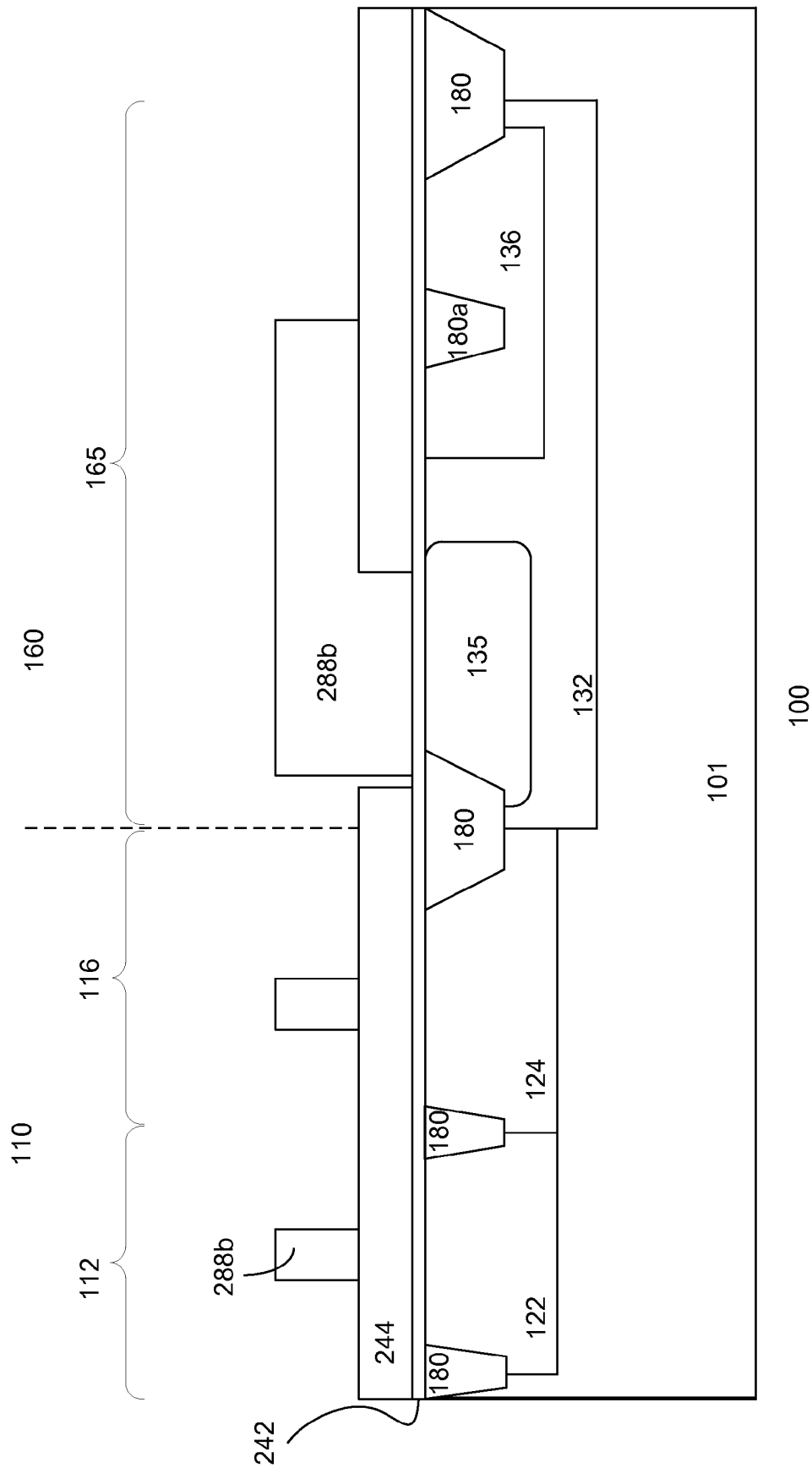


Fig. 29

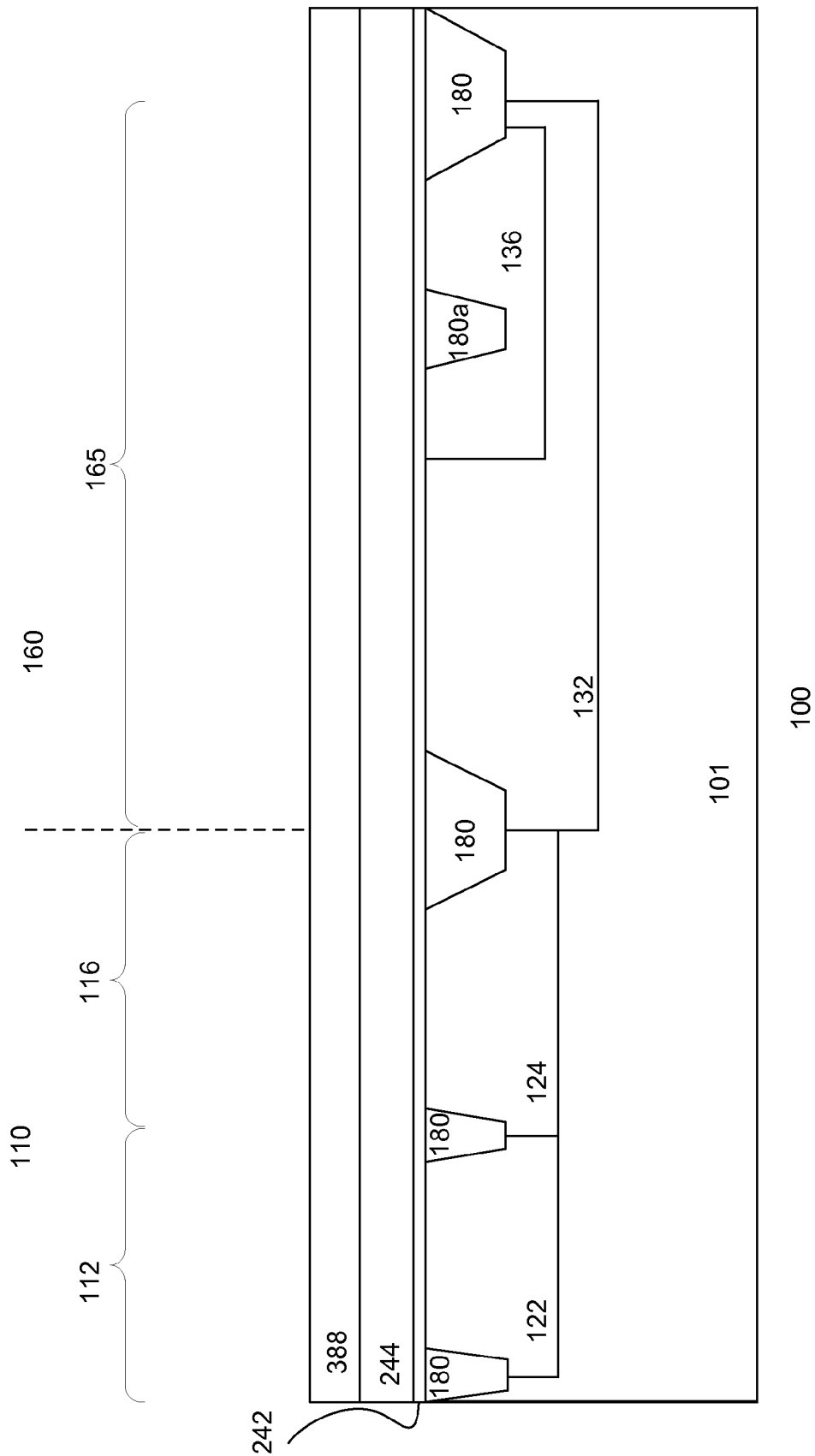


Fig. 3a

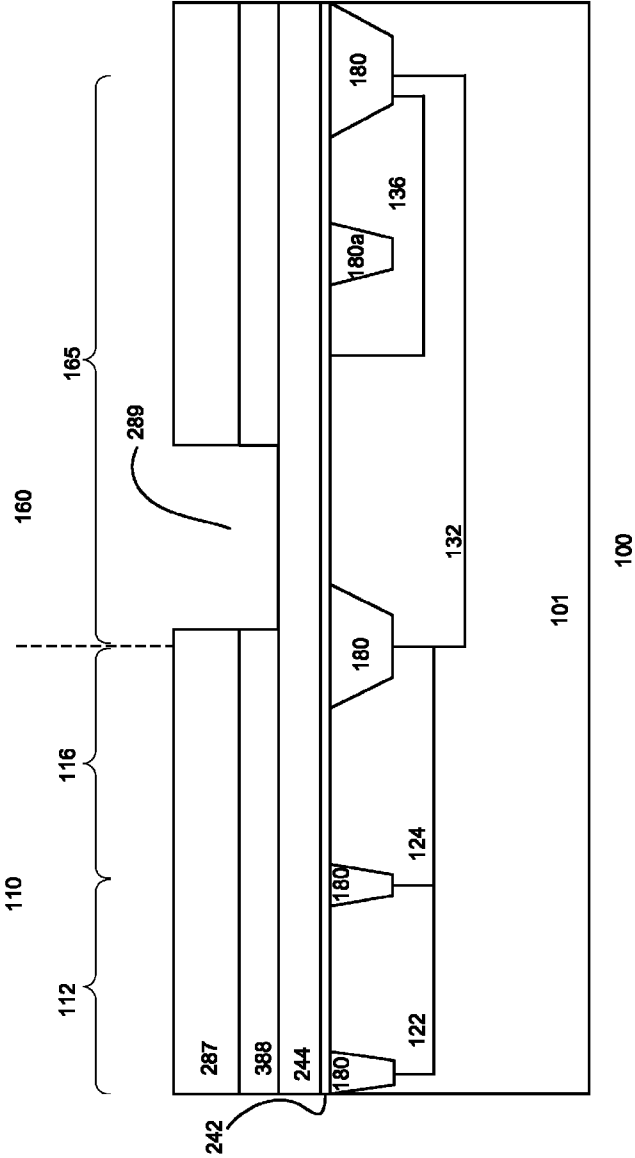


Fig. 3b

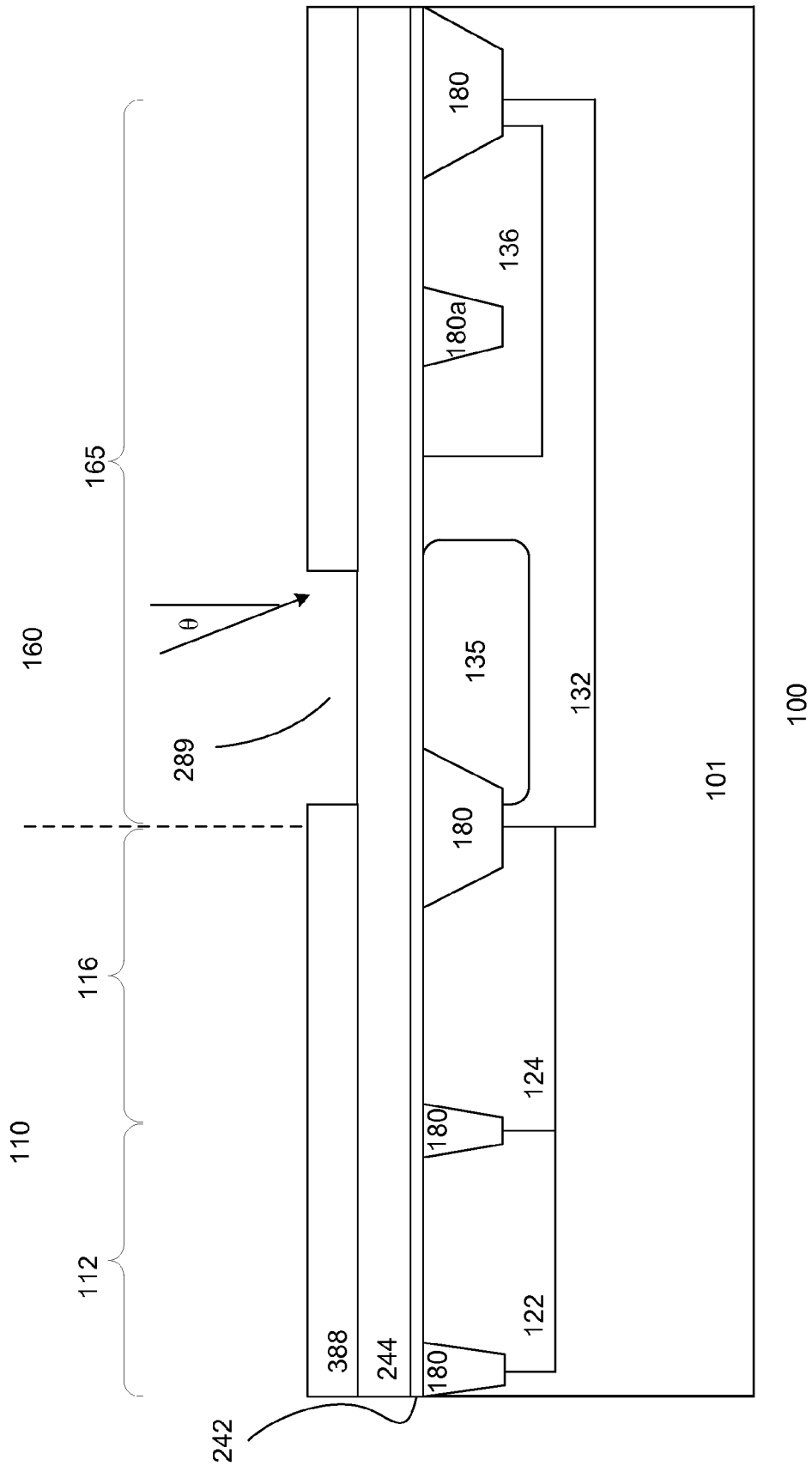


Fig. 3c

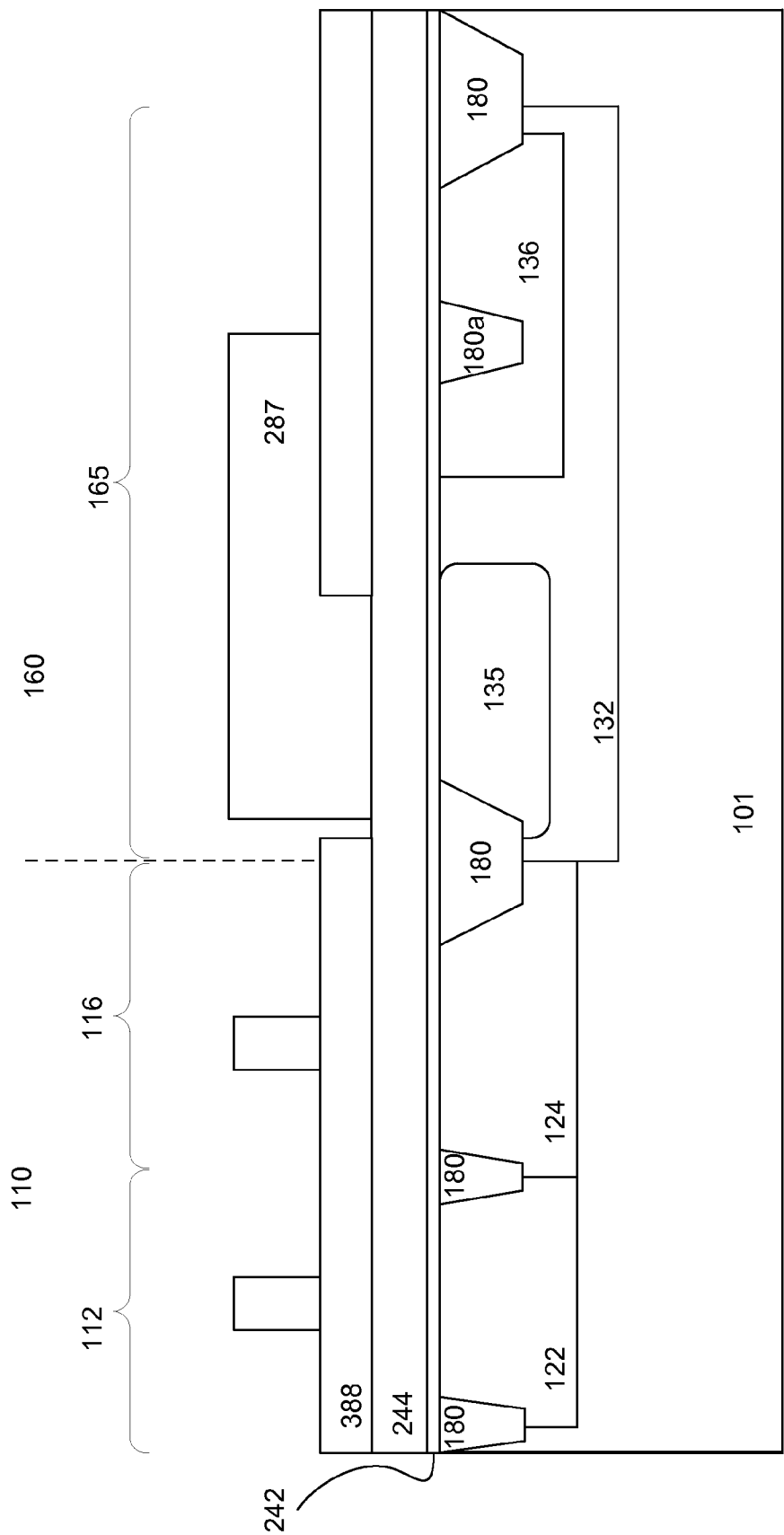


Fig. 3d

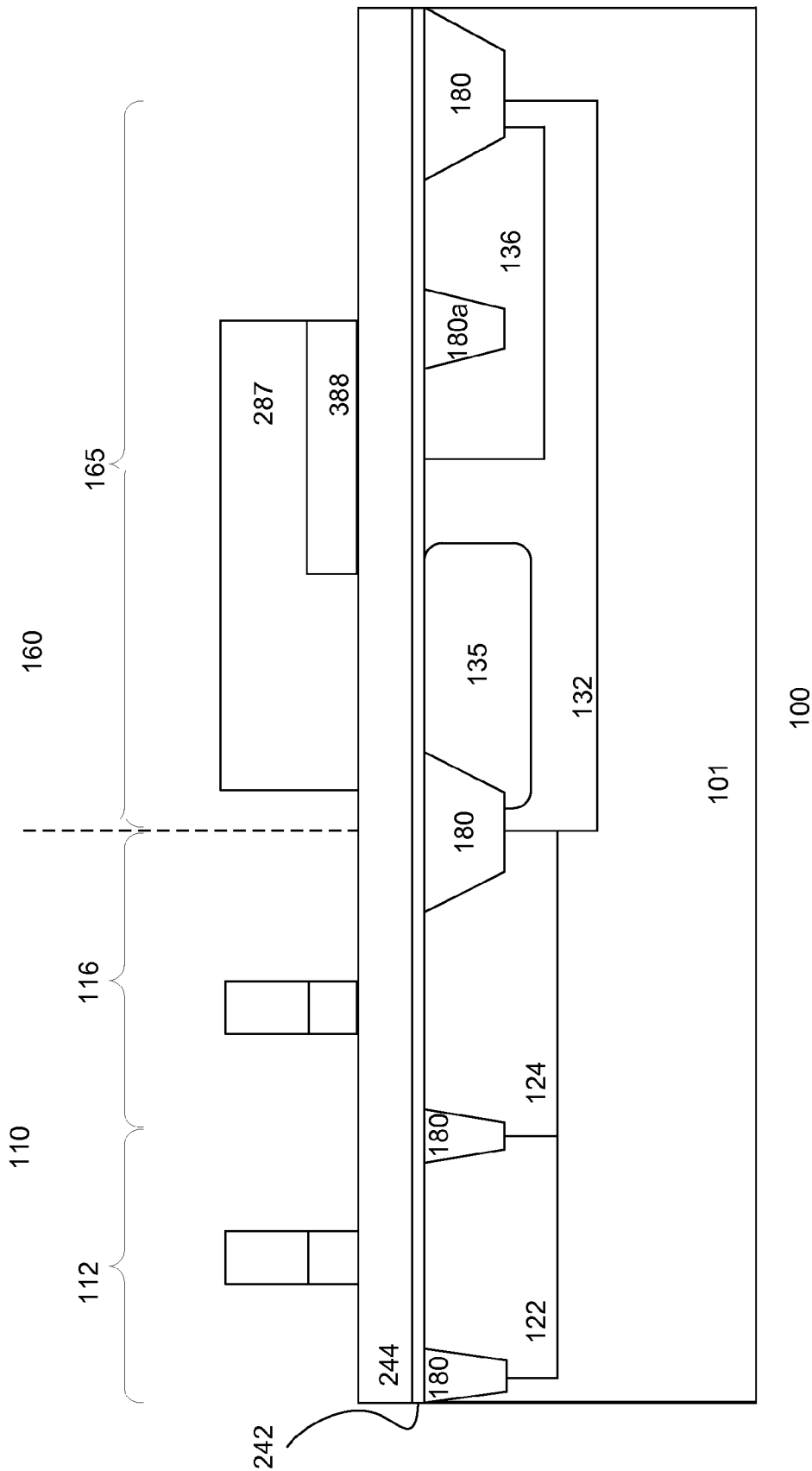


Fig. 3e

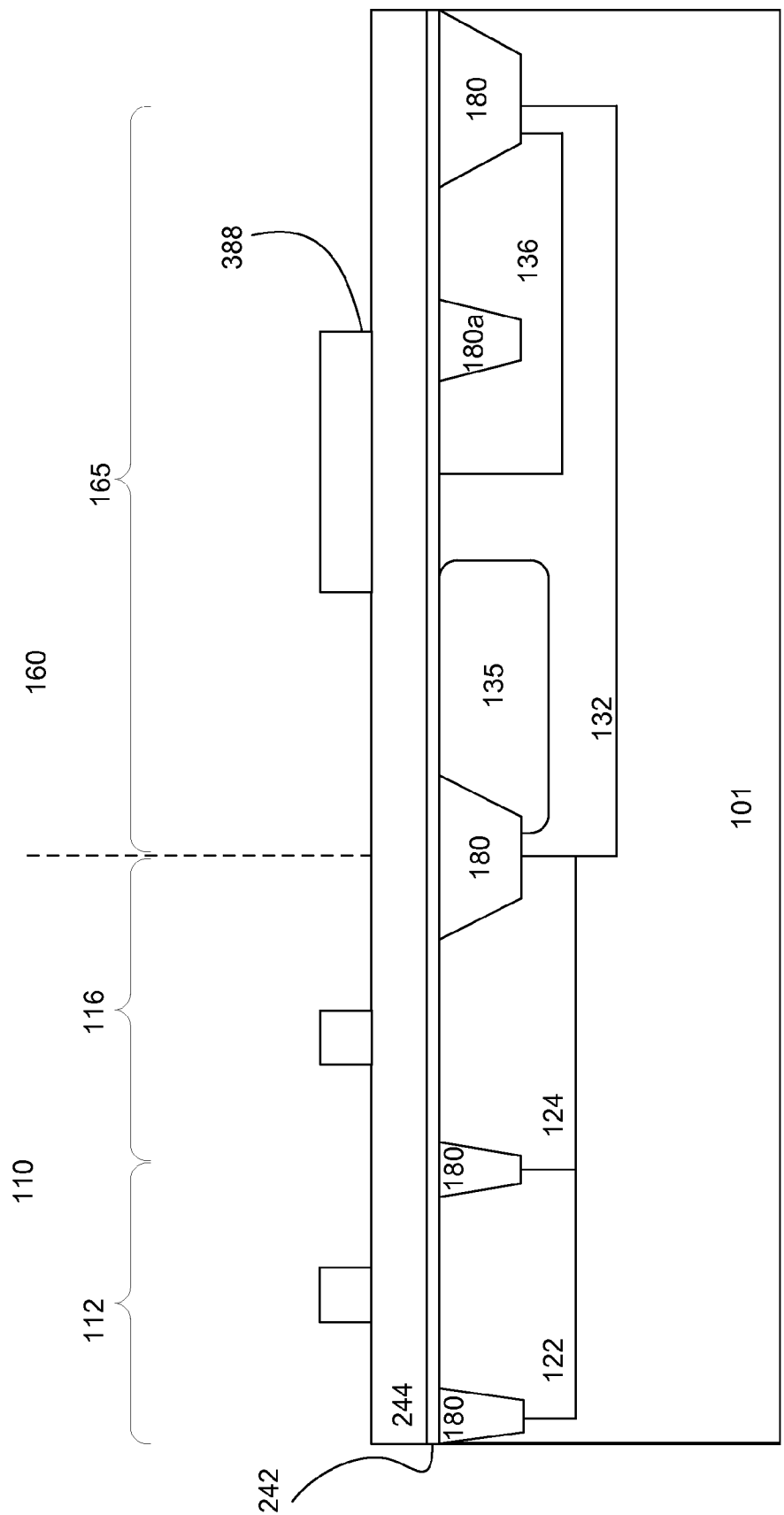
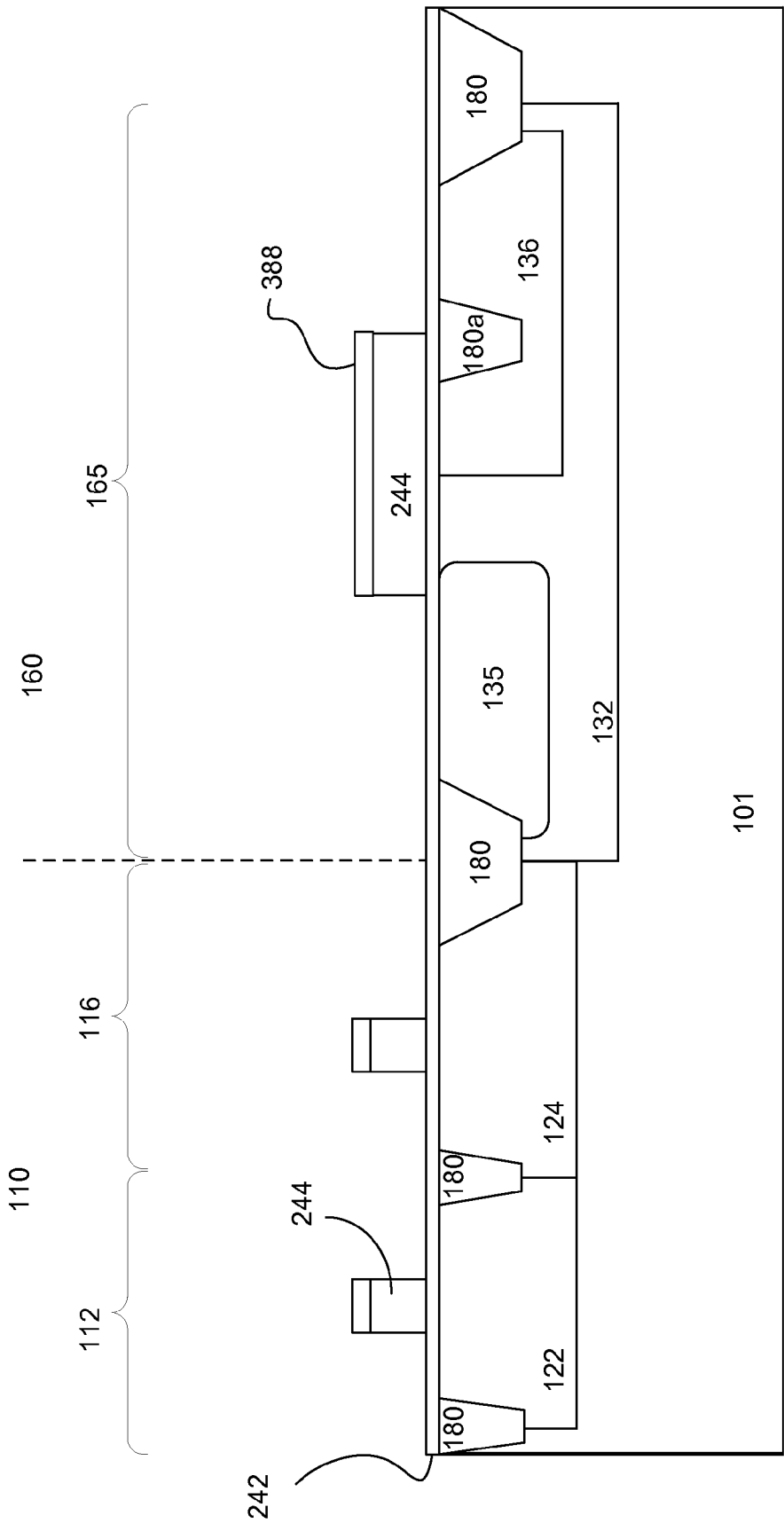


Fig. 3f



100

Fig. 3g

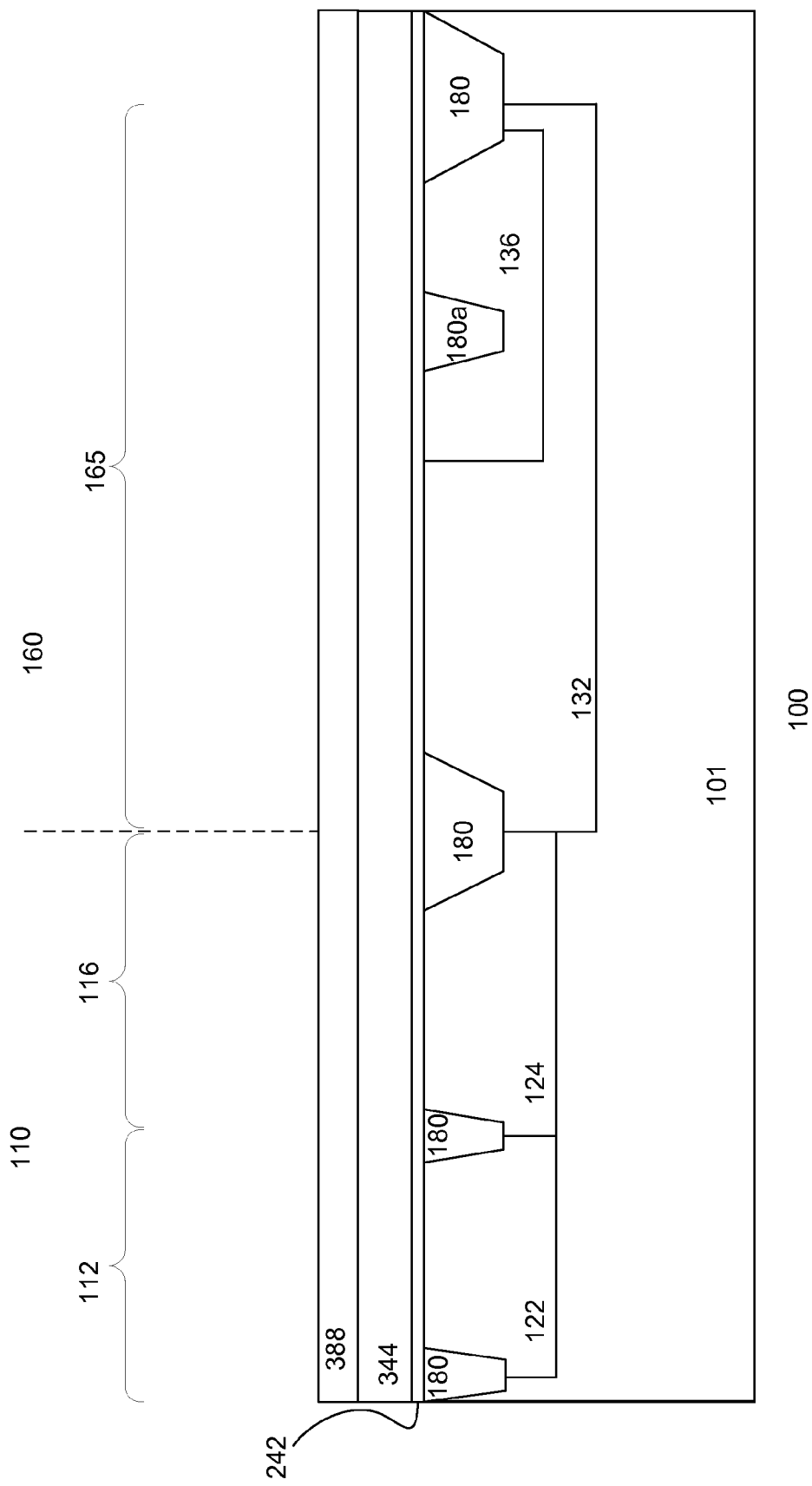


Fig. 4a

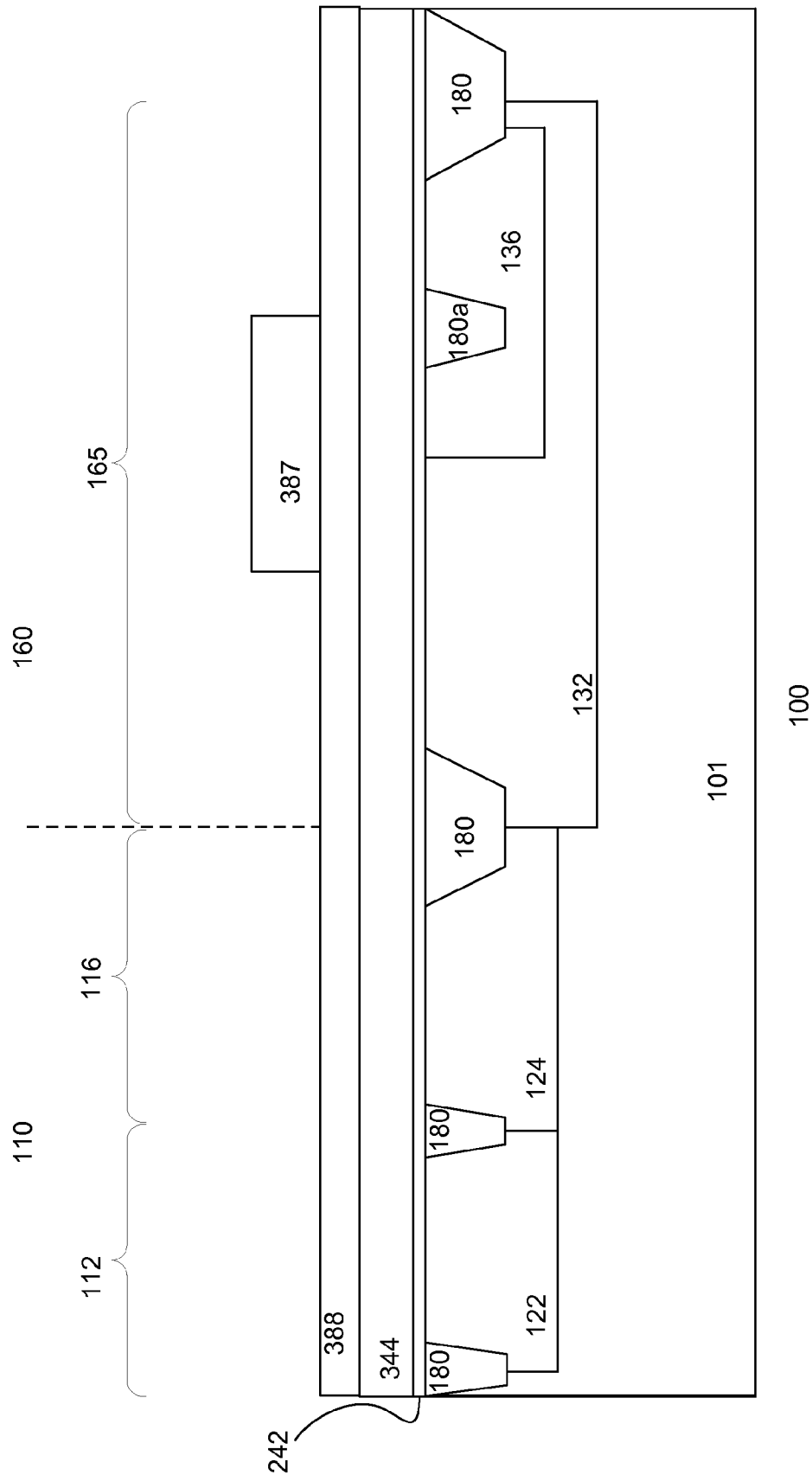


Fig. 4b

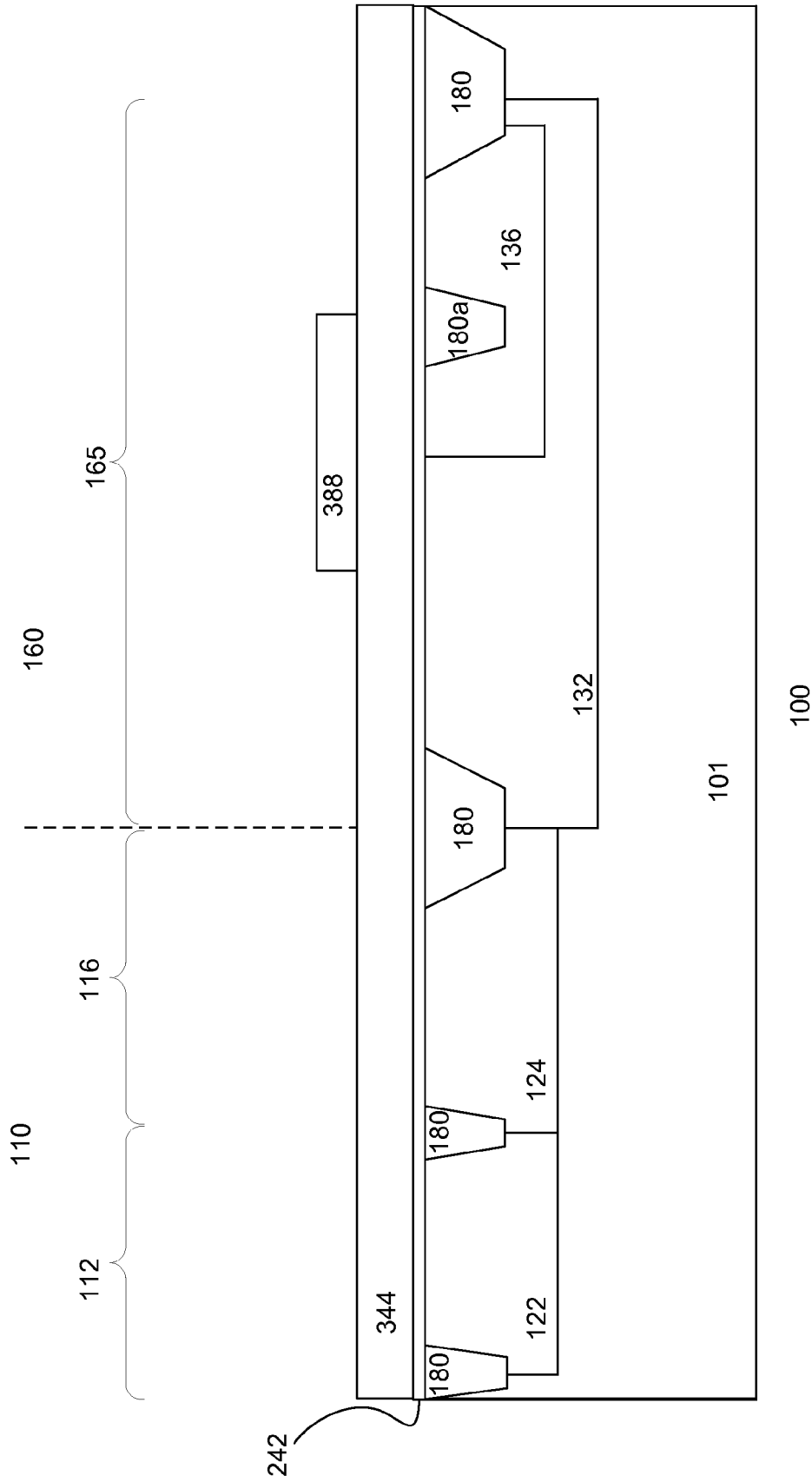


Fig. 4c

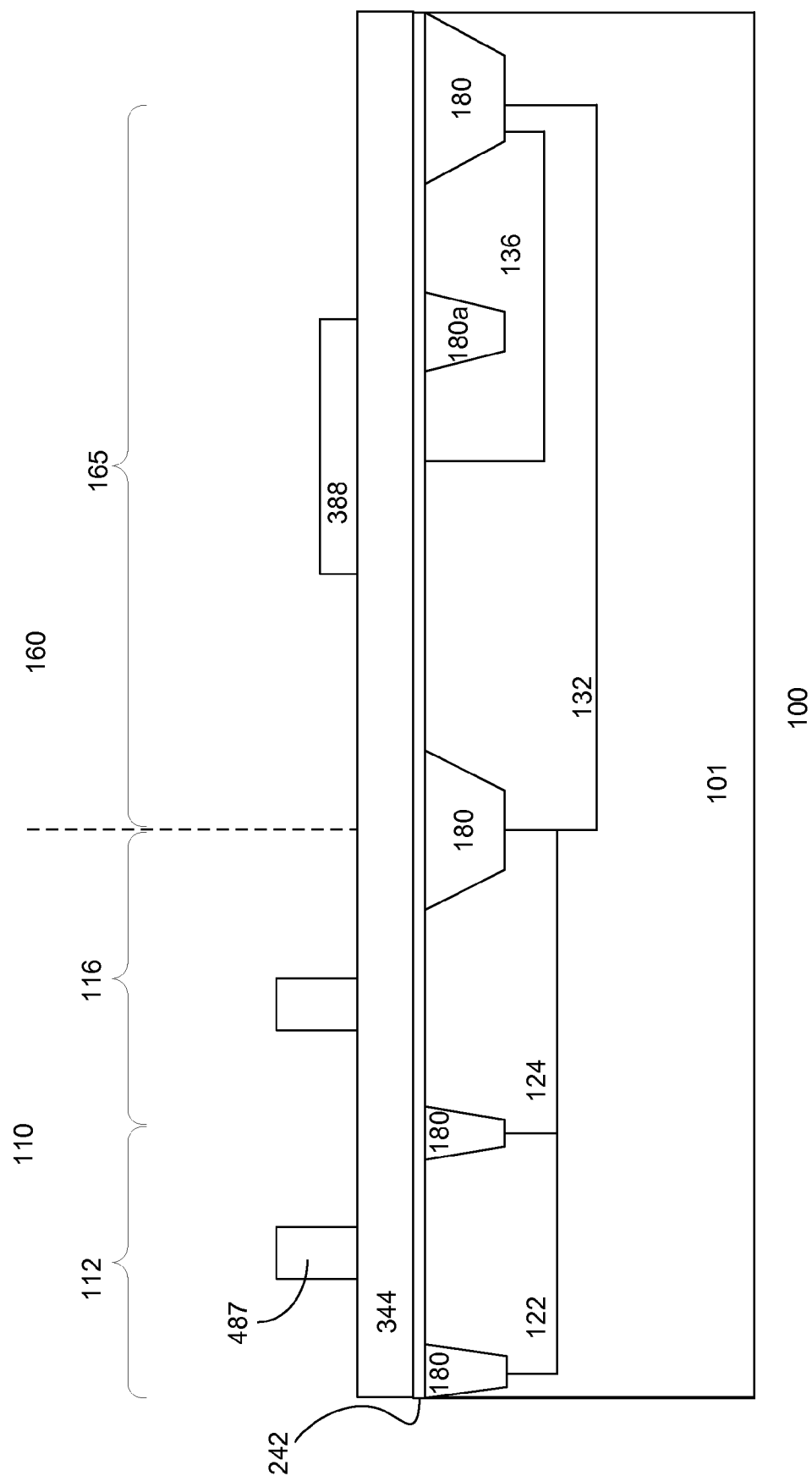


Fig. 4d

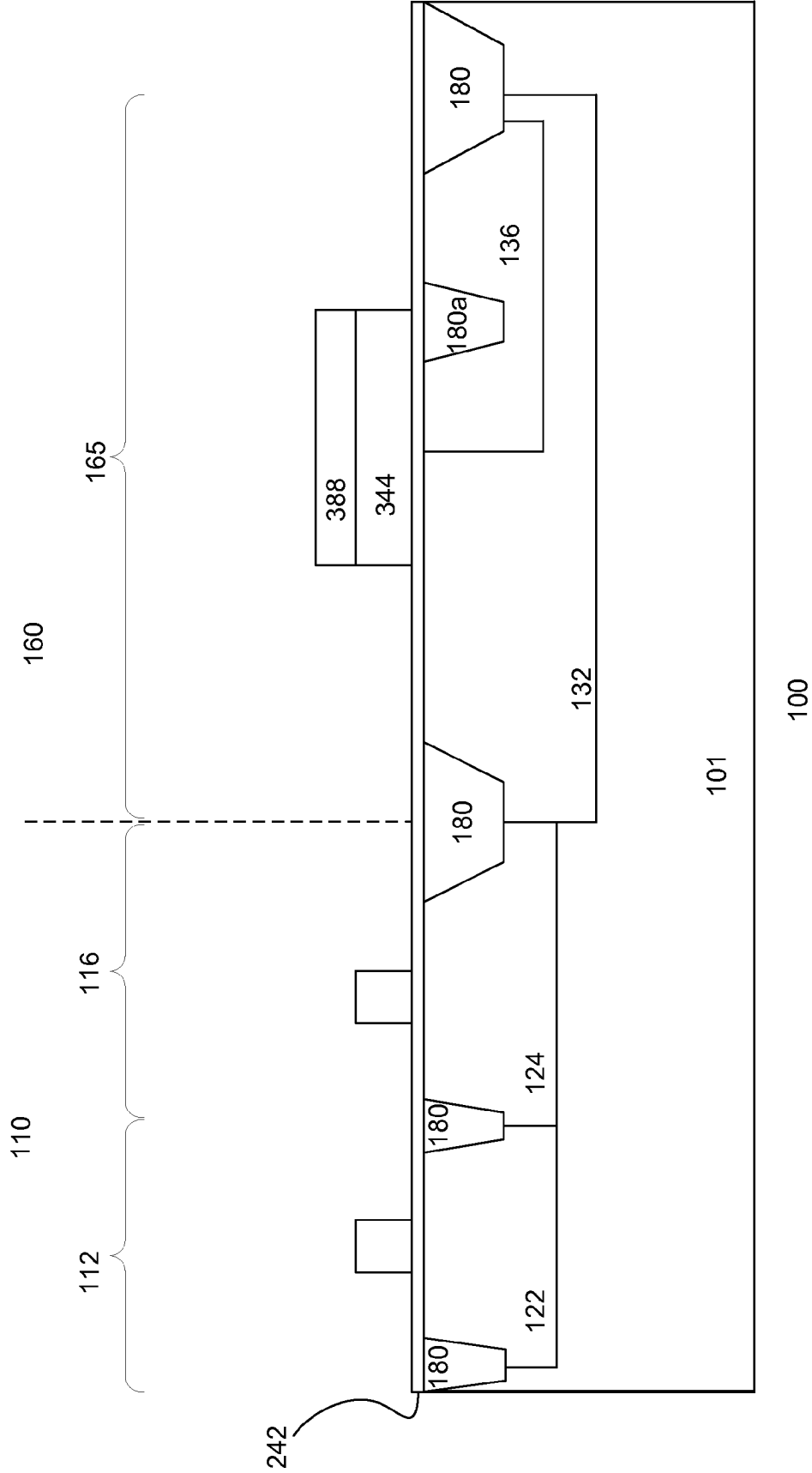


Fig. 4e

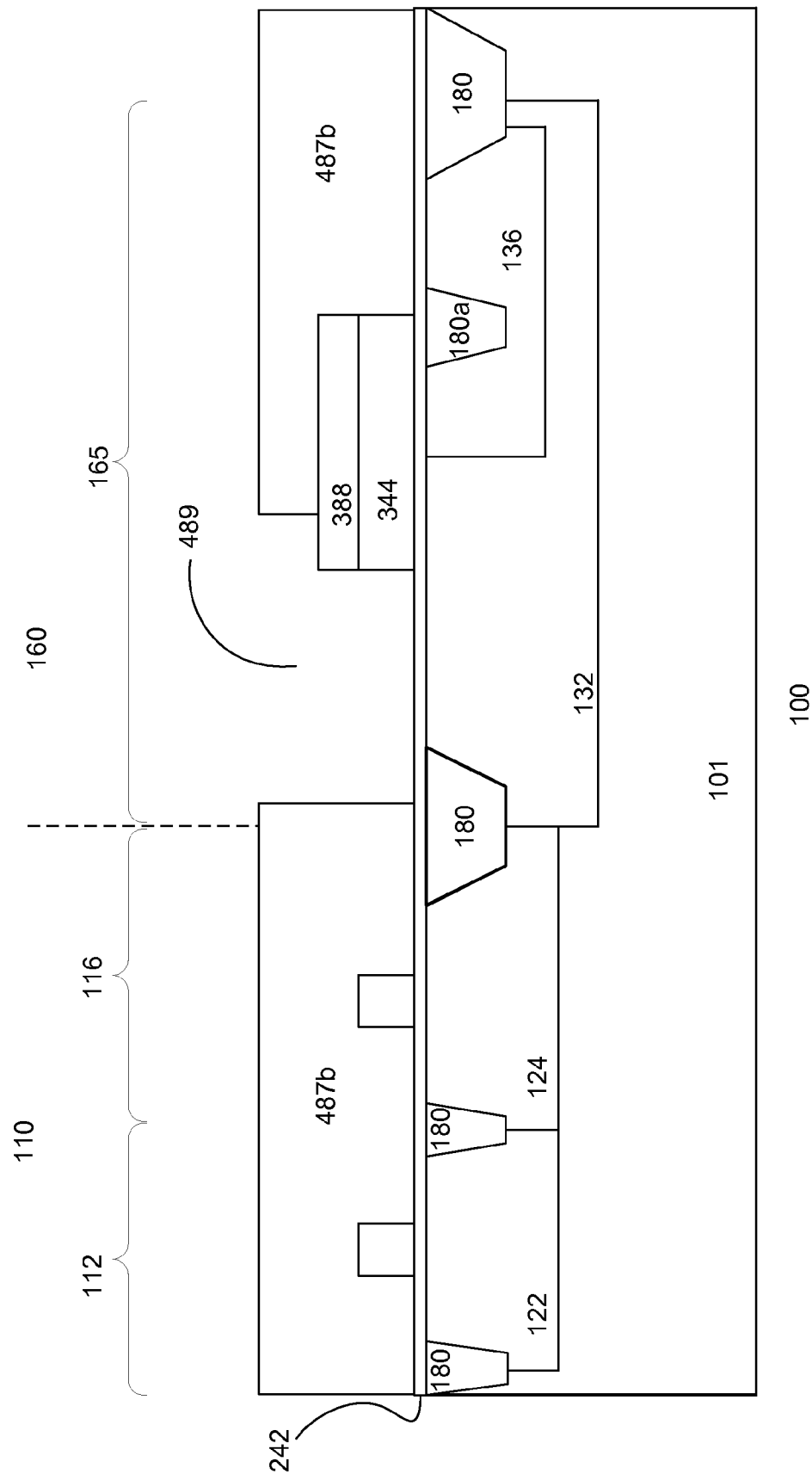


Fig. 4f

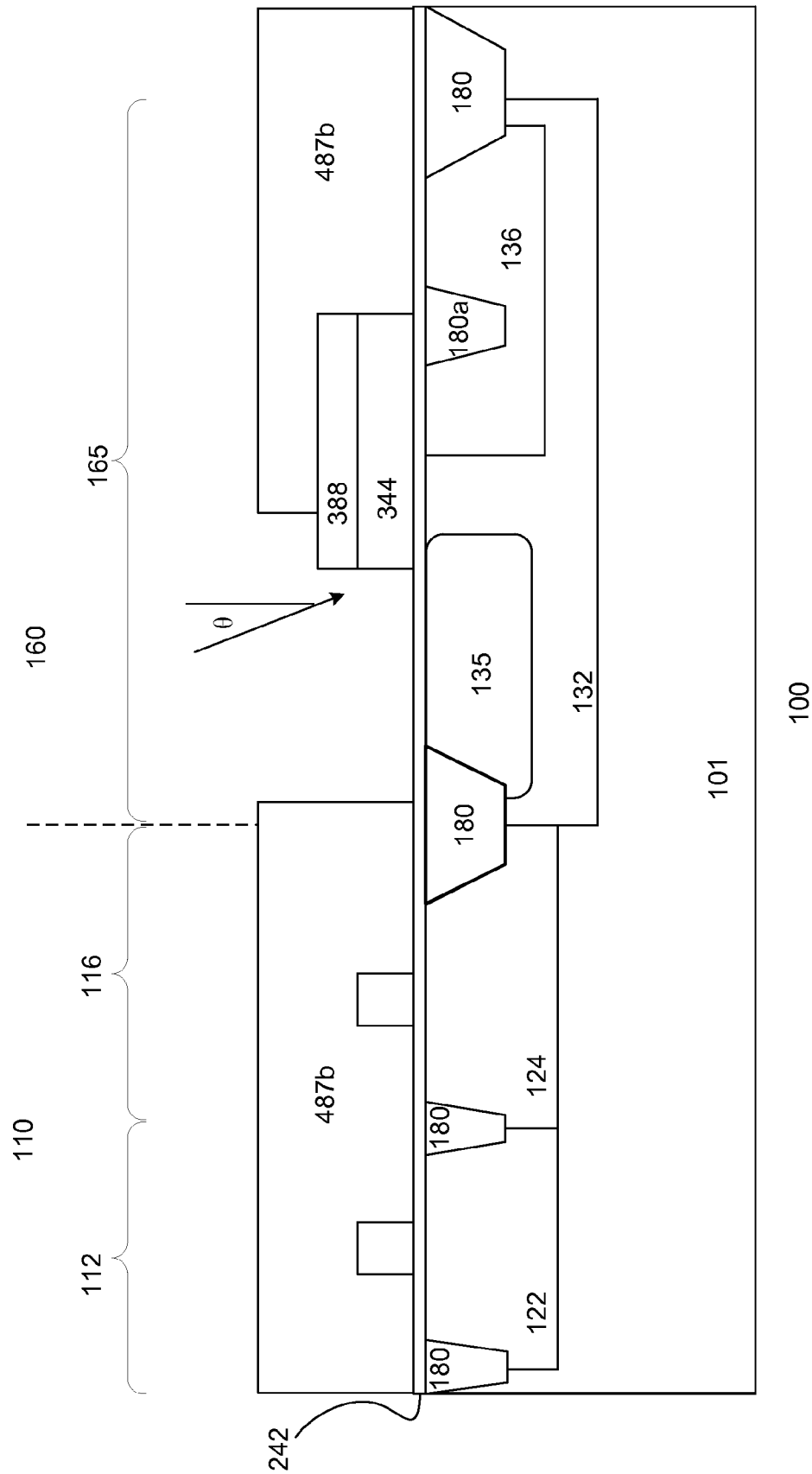


Fig. 4g

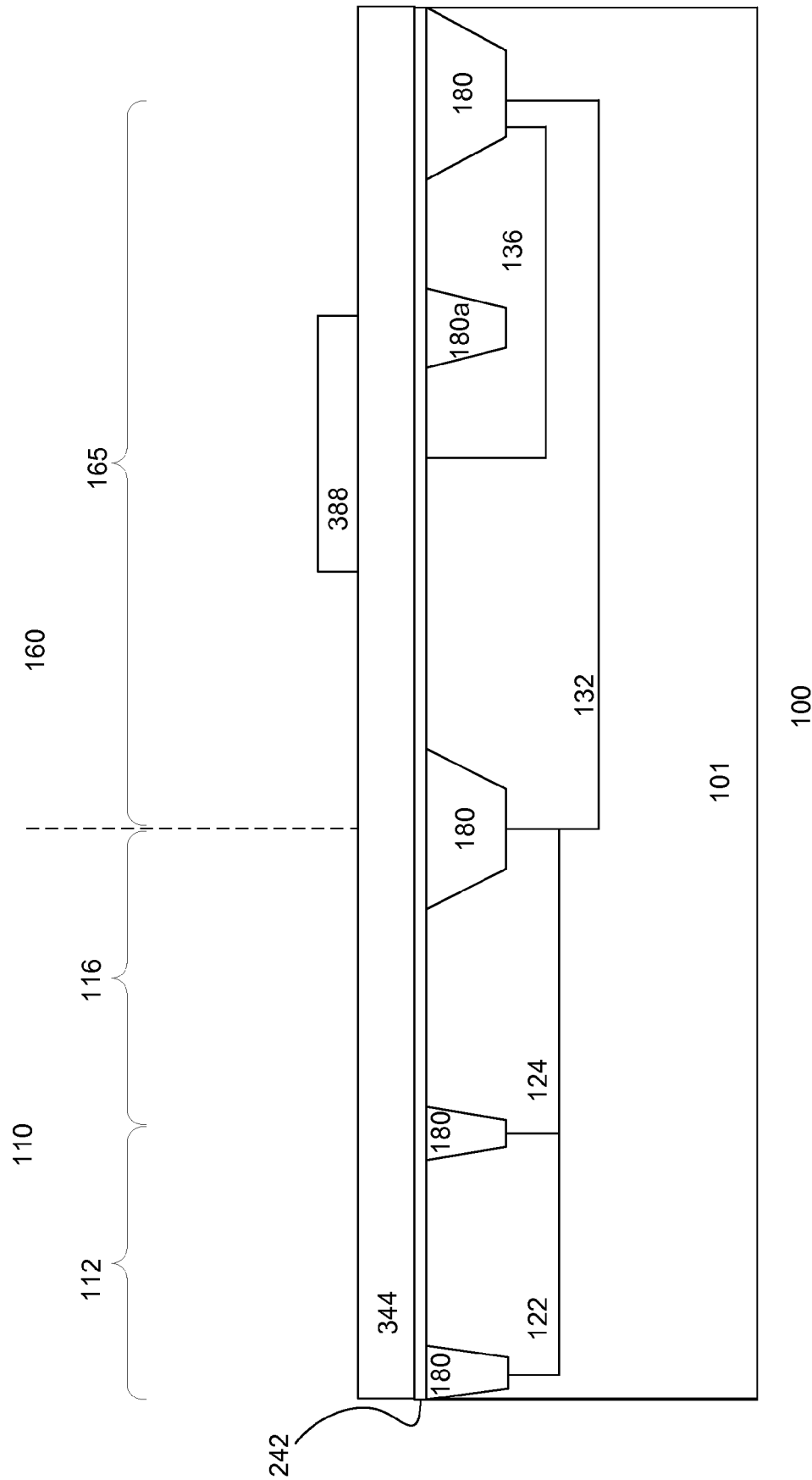


Fig. 5a

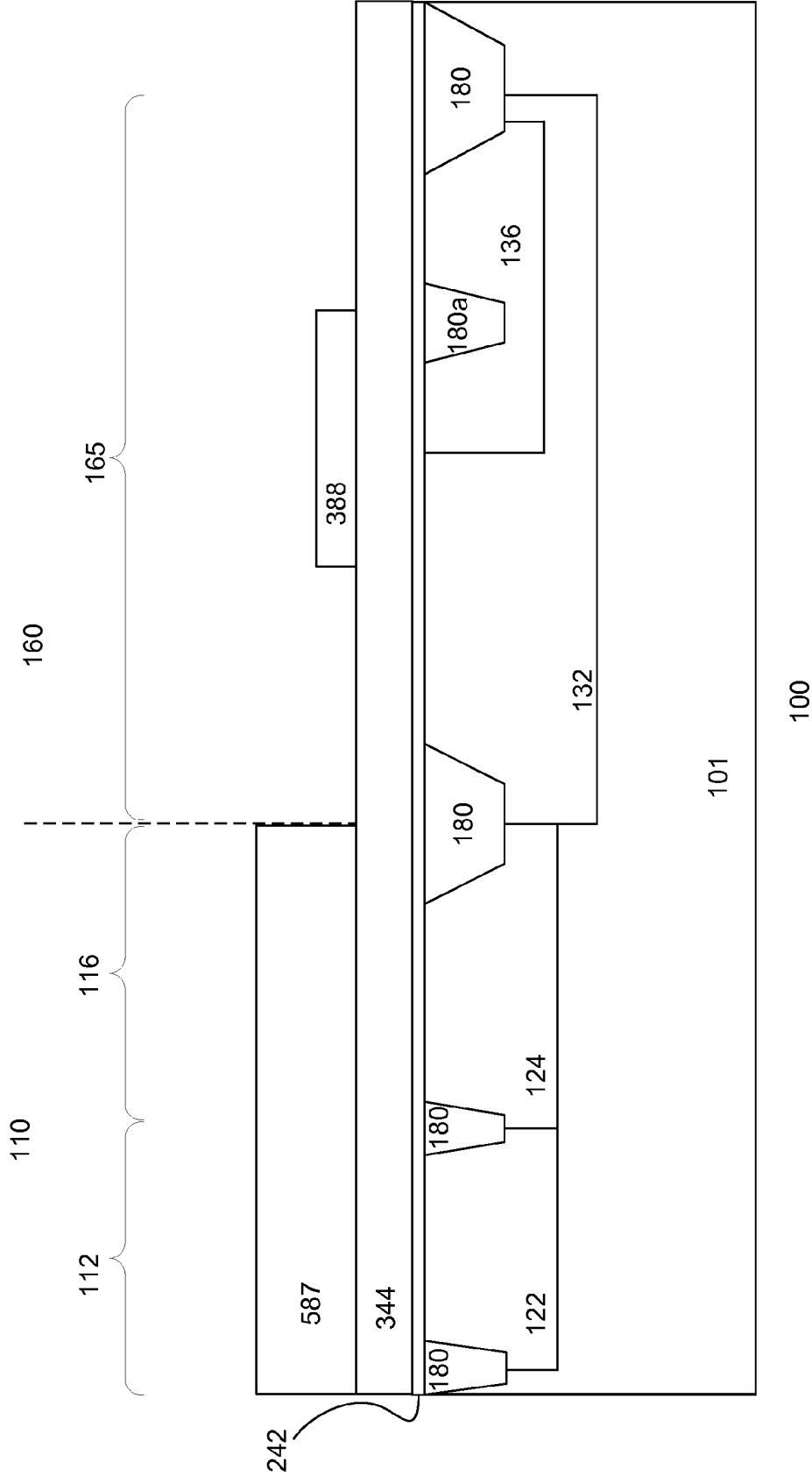


Fig. 5b

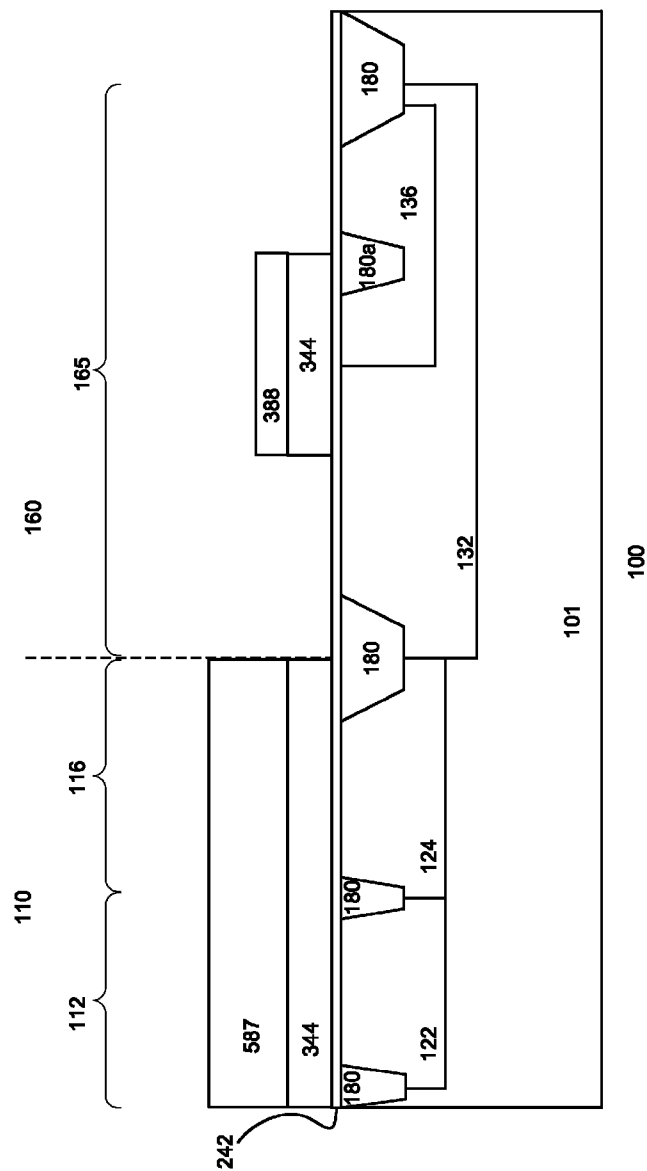


Fig. 5c

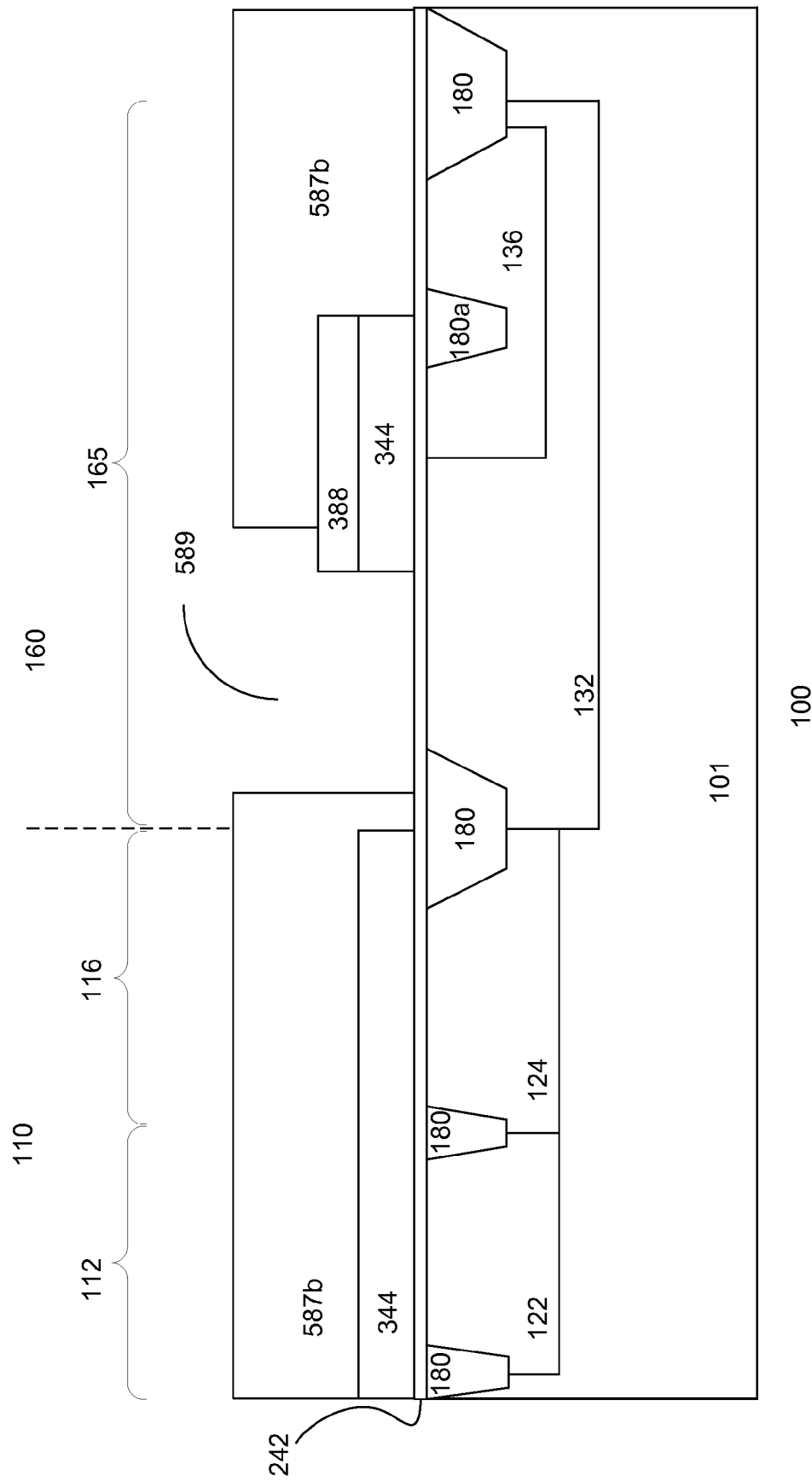


Fig. 5d

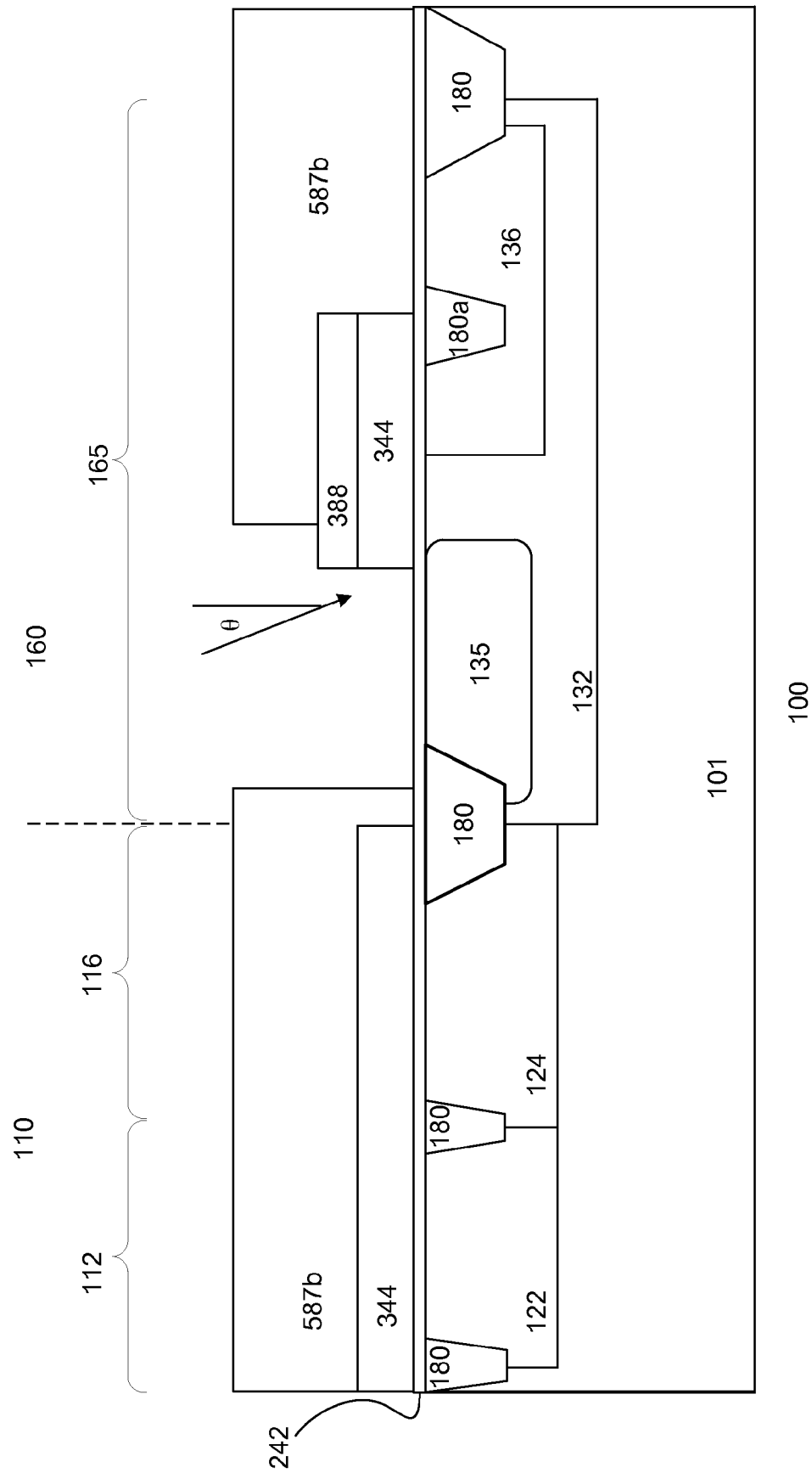


Fig. 5e

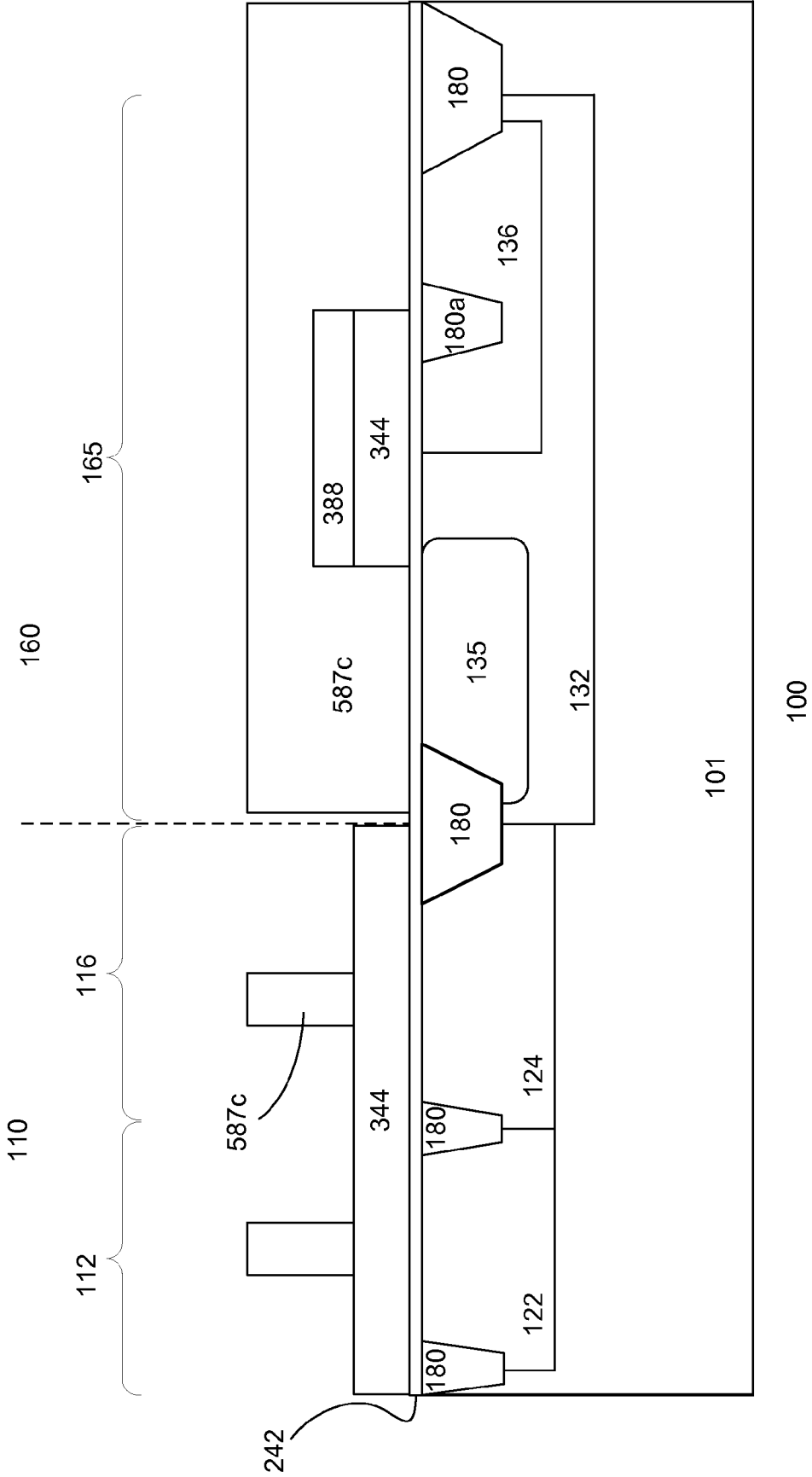


Fig. 5f

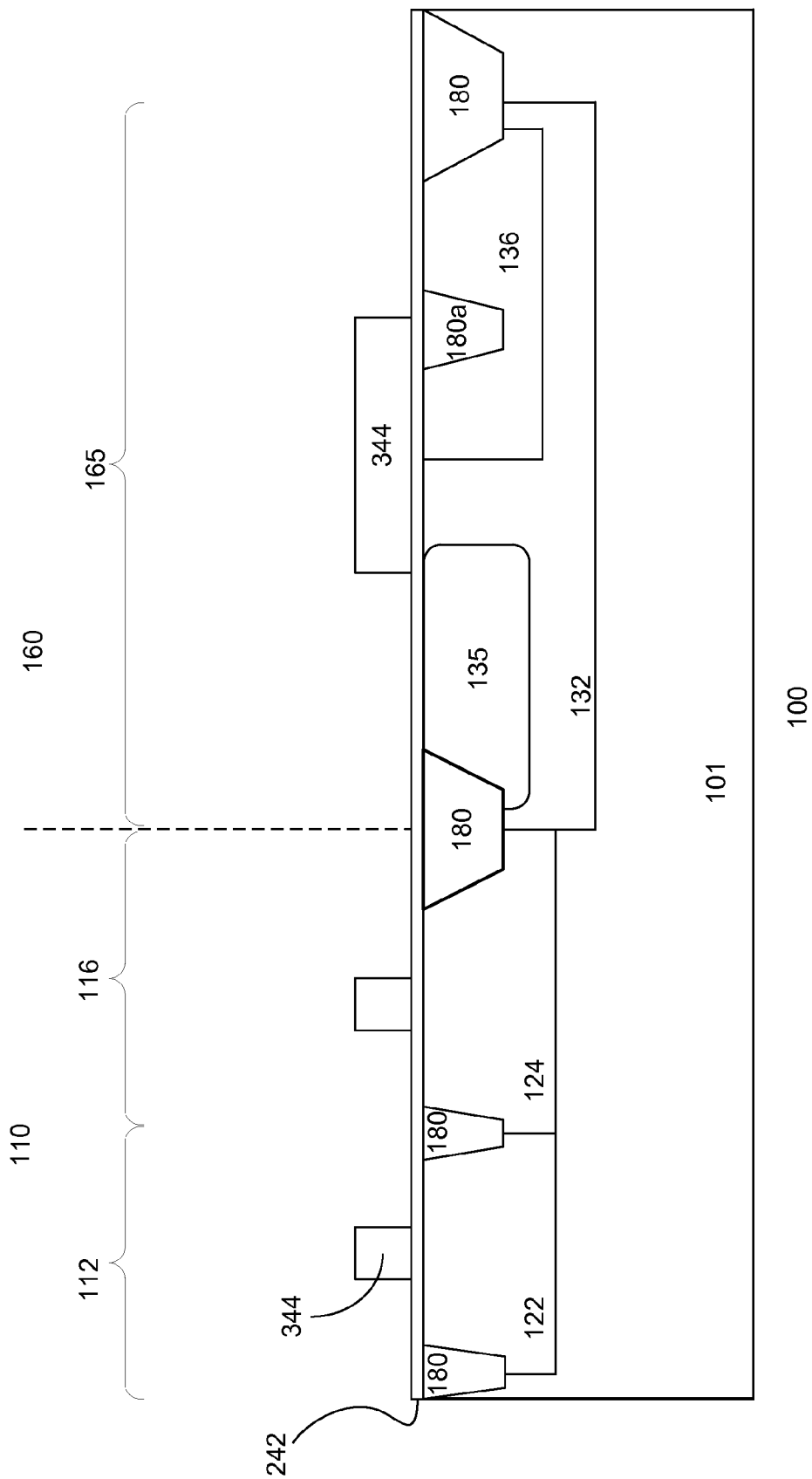


Fig. 5g

1

HIGH VOLTAGE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a divisional application which claims benefits of co-pending U.S. patent application Ser. No. 12/500,620, filed on Jul. 10, 2009, which is a continuation-in-part application of U.S. patent application Ser. No. 12/390,509, filed on Feb. 23, 2009, now issued as U.S. Pat. No. 8,053,319. All disclosures are incorporated herewith by reference.

BACKGROUND

Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications. For high performance LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability. To achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.

As technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate. Generally, the processes for forming LV devices, such as complementary metal-oxide-semiconductor (CMOS) processes, are used to form the HV devices. However, conventional CMOS processes are incompatible for forming HV transistors. For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.

SUMMARY

A device is disclosed. In one embodiment, the device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device further includes a doped drift well adjacent to a second edge of the gate.

In another embodiment, another device is disclosed. The device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. The device further includes a drift isolation region in the doped drift well.

2

In yet another embodiment, a semiconductor device is disclosed. The semiconductor device includes a substrate prepared with an active device region. The active device region includes a gate. The device also includes a doped channel well disposed in the substrate adjacent to a first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge defines an effective channel length of the device. The effective channel length is self-aligned to the gate. The device also includes a doped drift well adjacent to a second edge of the gate. A drift isolation region is disposed in the doped drift well. The device further includes first and second diffusion regions adjacent to the gate.

These and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

FIG. 1 shows an embodiment of a device;

FIGS. 2a-g show an embodiment of a process for forming a device;

FIGS. 3a-g show an alternative embodiment of a process for forming a device;

FIGS. 4a-g show another embodiment of a process for forming a device; and

FIGS. 5a-g show another embodiment of a process for forming a device.

DETAILED DESCRIPTION

Embodiments generally relate to semiconductor devices or integrated circuits. More particularly, some embodiments relate to high power devices. For example, high power devices include lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS). The high power devices can be employed as switching voltage regulators for power management applications. The LD transistors can be easily integrated into devices or ICs. The devices or ICs can be incorporated into or used with, for example, consumer electronic products, particularly portable consumer products, such as cell phones, laptop computers and personal digital assistants (PDAs).

FIG. 1 shows a portion 100 of an embodiment of a device. As shown, the portion includes a substrate 101. The substrate, for example, comprises a silicon substrate. The substrate can be lightly doped with p-type dopants. N-type or other types of substrates, such as silicon germanium or silicon-on-insulator (SOI), are also useful.

The substrate may be prepared with first and second regions 110 and 160. In one embodiment, the first region comprises a low voltage (LV) device region while the second region comprises a high voltage (HV) device region. The regions may be other types of device regions or additional device regions may be provided. For example, a dual gate oxide (DGO) region may be provided for DGO devices. The

DGO devices may be used for I/O circuitries. Alternatively, the device may include only a HV region.

The first region, in one embodiment, comprises first and second type active sub-regions **112** and **116**. The first and second types are complementary types, forming a complementary type device. For example, the complementary type device comprises a complementary metal oxide semiconductor (CMOS) device. Forming non-complementary or other types of devices is also useful.

The first type active sub-region, for example, comprises a n-type active sub-region and the second type active sub-region comprises a p-type active region. The first type active sub-region comprises a second type doped well **122**; the second type active sub-region comprises a first type doped well **124**. For example, the first type active sub-region comprises a p-type doped well and the second type active sub-region comprises a n-type doped well. Other configurations of active sub-regions are also useful. P-type dopants can include boron (B), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof.

A first type transistor **140** is provided in the first type active sub-region and a second type transistor **150** is provided in the second type active sub-region. For example, a n-type transistor is provided in the n-type active sub-region with a p-type doped well while a p-type transistor is provided in the p-type active sub-region with a n-type doped well. The transistors, in one embodiment, are MOS field effect transistors (MOS-FETs). Other types of transistors may also be useful.

A transistor, for example, comprises a gate having a gate electrode **144** over a gate dielectric **142**. The gate electrode, for example, comprises polysilicon. Other types of gate electrode materials, such as polysilicide or metal silicide, are also useful. Polysilicide or metal silicide gates can be formed by reacting the desired metal with polysilicon. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. Doping the gate electrode may depend on the technology. The gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the gate electrode with other dopant types is also useful. The gate dielectric can be silicon oxide. Other types of gate electrode or gate dielectric materials can also be useful. The material of the gate electrode and gate dielectric may depend on the technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

Sidewalls of the gate can be provided with sidewall spacers **148**. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

Adjacent to the gate are diffusion regions **146**. The diffusion regions serve as, for example, source and drain of the transistor. Extension diffusion regions **147** can be provided. The extension diffusion regions, for example, are shallow diffusion regions which extend under the spacers. The diffusion regions (including the extension diffusion regions) of the first type transistor are doped with the first type dopants and the diffusion regions of the second type transistor are doped with the second type dopants. For example p-type diffusion regions are provided for p-type transistors and n-type diffusion regions are provided for n-type transistors. The depth and dopant concentration of the diffusion regions may depend, for example, on the application, such as voltage requirements.

The HV region includes a HV active region **165**. In one embodiment, the HV active region comprises a first type active region. For example, the HV active region comprises a n-type active region. Providing a p-type HV active region is also useful. Furthermore, it is understood that the HV region is depicted with only one active region for purpose of simplification and that the HV region can include numerous HV active regions or sub-regions. For example, the HV active sub-regions can be complementary types of HV sub-regions for complementary types of HV devices.

Isolation regions **180**, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions. Other types of isolation regions are also useful.

In one embodiment, the first type HV active region comprises at least first and second portions. The first portion comprises a second type HV channel (or body) well **135** and the second portion comprises a first type HV drift well **136**. The HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well. In one embodiment, a third or middle portion separates the channel and drift wells. In other embodiments, the first and second portions are contiguous portions. For example, the HV channel and HV drift wells contact each other.

A deep well **132** may be provided in the HV active region, encompassing the HV channel and drift wells. Generally, the deep well comprises dopants of the first type or same as the HV device type. The deep well may be used to isolate the drift well from the substrate. For example, a p-type device will have a n-type deep well. In some cases, the deep well can be optional or the deep well can comprise dopants of the second or opposite type of the HV device type. For example, in the case of a n-type HV device, the deep well can be a p-type or n-type deep well. The different types of deep well selected may depend on the type of substrate. Generally, p-type doped substrates are used for fabricating devices. When p-type substrates are used, the n-type deep well may be optional. This is because the n-type drift well is already different from the p-type substrate. For a n-type substrate, a p-type deep well may be used to isolate the channel well from the substrate. Other configurations of deep wells may also be useful.

As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges **179a-b** or sidewalls.

The gate includes a HV gate electrode **174** over a HV gate dielectric **172**. The HV gate electrode, for example, comprises polysilicon. Other types of HV gate electrode materials, such as polysilicide or metal silicide, are also useful. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of HV gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. The HV gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the HV gate electrode with other dopant types is also useful. The HV gate dielectric can be silicon oxide. Other types of HV gate dielectric materials can also be useful.

In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process and/or technology. In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process for LV devices and/or technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be

used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

Sidewalls of the gate can be provided with sidewall spacers **178**. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

In one embodiment, the various gate elements of the HV gate, such as gate electrode, gate dielectric and spacers are of the same type of material as the LV devices in the LV region. This facilitates process compatibility for forming HV and LV devices.

The HV gate overlaps the channel and drift wells in the first and second portions of the HV active region. A drift isolation region **180a** may be provided within the drift well **136** on the high voltage side of the HV device. The drift isolation region **180a** can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region **180a** is disposed in the drift well **136** under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also be useful.

A first type drain diffusion region **176b** is provided between the drift isolation region **180a** and isolation region **180** which isolates the HV active region from other sub-active regions in the HV region. A first type source diffusion region **176a** is provided in the channel well **135** of the first type HV active region. A source extension region **177** can be provided which extends under the spacer on the channel side of the HV gate. The first type source and drain diffusion regions, for example, comprise a p-type diffusion regions for a p-type HV active region. Providing n-type diffusion regions for a n-type HV active region is also useful. The dopant depth and concentration of the diffusion regions may be selected depending on the application. In one embodiment, the dopant depth and concentration of diffusion regions may be the same as those of LV devices. Other dopant depths and concentrations are also useful.

A second type body contact region **171** may be provided in the first portion of the HV active region. As shown, the second type body contact region **171** is between the source diffusion region **176a** and the isolation region **180**. The second type body contact region comprises second type dopants. In one embodiment, dopant depth and concentration of the body contact region may be the same as the diffusion regions of the LV devices. Other dopant depths and concentrations are also useful. The body contact region **171** provides electrical coupling to the channel well **135**.

An edge **133** of the channel well **135** extends below the gate on the source side of the gate. The edge **133**, for example, can be referred to as the channel edge of the channel well. The channel edge of the channel well **135** and the first edge **179a** of the gate on the source side define a channel of the HV transistor having an effective channel length L .

The effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge. For example, dopant ions are implanted using an implant mask through the gate electrode or a hard mask and the gate electrode. In one embodiment, the implant mask is also used to pattern the channel edge of the HV gate. The effective channel length may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements. For example, shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.

With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low $R_{ds, on}$ can be achieved, enabling high current to pass through a small area. This is particularly useful for technologies beyond 0.35 μm , such as, for example, 0.25 μm , 0.18 μm and 0.13 μm technologies.

Smaller area requirements for power management applications with reduced power dissipation increases performance while reducing costs.

In one embodiment, the effective channel length L is less than 0.4 μm . In other embodiments, the effective channel length is less than 0.3 μm . In yet another embodiment, the effective channel length is less than 0.25 μm . Other effective channel lengths may also be useful, for example, depending on the application.

FIGS. 2a-g show cross-sectional views of an embodiment of a process for forming a device or IC **100**. Referring to FIG. 2a, a substrate **101** is provided. The substrate can comprise a silicon substrate, such as a lightly p-type doped substrate. Other types of substrates, including n-type doped substrates, silicon germanium or silicon-on-insulator (SOI), are also useful.

As shown, first and second device regions **110** and **160** are defined on the substrate. The first region, for example, comprises a LV device region while the second region comprises a HV device region. Other types of device regions or additional device regions may be provided. Alternatively, the device may include only a HV region.

The LV device region **110** can be defined with first and second type active sub-regions **112** and **116**. In one embodiment, the first and second type active sub-regions are complementary type active sub-regions to form a complementary device such as a CMOS device. Forming non-complementary or other types of devices is also useful.

As for the HV device region **160**, a HV active region **165** is defined therein. In one embodiment, the HV active region **165** can be a first or second type HV active region. Although only one HV active region is depicted, it is understood that the HV device region may include additional HV active regions or sub-regions. The HV active sub-regions can be complementary HV active sub-regions.

The substrate is also prepared with isolation regions **180** to separate the device regions as well as other active device sub-regions. In one embodiment, the isolation regions **180** comprise STIs. Various conventional processes can be employed to form the STI regions. For example, the substrate can be etched using conventional etch and mask techniques to form trenches which are then filled with dielectric materials such as silicon oxide. Chemical mechanical polishing (CMP) can be performed to remove the excess dielectric materials and provide a planar substrate top surface. Other processes or materials can also be used to form the STIs.

In FIG. 2b, the doped wells for the active regions of the device regions are formed. In one embodiment, a first type doped well **124** is formed in the second type LV active sub-region **116** and a second type doped well **122** is formed in the first type LV active sub-region **112**. For example, a n-type doped well is formed in the p-type active sub-region and a p-type doped well is formed in the n-type active sub-region.

In the HV active region **165**, it includes at least first and second portions. For example, the first portion corresponds to a drain portion and the second portion corresponds to a source portion. In some embodiments, a third or middle portion is provided to separate the first and second portions. Alternatively, the first and second portions are contiguous portions without a middle portion.

In one embodiment, a doped well **136** is formed in the drain portion of the HV active region. The doped well, for example, is a drift well of the HV device. The drift well comprises the same type as the HV active region. In the case of a first type HV active region, the drift well comprises a first type. For example, a n-type drift well is provided for a n-type HV active region.

A drift STI region **180a** may be optionally provided in the drift well **136**. The drift STI region is disposed in the drift well to improve reliability of high voltage devices. For example, the drift STI region improves breakdown reliability of the HV device. The drift STI region can be formed at the same time as the other STI regions.

A deep HV well **132** may be provided in the HV active region **165**. As shown, the deep HV well **132** is formed in the whole HV active region **165** and encompasses the first and second portions of the HV active region **165**. The deep HV well comprises an opposite type as the drift well and the HV active region. In the case of a first type HV active region, the deep well comprises a second type. For example, a p-type deep well is provided for a n-type HV active region.

The depth and concentration of dopants of the various wells may depend on, for example, application. For example, higher voltage applications may require lower doping concentration of dopants and greater depth of the wells. In one embodiment, the depth of the deep well **132** is about 3 μm with a dopant concentration of about $5 \times 10^{16} \text{ cm}^{-3}$. For the drift well **136**, it may have a depth of about 1.5 μm with a dopant concentration of about $5 \times 10^{17} \text{ cm}^{-3}$. Other depths and dopant concentrations are also useful for the wells.

As described, the STI regions are formed prior to forming the various doped wells. Other process schemes, such as forming the STI regions after the various doped wells are formed may also be useful.

The doped wells can be formed by ion implantation. P-type dopants can include boron (B), BF_2 (boron compounded with fluorine), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof. Generally, the first and second types of the active regions are formed selectively in separate processes. For example, the n-type regions can be implanted with n-type dopants while an implant mask prevents doping of the p-type regions.

In one embodiment, the wells of the HV region are formed separately from the LV regions. For example, the LV wells can be formed first followed by the HV wells. Forming the HV wells prior to the LV wells is also useful. The different types of LV wells can be formed in separate processes and the different types of HV wells can also be formed in separate processes. For example, the first type and second type wells in the LV region are formed in separate implant processes. Similarly, the different types of wells in the HV region are formed in separate implant processes.

A doped well can be formed by a single implant process or multiple implant processes. In multiple implant processes, dopants can be implanted at different energy levels to achieve the desired doping profiles. In the case of a single implant process, a high temperature drive-in process is performed to achieve the desired doping profile.

A single implant process may be employed to, for example, form deep wells. For a n-type deep well, phosphorus may be implanted with a dopant concentration of about $6 \times 10^{12} \text{ cm}^{-2}$ at an energy level of about 2000 KeV followed by a two-hour drive-in at a temperature of about 1100° C. Single implant processes may also be used to form other types of wells.

Forming wells using multiple implant processes may also be useful. For example, LV wells may be formed by multiple implant processes.

In FIG. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer **242** is formed over the substrate. The gate dielectric layer, for example, comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer **242** may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

A gate electrode layer **244** is deposited on the gate dielectric layer **242**. The gate electrode layer comprises, in one embodiment, polysilicon (poly). The gate electrode layer **244** can be formed as an amorphous or non-amorphous layer. For amorphous deposited layer, subsequent processing can be performed to crystallize it. Other types of gate electrode materials are also useful. For example, the polysilicon can be subsequently processed to form polysilicide or metal gates. The thickness of the gate electrode layer may be about 2000 Å or less. In another embodiment, the thickness of the gate electrode layer is about 2500 Å or less. In yet another embodiment, the thickness of the gate electrode layer can be about 4000 Å or less. For example, in the case of polysilicide gates, the polysilicon may be about 2000 Å while the tungsten is about 2000 Å. Other thicknesses are also useful. Various techniques can be used to form the gate electrode layer. For example, polysilicon can be deposited by CVD while metal can be deposited by sputtering. Other techniques, depending on the material, may also be useful.

Referring to FIG. 2d, a mask layer **288** is formed on the substrate, covering the gate electrode layer. In one embodiment, the mask layer comprises photoresist. The mask is patterned as desired. To pattern the mask layer, photolithography can be employed. For example, the mask can be selectively exposed to an exposure source through a lithographic mask. Depending on whether a positive or negative resist is used, the exposed or unexposed portions are removed by development. To enhance lithographic resolution, an ARC layer (not shown) may be provided beneath the mask layer.

The mask layer **288** is patterned to form an opening **289** to expose a portion of the gate electrode layer **244**. In one embodiment, the opening corresponds to the opening for the channel well implant. For example, the mask layer serves as the channel well implant mask. The ARC layer may be patterned with the mask layer. Subsequently patterning the ARC layer may also be useful.

In FIG. 2e, the substrate is implanted with dopants to form a channel well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region. In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

After forming the channel doped well **135**, exposed portions of the gate electrode layer **244** are removed, as shown in FIG. 2f. In one embodiment, an anisotropic etch, such as a reactive ion etch (RIE) is performed to remove exposed portions of the gate electrode layer. The removal of the exposed portions of the gate electrode layer, in one embodiment, forms a first edge of a HV gate. The first edge, in one embodiment, corresponds to the source side of the HV gate. As shown, the gate dielectric layer **242** serves as an etch stop for etching the gate electrode layer. Leaving the gate dielectric layer can be advantageous since it can serve as an implant mask for subsequent ion implantation to protect the substrate from implant damage. Alternatively, the exposed portion of the gate dielectric layer can be removed. Removing the gate dielectric layer **242** to expose the substrate may also be useful. After etching the gate electrode layer, the mask layer **288** is removed.

As described, the implant mask, in one embodiment, serves as a channel well implant mask as well as a mask for patterning the first edge (e.g., source side) of the HV gate. By using the same mask to form the channel well and source side of the gate, the channel is self-aligned to the gate edge. This enables the channel length of the HV device to be well controlled to produce very short effective channel lengths to reduce $R_{ds, on}$ performance. In one embodiment, the effective channel length L is less than $0.4 \mu\text{m}$. In other embodiments, the effective channel length is less than $0.3 \mu\text{m}$. In yet another embodiment, the effective channel length is less than $0.25 \mu\text{m}$.

Referring to FIG. 2g, another mask layer **288b** is formed on the substrate and patterned. The mask layer is patterned to expose portions of the gate electrode layer **244** to be removed, protecting portions corresponding to gates of the LV devices. For example, the mask serves as a gate mask. Additionally, the mask layer **288b** protects the source portion of the HV region and the gate of the HV device. The exposed portions of the gate electrode layer **244** are removed to form gates of both the LV and HV devices.

After the gates are formed, the process continues to form the device, as illustrated in FIG. 1. The process, for example, includes forming extension regions **147** by ion implantation, followed by forming spacers **148** and **178** and diffusion regions **146** and **176**. The different types of extension and diffusion regions can be formed in different processes. For example, n-type diffusion regions are formed in one implant process and p-type diffusion regions are formed in another process. Furthermore, the diffusion and extension regions of the different device regions can be formed separately. Silicide gates and contacts can be formed after the diffusion regions are formed. For example, metals, such as cobalt are deposited on the substrate **101** and reacted to form silicide contacts and gates. The contacts and gates can be formed in the same or different processes. Unreacted metals are removed. A pre-metal and an inter-level dielectric layer can be formed in

which contacts and interconnects are formed. Additional interconnect levels can be formed by, for example, dual damascene techniques. Final passivation, dicing, assembly and testing may be performed to complete the IC.

An alternative embodiment of a process of forming a device **100** is shown in FIGS. 3a-g. Referring to FIG. 3a, a partially processed device is shown. The partially processed device is similar to that shown in FIG. 2c. A hard mask **388** is formed on the gate electrode layer **244**. The hard mask, for example, comprises silicon oxide. In one embodiment, the hard mask **388** is formed by CVD. The thickness of the hard mask, for example, is about 40 nm. Other materials, techniques or thicknesses may also be useful. For example, the hard mask may be formed from silicon nitride or other types of materials.

Referring to FIG. 3b, a soft mask **287**, such as photoresist, is formed on the hard mask **388**. The soft mask can be patterned by photolithographic techniques to form an opening **289** which can serve as a channel implant mask. An ARC layer can be provided between the soft mask **287** and the hard mask **388**. The pattern of the soft mask is transferred to the hard mask by, for example, RIE.

In one embodiment, the soft mask layer is removed after patterning of the hard mask, as shown in FIG. 3c. The substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45° . Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

As shown in FIG. 3d, another soft mask layer **287** is deposited on the substrate, covering the hard mask layer and exposed gate electrode layer. The soft mask is patterned by lithography. The soft mask protects portions of the hard mask layer **388** corresponding to the transistor gates. A new ARC layer can be provided between the soft mask **287** and the hard mask **388**. In FIG. 3e, exposed portions of the hard mask are removed, exposing portions of the gate electrode layer. Patterning of the hard mask layer can be achieved by, for example, RIE. The patterned hard mask serves as a gate mask. The soft mask, as shown in FIG. 3f, is removed after patterning the hard mask.

Referring to FIG. 3g, the gate electrode layer **244** is patterned to form gates in the LV and HV region by, for example etch. Patterning of the gate electrode layer can be achieved using, for example, RIE. The patterning of the gate electrode layer also erodes the hard mask, reducing its thickness. The hard mask **388** can be removed by, for example, a clean step

11

after etching the gate electrode layer. The process continues as previously described to complete the device.

Another embodiment of a process for forming a device **100** is shown in FIGS. **4a-g**. Referring to FIG. **4a**, a partially processed device is shown. The partially processed device is similar to that shown in FIG. **3a**.

Referring to FIG. **4b**, a soft mask layer **387**, such as a photoresist layer, is formed on the hard mask **388**. An ARC layer can be provided between the soft mask and the hard mask. The soft mask layer is patterned to expose portions of the hard mask **388**. The remaining portion of the soft mask corresponds to the HV gate in the HV region **160**. Patterning of the soft mask layer can be achieved by photolithographic techniques. In one embodiment, the patterned soft mask layer also exposes the hard mask in the LV region **110**.

As shown in FIG. **4c**, the pattern of the soft mask is transferred to the hard mask by, for example, RIE. The RIE removes exposed portions of the hard mask to expose portions of the gate electrode layer **344** in the HV region which are to be removed. In one embodiment, the removal of the hard mask also exposes the gate electrode layer in the LV region. After patterning of the hard mask, the soft mask layer is removed.

In FIG. **4d**, another soft mask layer **487**, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose portions of the gate electrode layer **344** in the LV region which are to be removed, protecting portions corresponding to gates of the LV devices. For example, the patterned soft mask serves as a gate mask for the LV region. The soft mask layer is removed from the HV region.

The gate electrode layer is patterned in FIG. **4e**. In one embodiment, the gate electrode layer in the LV region is patterned using the soft mask while the hard mask is used in the HV region. The use of the soft mask advantageously facilitates forming gates of critical dimensions in the LV region.

Referring to FIG. **4f**, another soft mask layer **487b**, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask. The soft mask layer can be patterned by photolithographic techniques to form an opening **489**. The patterned soft mask layer serves as a channel implant mask.

As shown in FIG. **4g**, the substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can

12

comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

In one embodiment, the implant mask and hard mask are removed after forming the channel well. For example, the implant mask is first removed followed by the removal of the hard mask. In some embodiments, the hard mask is remained. The process continues as previously described to complete the device after the formation of the gates.

An alternative embodiment of a process for forming a device **100** is shown in FIGS. **5a-g**. Referring to FIG. **5a**, a partially processed device is shown. The partially processed device is similar to that shown in FIG. **4c**.

As shown in FIG. **5b**, a soft mask layer **587**, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the gate soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose the HV region, protecting the LV devices from being processed. As illustrated by FIG. **5c**, the gate electrode in the HV region is patterned by, for example, RIE, to form the gate using the hard mask as the gate mask. After patterning of the gate electrode in the HV region, the soft mask layer is removed.

Referring to FIG. **5d**, another soft mask layer **587b**, such as photoresist, is formed over the substrate. An ARC may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to form an opening **589**. The patterned soft mask layer serves as a channel implant mask.

As shown in FIG. **5e**, the substrate is implanted with dopants to form a channel doped well **135**. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{ cm}^{-2}$ at an energy level of about 150 KeV.

Referring to FIG. **5f**, another layer of soft mask layer **587c**, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer is patterned by photolithographic techniques to expose portions of the gate electrode layer **344** in the LV region to be removed. Additionally, the soft mask layer remains in the HV region to protect it from being processed. The exposed portions of the gate electrode layer **344** are removed to form the gates of the LV device, as illustrated in FIG. **5g**.

Similar to the embodiment of FIGS. **4a-g**, the LV gates are patterned using a soft mask while a hard mask is used to

13

pattern the HV gate. After forming the LV gates, the soft mask is removed followed by removal of the hard mask over the HV gate. In some embodiments, the hard mask can remain. The process continues to complete the device as described previously.

The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

What is claimed is:

1. A device comprising:
 - a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
 - a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; and a doped drift well adjacent to a second edge of the gate.
2. The device of claim 1 further comprising spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.
3. The device of claim 1 wherein the substrate comprises: a drift isolation region in the doped drift well; and a doped deep well in the substrate of the active device region encompassing the doped drift well.
4. The device of claim 3 wherein the drift isolation region is disposed under the gate.
5. The device of claim 4 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:
 - the first heavily doped region is disposed in the doped channel well on a source side of the device; and
 - the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.
6. The device of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.
7. The device of claim 1 wherein the channel having a channel length that is controlled by ion implantation.
8. The device of claim 7 wherein the ion implantation comprises a tilt angle implant.
9. The device of claim 8 wherein a tilt angle of the ion implantation is about 1-45°.
10. The device of claim 7 wherein the ion implantation comprises multiple tilt angle implants which are rotated about a plane of the substrate.
11. The device of claim 10 wherein a tilt angle of the implant is about 1-45°.

14

12. The device of claim 7 wherein the ion implantation comprises a quad tilt angle implant which is rotated about a plane of the substrate.

13. A device comprising:

- a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
- a doped channel well disposed in the substrate adjacent to a first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate;
- a doped drift well adjacent to a second edge of the gate; and a drift isolation region in the doped drift well.

14. The device of claim 13 further comprising a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region.

15. The device of claim 13 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to a drain edge of the gate of the device.

16. The device of claim 13 wherein the drift isolation region is disposed under the gate.

17. The device of claim 13 wherein the first and a second heavily doped regions are provided adjacent to the gate, wherein:

- the first heavily doped region is disposed in the doped channel well on a source side of the device; and
- the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device.

18. The device of claim 13 wherein the channel having a channel length that is controlled by ion implantation.

19. The device of claim 18 wherein the ion implantation comprises a tilt angle implant, multiple tilt angle implants or quad tilt angle implant.

20. A semiconductor device comprising:

- a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric;
- first and second heavily doped regions adjacent to first and second edges of the gate;
- a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from the first heavily doped region adjacent to the first edge of the gate; and
- a doped drift well adjacent to a second edge of the gate, wherein a drift isolation region is disposed in the doped drift well.

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EXHIBIT D

**REDACTED IN
ITS ENTIRETY**

EXHIBIT E

**REDACTED IN
ITS ENTIRETY**